NONLINEAR SIMULATIONS AND THEORY OF THE FARLEY-BUNEMAN INSTABILITY IN THE *E*-REGION IONOSPHERE

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Meers Maxwell Oppenheim

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NONLINEAR SIMULATIONS AND THEORY OF THE FARLEY-BUNEMAN INSTABILITY IN THE *E*-REGION IONOSPHERE

Meers Maxwell Oppenheim, Ph.D.
Cornell University 1995

The Farley-Buneman instability is a collisional two-stream instability which often occurs in the equatorial electrojet, a stream of current that travels above the equator at an altitude of 90 - 120 km (the E region). In 1963, Farley and Buneman applied linear theory to show that this instability develops when the electrojet current exceeds a specific threshold. This thesis uses simulations and theory to explore the nonlinear behavior of this instability and the resulting waves. The simulations model wave behavior in the plane perpendicular to the Earth's magnetic field. They apply either a particle or a fluid model to describe ion behavior and use a fluid model for electron behavior. The results show the growth, saturation and nonlinear behavior of the instability. Our technique follows the saturated state of the instability for a much longer period of time than did the pure particle codes that preceded it. The simulations show: (1) spectra similar to those observed by radar, (2) nonlinear coupling to waves propagating perpendicular to the electron drift direction, (3) a saturated wave phase velocity below the velocity predicted by linear theory but above the acoustic speed, (4) substantial energy associated with waves traveling obliquely as well as parallel to the electron drift direction and (5) spectral asymmetries with respect to elevation angle which correspond to observed asymmetries. Nonlinear $\mathbf{E}\times\mathbf{B}_0$ electron motion dominates the behavior of the saturated waves. Motion pictures of the density waves show secondary waves traveling upward along the crests of the principal waves and downward in the valleys while eddies form at the edges. This thesis also develops a perturbation theory which explains the growth of these secondary waves.

Biographical Sketch

Meers Oppenheim was born in Bethesda, Maryland on November 2, 1962. His mother, Elizabeth Kaner Oppenheim escaped a small town in northern Wisconsin to attend graduate school at Columbia University in New York. His father, Joost Oppenheim, survived the holocaust by hiding with his brother in the attic of a Dutch family and later moved to New York and attended medical school. They were married in Seattle in 1961 and moved to Bethesda soon thereafter so that Joe could begin a career at the National Institutes of Health. He became one of the world's leading researchers in immunology.

Meers spent most of his childhood in Bethesda with his younger brothers, Monty and Matthew, and his sister, Emia. After growing to the excessive height of 4' 29", he graduated from Walt Whitman high school in 1980 and moved to Ithaca, New York to attend Cornell University. There he studied physics, history and volleyball. As a junior, he met an intelligent, athletic and vivacious young woman, Yvonne Bopp. After graduating and spending a summer bicycling around Europe, they moved to Northern California where Meers found a job in an engineering firm. They married in 1985 in Maryland and honeymooned in a tent in the Grand Canyon before returning to California.

In 1986, Yvonne and Meers moved back to Ithaca, so that Yvonne could attend the New York State College of Veterinary Medicine. There she would specialize in the medicine of exotic animals, particularly birds of prey. In need of more stimulating work than his lonely occupation as a consultant, Meers entered graduate school. After almost seven years and many trials and tribulations, he graduated with a Ph.D. in space plasma physics and a greatly improved tennis game.

I dedicate this thesis to my closest friend
Yvonne Oppenheim

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Chapter 1

Introduction

A Severe Strain on the Credulity

As a method of sending a missile to the higher, and even to the highest parts of the Earth's atmospheric envelope, Professor Goddard's rocket is a practicable and therefore promising device. It is when one considers the multiple-charge rocket as a traveler to the moon that one begins to doubt ... for after the rocket quits our air and really starts on its journey, its flight would be neither accelerated nor maintained by the explosion of the charges it then might have left. Professor Goddard, with his "chair" in Clark College and countenancing of the Smithsonian Institution, does not know the relation of action to re-action, and of the need to have something better than a vacuum against which to react ... Of course he only seems to lack the knowledge ladled out daily in high schools.

– New York Times Editorial, 1920

The Sun's energy drives winds at all elevations and latitudes in the atmosphere, causing a worldwide collection of atmospheric tides and dynamos. Above an altitude of 80 kilometers, the Sun's ultraviolet radiation ionizes an appreciable number of particles, creating the ionosphere. While neutral winds respond principally to temperature and pressure effects, charged particles also respond to electric fields and the Earth's magnetic field causing them to display even more complex behavior than the neutral particles. Charged particles also respond to electromagnetic radiation, making them important because of our use of these waves for communication and observation. This reactive behavior also makes charged particles far easier to study from the ground than neutral particles. The combination of easy

detection and complex behavior has made the study of charge particle dynamics in the atmosphere an important topic in geophysics for the last fifty years.

This thesis presents a detailed study of a single type of ionospheric plasma density wave observed frequently a few years after the invention of radar in the 1930s. While these waves exist between 90 and 120 kilometers altitude near the equator and the poles, we limit ourselves to studying those that occur near the equator and at high altitudes ($> 100km$). We further limit ourselves to studying those aspects of these waves which can be simulated by the computational techniques we developed. Within these strict limits, we endeavor to make a substantial contribution to the study of ionospheric physics.

1.1 Organization of the Thesis

We hope that this thesis is intelligible to anyone with a graduate level physics, astronomy or electrical engineering education. To that end, the first chapter presents much of the background material necessary for understanding the later chapters. This chapter begins with a brief historical introduction to our topic, followed by an explanation of the general characteristics of the ionosphere, the special nature of the E region, the specifics of the equatorial electrojet, and a brief, non-mathematical, explanation of the basic features of the Farley-Buneman instability. We follow this with a survey of the published literature with an emphasis on those works directly relevant to our simulations. A brief list of what is known about the Farley-Buneman instability and also of questions that remain unanswered follows. The last section of this chapter summarizes the simulations and theories presented in later chapters.

Chapter 2 explains and derives the linear results with the detail necessary for understanding the simulations and theories presented in later chapters. Readers familiar with the Farley-Buneman instability, will probably wish to skip or skim this chapter, returning to it from later chapters only when necessary. Chapter 3 describes the simulation methods with an emphasis on original techniques. The reader interested in numerical methods of modeling E-region phenomena should find this chapter useful. Chapter 4 summarizes the results of our simulations, distilling a huge amount of information into a limited number of words and figures. We emphasize looking at the simulations in a manner comparable to experiments

and theories. Chapter 5 uses analytical techniques to explain a number of features seen in the simulations with nonlinear theories. Chapter 6 further discusses the literature, comparing the results obtained from our simulations to observations and theory.

1.2 Historical Background

As early as the late middle ages, navigators observed that magnetic compass readings taken near the equator varied slightly, depending on the time of day. In 1839, Friedrich Gauss speculated that these fluctuations resulted from the presence of large currents in the atmosphere. Balfour Stewart proposed in 1882 that these currents resulted from a solar driven dynamo in an ionized region of the upper atmosphere which he called the ionosphere. Studies of this region received a tremendous boost from the invention of radio by the 21 year old Italian, Guglielmo Marconi, in 1895. In 1901, he astonished both scientists and the public by transmitting a Morse code from England to Newfoundland, a distance of over 2000km. A year later, Arthur Kennelly and Oliver Heaviside proposed that the waves overcame the curvature of the Earth by reflecting from a layer of ionized particles at 80km altitude.

With the advent of widespread use of radio in the 1910s and 1920s, attempts to characterize the ionosphere began in earnest. *Appleton and Barrett* [1925] demonstrated that radio waves bounced off an electrically active layer at 100km which they called the E region (E for electric). Later, Appleton used interference patterns to identify electrically active layers above and below the E region, which he labeled the F and D regions respectively. Appleton received the 1947 Noble prize for this work.

By the late 1920s, attempts to understand the ionosphere's fundamental properties and determine the plasma density as a function of height were fully underway. *Pedersen* [1927*a*] and *Chapman* [1931] theorized that these layers were formed by solar ultraviolet radiation. Also, at this point, *Chapman* [1931], *Pedersen* [1927*b*] and others mistakenly believed that large numbers of negatively charged ions were responsible for much of the behavior of this region. It was not until the mid 1940s that this was belief was overturned. [*Bates and Massey*, 1946; *Bates*, 1994; *Fukushima*, 1994]

Rockets can obtain direct measurements of ionospheric quantities which can only be inferred by ground based detectors. After World War II, the allies captured numerous German V-2 rockets. Many of these were brought to the White Sands Missile Range in California where they were used to study the atmosphere above the altitude accessible to balloons or airplanes. Using ten tons of alcohol and liquid oxygen these rockets could ascend to 170km carrying a multiple tool instrument pack. The first flights of these rockets ended with a crash landing which caused the complete disintegration of the instrument packs and, typically, created a crater 20 meters in diameter and 9 meters deep. After overcoming these and other problems, the V-2 rockets and their antecedents carried instruments to high altitudes, giving researchers a direct measure of the composition and dynamics of the atmosphere, the ionosphere and beyond. [*Friedman*, 1994]

1.3 The Ionosphere

Remote sensing by radars and *in-situ* measurements by rockets remain the principal tools with which we investigate the ionosphere today. However, we now send and receive radio waves with antenna arrays sometimes as large as a mountain valley. Modern rockets sample electric fields, currents and conductivities as well as the density and composition of the ionosphere. A number of other tools have been added to our diagnostic tools. Satellites probe the upper elevations of the ionosphere (above $\approx$250km) with a repetition that enables us to build a detailed understanding of processes in the upper ionosphere and above. We also study the ionosphere by modifying it. Megawatt radio frequency oscillators heat local regions of the ionosphere. Rockets disperse probing material such as barium into it. These tools have enabled us to characterize the ionosphere in detail. Figure 1.1 shows the temperature and plasma density of the atmosphere as a function of height. Figure 1.2 shows a modern understanding of atmospheric composition.

Two regions of the upper atmosphere received the most attention from early observers. The polar regions drew scientist's interest with displays of aurora and magnetic variability. Likewise, the equatorial regions showed substantial magnetic activity by causing compass needles to waver. The explanation for these equatorial variations lies with a region called the equatorial electrojet. This thesis attempts to expand out understanding of the physics underlying meter-scale density irregu-

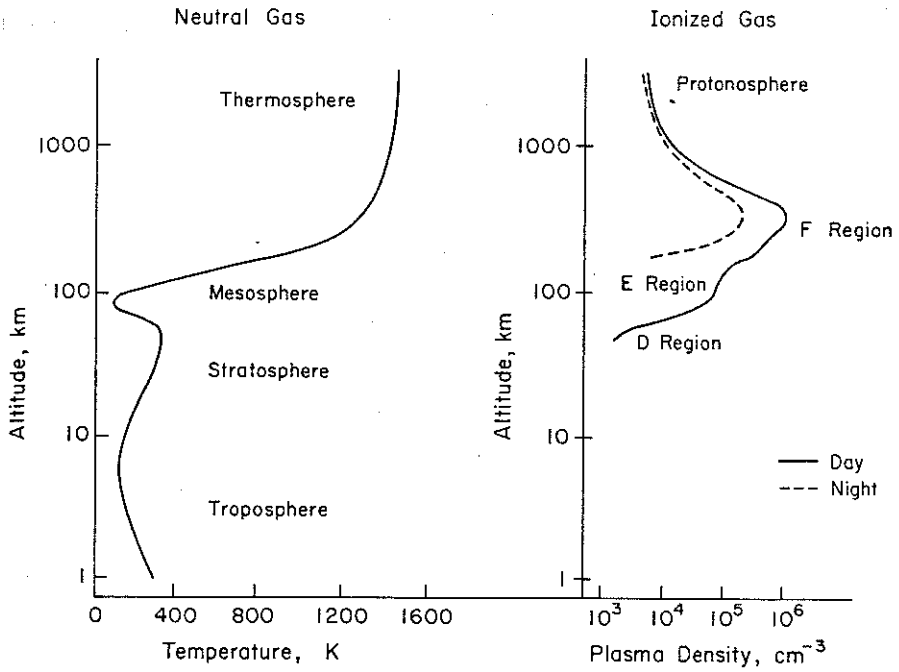


Figure 1.1: Typical equatorial daytime temperature and plasma density as a function of height. Reprinted from *Kelley* [1989].

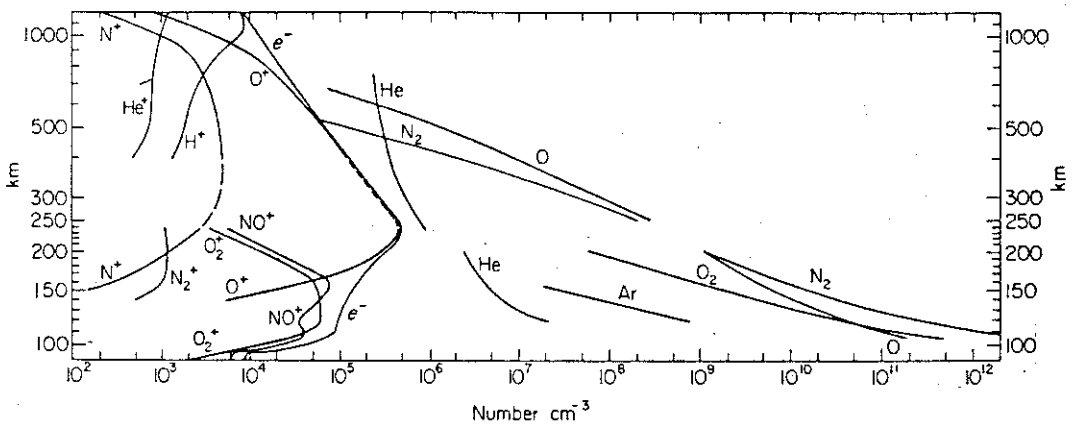


Figure 1.2: Major components of the atmosphere as a function of height. These were measured with a mass spectrometer above White Sands, New Mexico in day time and quiet solar conditions. (After *Johnson* [1969]).

larities in this region.

1.4 The Equatorial Electrojet

The Sun's heating combined with the Earth's rotation causes pressure gradients and winds at all elevations in the atmosphere. At an altitude of approximately 100km a complex process transforms some of this wind energy into currents. Ground based magnetometers attest to the strength of these currents. Their source lies in the differing responses of ions and electrons to the neutral wind and in the change in ionospheric conductivity with altitude. These physical characteristics result in a large episodic current called the equatorial electrojet. This section briefly outlines the processes leading to these currents. *Kelley* [1989] provides a more detailed explanation of the physics of this system and other ionospheric systems. For a more complete review of the models describing equatorial electrojet physics, see *Forbes and Lindzen* [1976], *Forbes* [1981] or *Fejer* [1981]. For a good review of ionospheric waves and other irregularities see *Fejer and Kelley* [1980].

Near the magnetic equator, neutral winds drive currents which lead, ultimately, to the electrojet and the Farley-Buneman instability. These winds arise primarily from diurnal and semidiurnal tides driven by solar heating, while lunar gravitational tides play a smaller role. For a discussion of the complex behavior of winds in the *E*-region see *Chapman and Lindzen* [1970].

Around 100km altitude, O_2^+ and NO^+ ions make up the vast majority of the positively charge particles. These ions are about 50,000 times as massive as the electrons. Further, they collide with neutral particles far more frequently than they rotate around the geomagnetic field lines, so that when a neutral wind blows, it sweeps along the ions essentially as if the Earth's magnetic field were absent. The electrons, at this altitude, revolve around the Earth's magnetic field lines about 16 times for every time they collide with a neutral particle. This means they are, largely, stuck on the Earth's magnetic field lines. The resulting difference between ion and electron velocities creates a current. This $0 - 50 m/s$ current is not strong enough to explain the currents seen along the equator. Another, subtler process leads to a major amplification of this current creating the electrojet.

Wind-driven currents cause a horizontal electric field. When a current travels through a medium having a finite conductivity, Ohm's law, $\mathbf{J} = \sigma_c \mathbf{E}$, requires

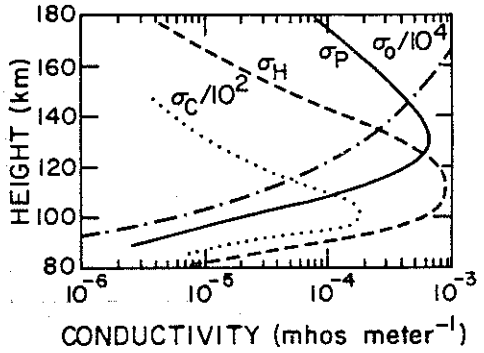


Figure 1.3: Conductivities as a function of height. (After *Forbes and Lindzen* [1976]).

an electric field to develop. In the electrojet, the conductivity, σ_c, is called the Cowling conductivity, a combination of the Pederson and Hall conductivities. This east-west pointing electric field and the Earth's north-south pointing magnetic field influence electron behavior, causing the electrons to $\mathbf{E}\times\mathbf{B}$ drift in an up-down direction. The ions do not $\mathbf{E}\times\mathbf{B}$ drift because of their high mass and frequent collisions with neutral particles.

Above and below the *E*-region lie regions of reduced conductivity as shown in figure 1.3. Currents flowing into these regions encounter reduced conductivities and a net charge accumulates. This charge imbalance causes a vertical electric field as high as 20mV/m across the E region. This field is an order of magnitude larger than the horizontal field and it drives the electrons to $\mathbf{E}\times\mathbf{B}$ drift in an east-west direction. This electron drift speed can be as high as 800m/s which means that the equatorially directed (east-west) currents are also an order of magnitude higher than the initial wind driven currents. These currents are called the electrojet. Figure 1.4 illustrates the geometry of the electrojet.

The electrojet occurs only along the Earth's magnetic equator for two reasons. First, along the magnetic equator, the dominant winds blow perpendicularly to the Earth's magnetic field lines. The ions are swept along but the electrons remain tied to the field lines. If a significant component of the wind blows parallel to the magnetic field, the highly mobile electrons will travel along the field lines with the ions and no current will develop. Second, the development of the electrojet requires the a vertical charge separation across the *E*-region. Along the magnetic equator the field lines are horizontal, preventing electrons from traveling vertically

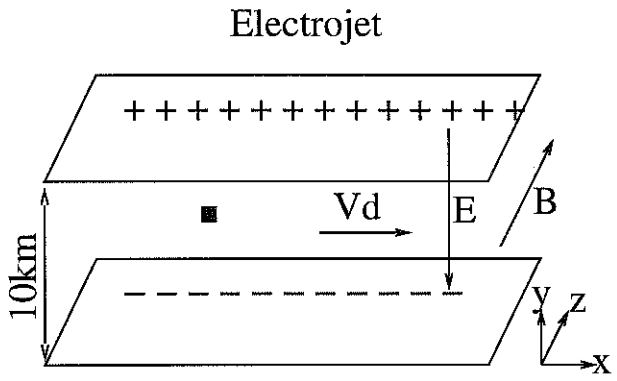


Figure 1.4: Geometry of the electrojet at, roughly, 100km. The little black box represents a two dimensional area which is more than ten times the area modeled in our simulations.

to cancel the accumulated charge separation.

1.5 The Farley-Buneman Instability

The Farley-Buneman instability [*Farley*, 1963; *Buneman*, 1963], also called the two-stream instability, provides a mechanism to excite the meter scale irregularities observed in the ionosphere considered responsible for the type I radar echoes detected during strong electrojet conditions. These echoes have typically narrow Doppler spectra with mean frequency shifts corresponding to wave phase velocities at or slightly above the ion acoustic speed in the plasma. Figure 1.5 shows an example of these waves.

The source of free energy for this instability comes from the differing motions of the electron and ion fluids found in the electrojet. If the relative electron drift speed, V_d, exceeds the ion acoustic speed, C_s, by a fraction dependent on other physical parameters, then the Farley-Buneman instability generates meter scale electrostatic plasma waves in the plane perpendicular to the ambient magnetic field. This threshold condition is easily met in both the equatorial and auroral E regions. Chapter 2 explains, in detail, the linear mechanisms leading to the this instability.

The huge radar facility at Jicamarca, Peru, periodically observes type I echoes from waves traveling vertically in the equatorial electrojet. Normally, one expects Farley-Buneman waves to travel in the same direction as the electrojet, east to

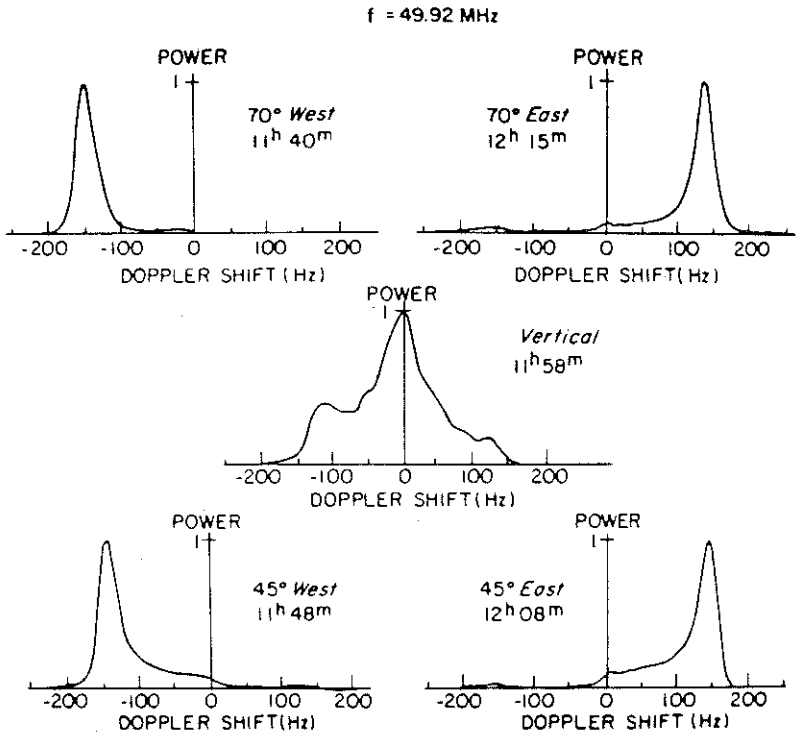


Figure 1.5: Type I radar echoes obtained at Jicamarca, Peru at a number of angles with respect to vertical (elevation angle) and perpendicular to the Earth's magnetic field. Note that the Doppler shift exceeds the spectral width and that the vertical echo has no net Doppler shift. (After *Cohen and Bowles* [1967])

west or, occasionally, west to east. These vertically traveling waves result from a coupling between the Farley-Buneman instability and a related instability called the gradient drift instability (described in *Ronchi* [1990]). Typically, gradient drift waves occur in the lower portion of the electrojet where the vertical electron density gradient is sufficient to initiate the instability. These waves occur more frequently and have far longer wavelengths than do two-stream waves. One result of these gradient drift waves is a horizontal electric field sufficient to initiate vertically traveling two-stream waves. Radars sensitive to type I echoes frequently observe these secondary two-stream waves.

In the upper part of the electrojet, the electron density gradient drops below the necessary threshold for initiating the gradient drift instability. In this region, one expects "pure" two-stream waves, independent of gradient drift waves. The simulations and theories presented in this thesis focus on these pure two-stream waves.

1.6 Review and Discussion of the Published Literature

Amateur radio operators probably were probably the first people to observe the strong echoes and distinct Doppler shifts of what we now call type I radar echoes. The first reference to these echoes in the scientific literature which we found was by *Bowles* [1954], who used a VHF radar in the polar ionosphere to measure a Doppler shifted radar spectrum. Five years later *Bowles et al.* [1960] saw similar echoes in the equatorial electrojet. Shortly after this *Farley* [1963] and *Buneman* [1963] presented their theories explaining the origin of this phenomenon. Since that time, hundreds of papers describing observations and theories have been published. Numerous review papers and books, including *Farley and Balsley* [1973], *Farley* [1979], *Fejer and Kelley* [1980], *Farley* [1985] and *Kelley* [1989] survey this topic with varying degrees of detail. In the following sections, we summarize some of the more important results presented in a large number of papers. We focus on features of the electrojet directly relevant to the equatorial Farley-Buneman instability and the simulations presented in this thesis.

1.6.1 The Observational Literature

Bowles et al. [1960] pointed out many of the features of equatorial electrojet radar echoes. Radars observe, principally, two types of echoes. A small, relatively insensitive radar can detect Type I echoes. These echoes occur only periodically and have pronounced Doppler shifts and narrow spectra. We now attribute these waves to the two-stream or Farley-Buneman instability. Type II echoes were identified after the construction of the large radar antenna at Jicamarca, Peru [*Cohen and Bowles*, 1967; *Balsley*, 1969]. These echoes appear more frequently than type I echoes and have broader spectra and small Doppler shifts. We now attribute these waves to the gradient drift instability [*Maeda et al.*, 1963; *Rogister and D'Angelo*, 1970; *Sudan et al.*, 1973]. For a general review and an explanation of relatively recent theoretical studies of type II phenomena, see *Ronchi* [1990] or *Ronchi et al.* [1991].

Radars detect only plasma density irregularities having wavelengths of exactly one half the radar wavelength and wave vectors aligned exactly with the radar propagation direction (i.e., $\mathbf{k}_{wave} = -2\mathbf{k}_{radar}$). This makes radars very sensitive to the alignment of the irregularities. *Bowles et al.* [1963] observed that both type I and II echoes represent density irregularities strictly aligned perpendicular to the Earth's magnetic field. When the radar beam was shifted only 2° away from perpendicular, the echoes disappeared.

The classic example of equatorial type I radar echoes was published by *Cohen and Bowles* [1967]. Figure 1.5 shows a number of spectra from their observations made at Jicamarca, Peru. These coherent radar echoes have a Doppler shift which exceeds the spectral width. They point out that this Doppler shift is less than that predicted by linear theory. This observation has been corroborated by numerous radar studies around the equator including, *Balsley and Farley* [1971], *Prakash et al.* [1974], *Crochet et al.* [1979], *Hanuise and Crochet* [1981], [*Kudeki et al.*, 1987], *Kudeki and Farley* [1989] and others. *Cohen and Bowles* [1967] also pointed out that their results indicate an asymmetry in the spectra not predicted by the theories of *Farley* [1963]. *Dougherty and Farley* [1967] also observed this asymmetry, stating that "the *upward* irregularities (observed at angles of incidence east of the zenith) are stronger than the *downward* irregularities (observed at angles of incidence east of the zenith)."

In 1971, *Balsley and Farley* [1971] used three radar frequencies (16MHz, 50MHZ and 146MHz) to probe the electrojet. They also detected type I instabilities with Doppler velocities corresponding to the acoustic velocity. The relationship between the velocities at the different frequencies roughly matched linear kinetic theory. These observations were confirmed in Africa by *Crochet et al.* [1979] and *Hanuise and Crochet* [1981] using HF and VHF radars and in India by *Prakash et al.* [1974]. Both teams observed the waves traveling at velocities close to the acoustic speed at all elevation angles rather than the drift velocity times the cosine of the elevation angle as predicted by linear theory. *Ierkic et al.* [1980] quantified this observation, asserting that the power received from type I echoes drops 0.3 dB/degree as the elevation angle increases from horizontal for 3m waves.

Before 1983, rockets designed to study the ionosphere flew through the electrojet at such high velocities that few measurements could be made, though *Prakash et al.* [1972] did manage to obtain some useful data. On March 12, 1983, the CONDOR rockets were launched from Peru with a flight trajectory designed to maximize the time spent in the electrojet [*Pfaff et al.*, 1987*a*]. These *in-situ* measurements were taken in conjunction with radar observations [*Kudeki et al.*, 1987]. *Pfaff et al.* [1987*a*] summarized their results about primary two-stream waves (i.e., two-stream waves of the upper electrojet, not coupled with gradient drift waves) by listing the following seven points. (1) At 108km, coherent oscillations of electric field and density indicate the presence of two-stream waves. (2) These waves travel principally horizontally and have an altitude range of 2km. (3) The wavelength observed to have the largest amplitude matches the wavelength which linear theory shows to have the highest growth rate. (4) The phase velocity of these waves "may have been comparable to the linear theory result." (5) "Because the primary two-stream waves propagate horizontally along the direction of the current, VHF backscatter radars looking vertically or obliquely would not be expected to observe these waves, as predicted by *Lee et al.* [1974]. The constant phase velocities observed by the radars are not in contradiction with the horizontal phase velocities observed *in-situ*, since the oblique and vertical two-stream waves observed by backscatter radars at the equator could originate primarily in the interaction region of large-scale waves and not from the laminar topside layer." (6) The phase velocity of the horizontally propagating waves does not decrease (as one would expect) at the highest elevations where the current density drops. (7) "The strong compo-

nent of vertical waves associated with the "pure" two-stream region observed both by the Jicamarca radar and the in-situ electric field instruments may possibly be explained by" coupling to the gradient drift instability as explained in *Pfaff et al.* [1987*b*]...

Measurements made by rockets have a number of important limitations. *Pfaff et al.* [1987*a*], shows a series of power versus frequency measurements where the frequency was measured in the rocket's frame of reference. Unfortunately, the rocket accelerates, spins and, sometimes, precesses. None of these effects have been removed from the data. This makes it difficult to determine the wavelength or wave velocity relative to the ground. Also, rockets have a very hard time measuring temperatures. This means we don't know the acoustic speed which, according to linear theory, plays a role in setting the threshold at which two-stream waves begin grow.

Recently, *Kudeki and Farley* [1989] measured the deviation of type I echoes from alignment perpendicular to the Earth's magnetic field. These measurements, performed with the multi-baseline radar interferometer at Jicamarca, Peru, show the existence of strong echoes from short wavelength modes propagating in slightly off-perpendicular directions at aspect angles below 0.4°. The aspect angle is defined as the smallest angle between a given line and any line perpendicular to the Earth's magnetic field (see figure 2.1 in Chapter 2). These observations seem to agree with the linear theories of *Ossakow et al.* [1975].

In 1994, the Guara rocket campaign sent the CUPRI mobile radar unit to Brazil to collaborate with R. Pfaff and others in making more accurate *in-situ* and radar measurements of electrojet waves [*Swartz and Pfaff*, 1995]. Though the results of these measurements are currently being analyzed, *Pfaff et al.* [1995] showed preliminary results of a rocket flying through a pure two-stream instability with approximately $600 m/s$ electron drift speed.

1.6.2 Review and Discussion of the Theoretical Literature

This section reviews the development of our theoretical understanding of the Farley-Buneman instability starting with the work of *Farley* [1963] and *Buneman* [1963] and extending to the elaborate kinetic studies of *Dimant and Sudan* [1995].

In 1963, *Farley* [1963] attributed type I radar echoes to a two-stream instabil-

ity. Using a fully kinetic theory, he demonstrated that large electrojet currents become unstable and develop meter scale-length waves. He described the underlying mechanism as a streaming instability analogous to the instability which occurs when two beams of electrons pass through each other (see, for instance, *Nicholson* [1983]).

Farley's explanation implies that the two-stream instability is fundamentally kinetic, which it is not. In 1963, Buneman published a fluid theory explaining the same waves using a simpler, pure fluid approach. This explanation also has problems. While the kinetic approach shows a peak growth rate at a finite wavenumber, Buneman's theory predicts the instability growth rate peaking for $k_x \to \infty$. Clearly, at least a partially kinetic model is needed. *Schmidt and Gary* [1973] used a kinetic model for ions and a fluid model for electrons to show that Farley's kinetic description of the two-stream instability predicts a finite wavelength of maximum growth. *Kelley* [1989] used these calculations and the results of *Pfaff et al.* [1987*b*] to show that the dominant wavelengths observed by rockets match the growing wavelengths predicted by Schmidt and Gary. This comparison depends on correctly determining wavelengths from rocket data, a task made difficult by the complications discussed above. Section 2.3 contains a derivation of the linear kinetic dispersion relation which results in a wavelength of maximum growth.

Ossakow et al. [1975] extended the linear kinetic theory to demonstrate that the peak growth rate occurs for waves with a small wavevector component parallel to the Earth's magnetic field. These waves have an aspect angle of only a fraction of a degree, small enough to have been overlooked. Even the largest radars have finite beam widths of at least 1° degree and sensitivity in side lobes. This makes them sensitive to field aligned density irregularities even when they are not looking directly perpendicular to the magnetic field. Not until *Kudeki and Farley* [1989] studied type I waves with sufficient sensitivity to the aspect angle was this result corroborated.

Type I echoes are not the only radar reflections resulting from waves in the *E*-region. *Maeda et al.* [1963] first described the mechanism underlying type II echoes, the other common radar signal which originates in the equatorial electrojet. In contrast to the Farley-Buneman instability, the gradient in the plasma density plays a crucial role in the dispersion relation of this instability, now called the gradient drift instability. *Rogister and D'Angelo* [1970] showed that this instability

occurs at considerably longer wavelengths than the Farley-Buneman instability. It is presumed that radars sensitive to 3m waves detect type II echoes because energy turbulently cascades from the longer, dominant wavelengths to shorter, measurable wavelengths. The gradient drift instability has no threshold which the electron drift velocity must exceed to initiate wave growth, unless one accounts for the influence of ion recombination. *Fejer et al.* [1975] added this effect and also showed the effects of adding a zero-order ion drift velocity.

Sudan et al. [1973] extended the theory of electrojet instabilities to include coupling between gradient drift and two-stream waves. The electric fields which result from long wavelength gradient drift waves may be sufficient to initiate two-stream waves. These secondary two-stream waves travel vertically instead of horizontally as do the "pure" two-stream waves. Simulations of weak electrojet conditions by *McDonald et al.* [1975], *Ferch and Sudan* [1977] and *Keskinen et al.* [1979] support this idea of a two-step mechanism to generate vertically oriented type I waves. The large radar at Jicamarca, Peru points almost directly upward, measuring vertically oriented irregularities far more readily than horizontal ones. This radar detects type II echoes resulting from gradient drift waves, type I echoes resulting from secondary two-stream waves, and vertical irregularities associated with pure two stream waves in the upper electrojet (see 1.5) [*Kudeki*, 1983].

Radars have a difficult time isolating secondary two-stream waves from pure two-stream waves because they average together large volumes of space. The huge radar at Jicamarca has a beam width of one half a degree (full-width at half-max) while the much smaller Cornell University Portable Radar (CUPRI) spans five degrees. At 100km range, Jicamarca averages the waves contained in a box roughly ten kilometer on a side while CUPRI averages ten times that area. Generally, only smaller radars can look in a direction which allows them to detect horizontally oriented irregularities. These radars average together such a huge swath of the *E*-region that they probably detect both secondary echoes from the middle electrojet and pure type I echoes from the upper electrojet simultaneously.

The rocket data does not seem to agree with the radar measurements. These *in-situ* measurements indicate that, in the lower electrojet, type II echoes exist independently of type I echoes. In other words, no secondary two-stream waves were seen by *Pfaff et al.* [1987*a*]. Additionally, the most recent results of the Guara rocket campaign, conducted in the fall of 1994, indicate that pure two-

stream waves, seen in the upper electrojet, appear to have higher phase velocities and longer dominant wavelengths than those seen by radar.

Various mechanisms have been proposed to explain saturation of the two-stream instability. [*Rogister*, 1972] presented a quasilinear theory leading to a mechanism which saturates the irregularity amplitude at a few percent. The suggested mechanism effectively increases the acoustic speed until the instability becomes marginally stable. *Weinstock and Sleeper* [1972] challenge the assumptions of this paper by pointing out that a one dimensional model inadequately describes the nonlinear behavior of the Farley-Buneman instability. They propose an alternate mechanism in which electron trapping transfers energy from the wave to the electrons. Neither of these papers take into account the nonlinear electron drifts which the fluid equations imply should play a critical role in any nonlinear theory. *Sudan* [1983] points out this failing and proposes that short wavelength waves perturb the orbits of the electrons nonlinearly, causing an 'anomalous' or 'wave-induced' diffusion. This causes a nonlinear modification of the wave's phase velocity, eventually bringing about their saturation. Although Sudan's theory accounts nicely for the observed saturation at the acoustic velocity, the question of whether or not anomalous diffusion dominates the nonlinear processes is still not settled.

Kudeki and Farley [1989] proposed an alternative mechanism for saturation. Nonlinear mode coupling of energy from linearly excited modes to highly damped modes parallel to the magnetic field could play a substantial role in saturating the system. They provide a limited amount of observational support and a minimal theoretic framework for this plausible hypothesis. Unfortunately, the simulations discussed in this thesis are unable to evaluate this hypothesis since they are limited to the two-dimensional plane perpendicular to the magnetic field.

The recent papers of *Hamza and St.-Maurice* [1993*b*], *Hamza and St.-Maurice* [1993*a*] and *St.-Maurice et al.* [1994] study the Farley-Buneman instability in the polar zone using strong turbulence theory. They build on the work of *Sudan* [1983] and *Sahr* [1990] to study the effects of mode coupling on two-stream waves. By assuming that the ion-neutral collision rate exceeds the dominant mode frequency of the two-stream waves, they limit themselves to a regime inappropriate for the equatorial two-stream. Nonetheless, these papers use relatively easily understood arguments about mode coupling in an *E*-region fluid. *Hamza and St.-Maurice* [1993*b*] also contains a useful summary of recent theoretical developments in the

areas of quasi-linear and weak turbulence theories. This paper predicts that the saturated two-stream waves should travel at a velocity below the acoustic speed, a result not supported by observations. Nonetheless, techniques similar to those used in these papers may prove useful in explaining type I echoes, though we have not done this analysis.

In the last decade, a number of theorists have attempted to refine the linear model of the Farley-Buneman instability, improving the collision operators of the Fokker-Planck equation which, previously, were modeled using a simple BGK model of electron collision effects [*Bhatnagar et al.*, 1954; *Greene*, 1973]. *Stubbe* [1989] found that such modifications affect the dispersion relation in a potentially relevant manner, though this paper is so mathematically complex that drawing any conclusions from it has proven difficult. *Dimant and Sudan* [1995] take these improvements one step further by using the complete Boltzmann collision operator. They focus principally on complex electron kinetic effects and have seen a number of new, long wavelength effects in the resulting dispersion relations. These new results mainly apply to the extreme conditions found at the poles or at low altitudes. All these studies are linear and, therefore, unable to discuss saturation.

1.6.3 Review and Discussion of the Computational Literature

One might think that recent theoretical developments are inseparable from recent numerical developments. However, in this field, the theoretical developments have proceeded largely independently of simulations. This may have been due to the difficulty of obtaining useful numerical results. Nonetheless, I shall briefly describe the simulations that precede those described in this thesis.

The earliest simulations of the Farley-Buneman instability were done in the late 1970s by Newman and Ott [*Newman*, 1979; *Newman and Ott*, 1981]. These were pure fluid simulations with an enhanced, wavelength-dependent viscosity to model the important Landau damping effect. This approach is not vastly different from our own two-fluid model, except that the algorithm which they applied to solve for the electron behavior has an important flaw. We describe the origin of this flaw and how to correct it in section 3.1. This numerical flaw generated a substantial amount of short wavelength noise which they eliminated by filtering. Even so, they were unable to model the two-stream wave behavior in saturation, partially due

to the limited computational power available at the time. The principal result of their work was to identify the turning of the wave propagation direction away from the direction of the electron drift.

The next set of simulations was conducted in the mid-1980's by *Machida and Goertz* [1988]. They used a pure particle-in cell (PIC) [*Birdsall and Langdon*, 1985] code to model the polar Farley-Buneman instability. Much larger electric fields exist in the polar regions than in the equatorial electrojet, causing larger two-stream density perturbations at the poles. In the early 1980s, *Schlegel and St.-Maurice* [1981] and *Nielsen and Schlegel* [1985] observed electron heating associated with polar two-stream waves. Modeling this requires a numerical method, such as a PIC method, which incorporates electron kinetic effects. However, pure PIC codes also suffer from a number of problems as described in section 3.2.1.

The principal purpose of these simulations was to distinguish between proposed electron heating mechanisms, a topic upon which our work only touches. Machida and Goertz concluded that the mechanism they observed was compatible with *Primdahl* [1986], *Robinson* [1986] and *Sudan* [1983] but incompatible with the mechanism proposed by *Schlegel and St.-Maurice* [1981].

These simulations modeled wave activity in a plane parallel to the Earth's magnetic field and perpendicular to the plane used by *Janhunen* [1994*b*] and ourselves. This geometry neglects the largest nonlinear component, those due to nonlinear $\delta\mathbf{E}\times\mathbf{B}_0$ effects. Nonetheless, these simulations show a number of interesting features of the Farley-Buneman instability not related to electron heating: (1) though *Machida and Goertz* [1988] observe the initially growing waves propagating with a non-zero aspect angle, as predicted by linear theory, nonlinear effects cause the dominant modes of the saturated system to parallel the electron drift direction, (2) they observe a wave phase velocity somewhere between the acoustic speed and the drift speed, though the necessarily low resolution of these simulations makes a clear evaluation of the wave speeds difficult, and (3) they observe a coupling to long wavelength modes. Again, these simulations neglect the dominant nonlinear term in the equations and this limits their usefulness.

The most recent PIC simulations were conducted by *Janhunen* [1994*b*]. These simulations look at the plane perpendicular to the Earth's magnetic field and include the dominant nonlinear term neglected by Machida and Goertz. They also suffer from the problems of pure PIC codes described in section 3.2.1. However,

Janhunen did see two-stream waves develop and propagate, though he was unable to reach saturation due to the computational expense of these simulations. The main result he observed was the turning of his waves so that $\mathbf{k}\cdot\mathbf{E}_0 < 0$ where $\mathbf{k}$ is the wavenumber and $\mathbf{E}_0$ is the driving vertical electric field. This result agrees with that of *Newman and Ott* [1981] and with our own results. His simulations also showed wave heating similar to that of *Machida and Goertz* [1988]. He also claims that the waves travel at or close to the linear phase velocity, a result which disagrees with our own observations. This probably results from the failure of his simulations to reach saturation or because of the problems with the numerical technique use. In a follow up paper, *Janhunen* [1994*a*] proposed that saturation results from the turning of the wave away from the drift direction to a direction which, according to linear theory, is slightly damped. However, he gives no explanation of why the waves turn, or what happens to the remaining horizontal waves where $\mathbf{k}\cdot\mathbf{E}_0 = 0$.

1.6.4 Resolved Issues and Unresolved Questions

After all these studies and articles, what do we know about the type I echoes of the equatorial E region? What is unresolved? With a reasonable degree of certainty, we know the following:

1. The underlying mechanism of type I echoes is the Farley-Buneman instability as described in section 1.5 and in chapter 2.

2. In the upper electrojet, pure two-stream waves propagate horizontally with spectral widths considerably smaller than the Doppler shift, as shown in the figure 1.5.

3. In the lower electrojet, coupling between gradient drift waves and two-stream waves make for a more complex picture.

4. The peak wavelength and range of wavelengths of type I echoes matches the predictions of the linear kinetic Farley-Buneman instability as shown by *Schmidt and Gary* [1973].

5. While electron heating plays a role in polar two-stream waves, its role is less pronounced and possibly absent from equatorial two-stream waves.

The following lists many of the unresolved questions about the type I density irregularities:

1. What is the saturation mechanism?
2. Why do these waves travel at the acoustic velocity rather than the higher velocity predicted by linear theory?
3. Why does the phase velocity not change as the elevation angle changes?
4. Why are type I echoes associated with vertically oriented irregularities, when the linear theory tells us such irregularities are strongly damped?
5. Why does the radar echo power diminish with elevation angle by about 0.3 dB/degree for 3m waves?

1.7 Summary of Current Research

A numerical simulation is a controlled theoretical experiment with excellent diagnostic capabilities. It gives us the ability to ask specific questions about a particular theoretical model. Examples are what is the importance of nonlinear ion kinetic effects? Or, if we use a limited theoretical model, how do the results agree with the experimental data? Ultimately, the purpose of these simulations is to improve our understanding of the physical processes occurring in the E region ionosphere. Sophisticated numerical techniques and unprecedented computational power enables us to address a number of longstanding questions about two-stream waves. The following summarizes the main results described in this thesis.

Our code applies standard PIC methods to the ions with the addition of the following features. We calculate ion velocities in three dimensions while tracking ion positions and the resulting electric fields only in the plane perpendicular to the Earth's magnetic field. We model ion-neutral collisions by elastically colliding the ions with a thermal distribution of neutral particles and use a short wavelength filter to halve the particle noise.

Our code uses a fluid model of electron behavior which neglects electron inertial effects. This makes the shortest timescale in the simulation the electron-neutral collision rate. From the inertialess electron momentum equation, assuming an

electrostatic system and quasineutrality ($\nabla\cdot\mathbf{J} = 0$), one can derive the following equation in the plane perpendicular to the magnetic field ($\mathbf{B} = B_0\hat{\mathbf{x}}$):

$$n\nabla^2\phi + \nabla\phi\cdot\nabla n + \kappa\hat{\mathbf{z}}\cdot\nabla\phi\times\nabla n =$$
$$\mathbf{E}_0\cdot\nabla n + \kappa\hat{\mathbf{z}}\cdot\mathbf{E}_0\times\nabla n + (\gamma T_e/e)\nabla^2 n + \Gamma\nabla\cdot\mathbf{J_i}$$
$$\kappa \equiv \Omega_e/\nu_e(\gg 1), \quad \Gamma = (m_e/e^2)(\nu_e^2 + \Omega_e^2)/\nu_e \tag{1.1}$$

where all units are in MKS and ϕ is the potential, n is the density, Ω_e is the electron cyclotron frequency, ν_e is the electron-neutral collision rate, E_0 is the driving electric field, T_e is the electron temperature in Joules, m_e is the electron mass and J_i is the ion current determined from the ion particles.

The nonlinear term, $\kappa\hat{\mathbf{z}}\cdot\nabla\phi\times\nabla n$, in (1.1) contains the factor κ which, in the case of the E region, is a large number (≈ 100). This makes the nonlinear term the same order of magnitude as the linear terms even for fairly small density variations. This nonlinear term couples cross-propagating waves making the behavior of the electron fluid fundamentally two dimensional.

Numerically solving (1.1) for ϕ proved very difficult. Using finite difference equations, we convert equation (1.1) into a matrix problem, $\mathsf{Ax} = \mathbf{b}$ where A and $\mathbf{b}$ are known quantities. When periodic boundary conditions are applied in both directions, ϕ in (1.1) is determined up to an arbitrary constant. This freedom translates into a singularity in the matrix. This singularity may not be eliminated by simply fixing the potential at a single point (node) as done by *Newman and Ott* [1981], though a subtle modification of this approach does work. *Janhunen* [1992] approached this problem using a successive-over-relaxation technique which fails to converge when the nonlinear components of (1.1) become large, making the matrix highly non-diagonally dominant. We solved these problems by carefully reformulating the finite difference equations to eliminate the singularity and by using a variant on the biconjugate gradient method, called the quasi-minimal residual method [*Freund and Nachtigal*, 1994].

The results of our simulations agree well with the existing observations of type I waves. The phase velocities obtained from the numerical solutions are well below the velocity predicted by linear theory. This corroborates the observations of *Cohen and Bowles* [1967], *Hanuise and Crochet* [1981] and many others. We find the dominant waves to be quite monochromatic, a result confirmed by the rocket observation of *Pfaff et al.* [1987*a*]. The nonlinearly induced vertical waves we

see may explain the radar observations of *Kudeki et al.* [1987]. Additionally, the dominant waves do not travel strictly horizontally. The angle between the drift direction and the dominant wave of the system becomes larger as the driving electric field, E_{0y}, becomes more negative. These features are all caused, principally, by the nonlinear $\mathbf{E}\times\mathbf{B}_0$ term represented in the potential equation, (1.1), by the $\kappa\hat{\mathbf{z}}\cdot\nabla\phi\times\nabla n$. Dropping this term from the simulations has a profound effect on the numerical solutions, an effect far greater than that caused by dramatically varying the electron temperature or even the electron drift speed.

Previous numerical investigations of the FB instability by *Newman and Ott* [1981] and *Janhunen* [1994*b*] saw the development of waves whose dominant mode travels at an angle with respect to the electron drift. Additionally, we see from these simulations that this angle increases with driving electric field. *Janhunen* [1994*b*] claims to see FB waves traveling at the velocity predicted by linear theory, a result which disagrees with both the experimental database and a hybrid simulation we have done using his parameters. Since Janhunen modeled electron behavior with a computationally expensive PIC method, he had to use an unrealistic electron-ion mass ratio and could not run his simulations long enough to obtain waves in a saturated state. These facts may account for the discrepancy.

The simulations described in this thesis make it clear that nonlinear electron motion dominates the behavior of these waves, having effects far more profound than the linear ion kinetic effects and, very likely, the electron kinetic effects. Nonlinear $\delta\mathbf{E}\times\mathbf{B}$ causes small scale waves to travel upward along the crests of the principal, mostly horizontal, waves and downward in the valleys of the principal waves. A simple, nonlinear analysis shows that these secondary waves propagate and grow. It is extremely likely that this effect accounts for the vertically propagating component of type I echoes. Also, coupling between these secondary waves and the principle waves plays a role in turning the primary wave off the drift direction and probably also play a crucial role in reducing the effective phase velocity and saturating the primary waves.

These simulations indicate that ion kinetic effects are not essential for the saturation of the instability. They also indicate that electron temperature effects have a minor impact on the final saturated state. However, because this code does not deal with electron kinetic effects, we can not say that more subtle electron kinetic effects do not play an important role. We can state that many, if not all, of the ob-

served features of the two-stream instability are seen in these simulations without including the electron kinetic effects.

Chapter 2

Fluid and Linear Theories of the Farley-Buneman Instability

For every complex problem, there is a solution that is simple, neat, and wrong.

– H. L. Mencken

This chapter explains and derives the linear theory of the Farley-Buneman instability. While this material is not original, these theories lay the foundation upon which our original work, presented in subsequent chapters, builds. Most readers, familiar with the linear Farley-Buneman instability, will probably wish to skip or skim this chapter, returning to it from later chapters when we make a reference to it. For readers wishing to learn about two-stream waves, we hope this is an intelligible introduction.

All of the equations presented are in the MKS (SI) units system, waves travel as $\exp[i(\mathbf{k}\cdot\mathbf{x} - \omega t)]$. Since gravity has no appreciable effect on the small-scale waves we are studying, it makes more sense to align the magnetic field with $\hat{z}$, and define the plane perpendicular to that field as (x, y), than to have $\hat{z}$ point up and have the important perpendicular plane be defined as (x, z). Figure 2.1 shows the coordinate system in which we work and defines many of the geometric quantities we will be using.

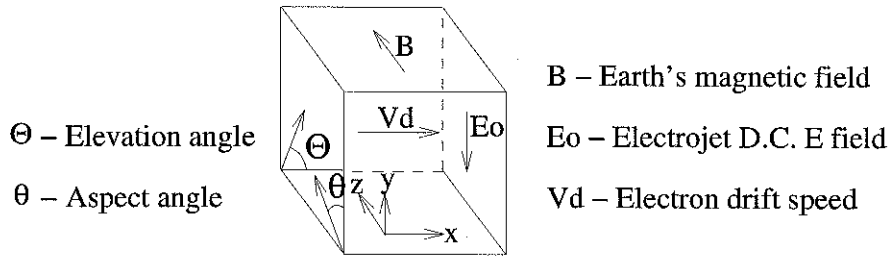


Figure 2.1: Coordinate system we use for the Farley-Buneman instability. B is the Earth's magnetic field, E_0 is the electrojet electrostatic field direction, V_d is the electron drift direction, Θ is the elevation angle relative to $\mathbf{V}_d$ and θ is the aspect angle relative to the vertical and perpendicular to $\mathbf{B}_0$.

2.1 The Two-fluid Theory of the Farley-Buneman Instability in Three Dimensions

To derive the fluid equations which govern the dominant processes in the electrojet, we make seven assumptions: (1) Perturbations in the magnetic field have no appreciable affect on the plasma, making the physics largely electrostatic. (2) The Earth's magnetic field has no influence on the ions because the ion-neutral collision rate (ν_i) substantially exceeds the ion cyclotron frequency, (Ω_i), such that each ion collides multiple times per cyclotron period, $(2\pi\Omega_i)^{-1}$. This implies that (3) the ion behavior is governed by an unmagnetized electrostatic ion momentum equation. (4) Given electric fields of millivolts per meter, meter-scale wavelengths, and charged particle densities on the order of $10^{11}m^{-3}$, the assumption that $n_i \approx n_e \approx n$ is valid to within one part per million. Hence, the assumption of quasineutrality applies and, instead of using Poisson's equation to determine the electric potential, we set $\nabla\cdot\mathbf{J} = 0$. Since n_i and n_e are essentially indistinguishable, we (5) have only one continuity equation, $\frac{\partial n}{\partial t} = \nabla\cdot(n\mathbf{v})$. (6) We treat the electrons as massless for three reasons: (a) the Farley-Buneman instability occurs on an ion collision frequency time scale which far exceeds the electron cyclotron or plasma frequency time scales, (b) the ions consist of O_2^+ and NO^+ which weigh about 50,000 times the electron's mass and (c) the assumption of quasineutrality means that the electrons are so light and responsive that, if they were displaced, they would reposition themselves on a time-scale which is much faster than any of the other time-scale in the system,

restoring $n_i \approx n_e$. (7) Hence, an inertialess electron momentum equation governs electron behavior.

The assumptions about ion behavior outlined above allow us to write an ion inertia equation as

$$m_i \left(\frac{\partial \mathbf{v}_i}{\partial t} + (\mathbf{v}_i \cdot \nabla) v_i) \right) = +e(\mathbf{E}_0 - \nabla\phi) - \nabla \cdot \mathsf{P}_i/n - m_i \nu_i \mathbf{v}_i \tag{2.1}$$

where all units are in MKS and $\mathbf{E}_0$ is the constant external field which drives the system, ϕ is the perturbed electric potential, n is the ion density, P_i is the ion pressure and ν_i is the ion-neutral collision rate. For the case where the ion pressure is a scalar quantity, we can reduce this vector equation into a scalar equation. Taking the curl of the ion inertia equation gives

$$\frac{\partial \nabla \times \mathbf{v}_i}{\partial t} + \nabla \times (\mathbf{v}_i \cdot \nabla)\mathbf{v}_i = -\nu_i \nabla \times \mathbf{v}_i. \tag{2.2}$$

If we assume that the advective term, $(\mathbf{v}_i \cdot \nabla)\mathbf{v}_i$, is a small second order term, then the remaining terms imply that the curl of the velocity (i.e., the vorticity) damps away. As a result, we assume the velocity is curl-free and write it as the divergence of a scalar,

$$\mathbf{v}_i \equiv \nabla\chi + \mathbf{V}_{id} \tag{2.3}$$

where $\mathbf{V}_{id}$ is the ion streaming velocity of $q/(m_i\nu_i)\mathbf{E}_0$, also called the ion Pederson drift speed. This makes the ion inertia equation

$$\frac{\partial \nabla\chi}{\partial t} = \frac{\nabla|\nabla\chi|^2}{2} - \frac{e}{m_i}\nabla\phi - \frac{\nabla \cdot \mathsf{P}_i}{m_i n} - \nu_i \nabla\chi \tag{2.4}$$

where $\nabla|\nabla\chi|^2$ derives from the advection. Since collisions with neutrals dominate ion motion, isotropic and isothermal approximations of ion pressure seems sensible. Hence, $\mathsf{P}_i = T_i n \mathsf{I}$, where T_i is the ion temperature in Joules and I is the identity matrix. By making this substitution and removing the gradient from all of the terms, we obtain

$$\frac{\partial \chi}{\partial t} = \frac{|\nabla\chi|^2}{2} - \frac{e}{m}\phi - \frac{T_i}{m_i}\ln n - \nu_i \chi \tag{2.5}$$

Quasineutrality implies that the continuity of the ions and the electrons can be written as

$$\frac{\partial n}{\partial t} = -\nabla \cdot J_i \equiv -\nabla \cdot (n \nabla\chi) \tag{2.6}$$

The Earth's magnetic fields makes the electron behavior far more complex than the ion behavior. Electron cyclotron motion and $\mathbf{E}\times\mathbf{B}$ drifting dominate the electron motion, yet electron collisions with neutrals still play an essential role in driving the Farley-Buneman instability. The electron motion derives from the inertialess electron momentum equation,

$$m_e\frac{d\mathbf{v}_e}{dt} = 0 = -e(\mathbf{E}_0 - \nabla\phi + \mathbf{v}_e\times B_0) - \frac{\nabla\cdot\mathsf{P}_e}{n} - m_e\nu_e\mathbf{v}_e \ . \tag{2.7}$$

Bringing the velocity terms to the left hand side of the equation yields, in matrix notation,

$$\mathsf{A}\mathbf{v}_e = -\frac{e}{\nu_e m_e}(\mathbf{E}_0 - \nabla\phi) - \frac{\nabla\cdot\mathsf{P}_e}{n\nu_e m_e} \tag{2.8}$$

where

$$\mathsf{A} = \begin{pmatrix} 1 & \kappa & 0 \\ -\kappa & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \mathsf{A}^{-1} = \begin{pmatrix} 1 & -\kappa & 0 \\ \kappa & 1 & 0 \\ 0 & 0 & 1+\kappa^2 \end{pmatrix} \frac{1}{1+\kappa^2} \tag{2.9}$$

where $\kappa \equiv \Omega_e/\nu_e$ and Ω_e is the electron cyclotron frequency in the Earth's magnetic field. The electron velocity can be rewritten as

$$\mathbf{v}_e = \mathsf{A}^{-1}[-\frac{e}{\nu_e m_e}(\mathbf{E}_0 - \nabla\phi) - \frac{\nabla\cdot\mathsf{P}_e}{n\nu_e m_e}] \ . \tag{2.10}$$

This equation shows why electrojet instabilities align almost exactly perpendicular to the magnetic field. In the lower E region, κ is a large number, on the order of 100, making the component of A^{-1} which parallels $\mathbf{B}_0$ at least κ times the perpendicular components. Any electric fields along $\mathbf{B}_0$ are therefore quickly neutralized.

The assumption of quasineutrality implies $\nabla\cdot J = 0$, which, after considerable algebra, yields

$$\begin{aligned} \nabla\cdot(n\nabla\chi) &= \nabla\cdot(n\mathbf{v}_e) \\ &= \frac{e(n\nabla_*^2\phi + (\nabla_*\phi - \mathbf{E}_0)\cdot\nabla_* n + \kappa(\nabla\phi - \mathbf{E}_0)\times\nabla n\cdot\hat{z}) - \nabla_*^2 P}{(1+\kappa^2)m_e\nu_e} \\ \nabla_* &\equiv \hat{x}\partial_x + \hat{y}\partial_y + \sqrt{1+\kappa^2}\hat{z}\partial_z \end{aligned} \tag{2.11}$$

where $\mathbf{E}_0$ is assumed to have no component in the $\hat{z}$ direction.

Assuming the electrons behave as an isothermal or adiabatic ideal gas yields $\mathsf{P}_e = T_e n^\gamma \mathsf{I}$ where T_e is in units of Joules, I is the identity matrix and γ is the adiabatic constant; $\gamma = 1$ for an isothermal electron gas and $\gamma = 2$ for a two

dimensional adiabatic electron gas. It is difficult to justify the assumption that electrons behave as either an adiabatic or isothermal gas. The electrons collide far more frequently with the neutrals than with each other. In elastic collisions, they do not transfer much energy to or from the neutrals because of the large mass difference between them. However, they do transfer large amounts of momentum, approximated by the viscosity term of equation (2.7). Inelastic collisions also play a role and transfer both momentum and energy. In the electrojet, where there is no evidence of electron heating, the isothermal or adiabatic assumption is probably fairly good. However, in the auroral electrojet, there is evidence of electron heating [*Schlegel and St.-Maurice*, 1981], which makes even the adiabatic assumption questionable.

This work assumes an adiabatic electron gas, making equation (2.11)

$$n\nabla_*^2\phi + \nabla_*\phi\cdot\nabla_* n + \kappa\nabla\phi\times\nabla n\cdot\hat{z} = \mathbf{E}_0\cdot\nabla_* n + \kappa\mathbf{E}_0\times\nabla n\cdot\hat{z}$$
$$+\frac{\gamma T_e}{e}\left[n^{\gamma-1}\nabla_*^2 n + (\gamma-1)n^{\gamma-2}|\nabla_* n|^2\right] + (1+\kappa^2)\frac{m_e\nu_e}{e}\nabla\cdot(n\nabla\chi) \qquad (2.12)$$

If we divide this equation and the continuity equation by the density, then all density terms can be written as the log of the density. Since the ion momentum equation also varies proportionally to the log of the density, this implies that the velocities and potentials vary proportionally to the log of the density instead of directly with the density. Making this substitution transforms the three equation system into

$$\frac{\partial\chi}{\partial t} = \frac{|\nabla\chi|^2}{2} - \frac{e}{m_i}\phi - \frac{T_i}{m_i}s - \nu_i\chi \qquad (2.13)$$

$$\frac{\partial s}{\partial t} = -\nabla^2\chi - \nabla s\cdot\nabla\chi \qquad (2.14)$$

$$\nabla_*^2\phi = -\nabla_*\phi\cdot\nabla_* s - \kappa\nabla\phi\times\nabla s\cdot\hat{z} + \mathbf{E}_0\cdot\nabla_* s + \kappa\mathbf{E}_0\times\nabla s\cdot\hat{z} \qquad (2.15)$$
$$+\frac{\gamma T_e}{e}e^{\gamma s}\left[\nabla_*^2 s + (\gamma-1)|\nabla_* s|^2\right] + (1+\kappa^2)\frac{m_e\nu_e}{e}(\nabla^2\chi - \nabla s\cdot\nabla\chi)$$

$$s \equiv \ln(n/n_0) \qquad (2.16)$$

These three equations should describe most waves in the electrojet which conform to the seven assumptions described in the beginning of this section.

2.2 Linear Fluid Theory

Linearizing the fluid equations, (2.13) - (2.15), enables us to solve them for simple wave propagation and growth. The growing waves found through this process are the two-stream instability and the gradient drift instability. The gradient drift instability requires the inclusion of a zero-order gradient in the density. Since the two-stream instability occurs on much shorter scale lengths than the scale-height of the ionosphere, we neglect that component here. This makes the linearized fluid equations,

$$\frac{\partial \chi}{\partial t} = -\frac{e}{m_i}\phi - \frac{T_i}{m_i}s - \nu_i \chi \tag{2.17}$$

$$\frac{\partial s}{\partial t} = -\nabla^2 \chi \tag{2.18}$$

$$\nabla_*^2 \phi = \mathbf{E}_0 \cdot \nabla_* s + \kappa \mathbf{E}_0 \times \nabla s \cdot \hat{z} + \frac{\gamma T_e}{e}\nabla_*^2 s + (1+\kappa^2)\frac{m_e \nu_e}{e}\nabla^2 \chi \tag{2.19}$$

For the purposes of linearization, we assume that the electrojet is driven by a static electric field pointing in the $\hat{y}$ direction. We will translate these equations into frequency and wavenumber space $(\omega, \mathbf{k})$ by assuming that s, χ and ϕ vary as $\exp[i(\mathbf{k}\cdot x - \omega t)]$. This notation differs from that of *Kelley* [1989] and many other papers which choose $\exp[i(\omega t - \mathbf{k}\cdot x)]$ instead. This transformation converts these three linear partial differential equations (PDEs) into three linear algebraic relations,

$$(-i\omega + \nu_i)\tilde{\chi} = -\frac{e}{m_i}\tilde{\phi} - \frac{T_i}{m_i}\tilde{s} \tag{2.20}$$

$$\tilde{s} = -\frac{k^2}{i\omega}\tilde{\chi} \tag{2.21}$$

$$\tilde{\phi} = \left(-iE_{0y}\frac{k_y}{k_*^2} + \kappa i E_{0y}\frac{k_x}{k_*^2} + \frac{\gamma T_e}{e}\right)\tilde{s} + (1+\kappa^2)\frac{m_e \nu_e}{e}\frac{k^2}{k_*^2}\tilde{\chi} \tag{2.22}$$

where $k_*^2 \equiv k_x^2 + k_y^2 + (1+\kappa^2)k_z^2$ and $\tilde{\chi}$, $\tilde{s}$ and $\tilde{\phi}$ are the Fourier amplitudes of χ, s and ϕ.

2.2.1 One and Two Dimensional Linear Fluid Theory

Reducing the linear fluid equations, (2.17) - (2.19), to the two dimensions perpendicular to the magnetic field and transforming these equations into $(\omega, \mathbf{k})$ yields a

dispersion relation,

$$\left(\omega^2 + i\omega\nu_i - C_s^2 k^2\right) - iV_d\Omega_i(\kappa k_x - k_y) + \frac{m_e}{m_i}\nu_e(1+\kappa^2)i\omega = 0 \qquad (2.23)$$

where $V_d \equiv E_{0y}/B_0$, the electron drift velocity and $C_s^2 \equiv (T_i + \gamma T_e)/m_i$, the acoustic velocity.

We may solve for the complex mode frequency, ω, without resorting to the quadratic equation by making the assumption that the growth rate, ω_i, is small compared to the frequency, ω_r, and the electron-neutral collision rate, ν_i. This, plus the assumption, that $\kappa \gg 1$ gives the real and imaginary components of the frequency as

$$\omega_r = \frac{V_d(k_x - k_y/\kappa)}{\Psi_0 + 1} \qquad (2.24)$$

$$\omega_i = \frac{\omega_r^2 - C_s^2 k^2}{\nu_i(1 + \Psi_0^{-1})} \qquad (2.25)$$

$$\Psi_0 \equiv \frac{\nu_e\nu_i}{\Omega_e\Omega_i} \, . \qquad (2.26)$$

The conditions for growth are

$$\omega_r^2 > C_s^2 k^2 \;\Rightarrow\; V_d > C_s(1+\Psi_0) \, . \qquad (2.27)$$

These relations predict that as k_x increases, ω_i increases monotonically and the shortest wavelengths dominate. This does not agree with the experimental evidence [*Schmidt and Gary*, 1973] which shows that the shortest wavelengths in the system are on the order of a meter. We will see later that ion kinetic effects damp out the shortest wavelength modes. Likewise, the nonlinear effects seem to damp short wavelength modes.

A physical picture the one dimensional Farley-Buneman instability

A number of features of the linear theory make the Farley-Buneman instability difficult to understand. First, assuming quasineutrality, $\nabla \cdot \mathbf{J} = 0$, in place of Poisson's equation, $\nabla^2\phi = -\rho/\epsilon_0$, makes understanding the origins and effects of the electric field less obvious. Second, the electrons have a zero-order horizontal drift, $\mathbf{V}_d = \mathbf{E}_0 \times \mathbf{B}_0/B_0^2$, while the ions have a small vertical Pederson drift, $\mathbf{V}_{id} = (e/m_i)\mathbf{E}_0/\nu_i$. Third, the presence of Ω_i in the fluid dispersion relationship, 2.24 - 2.27, incorrectly suggests the importance of ion cyclotron behavior. Fourth,

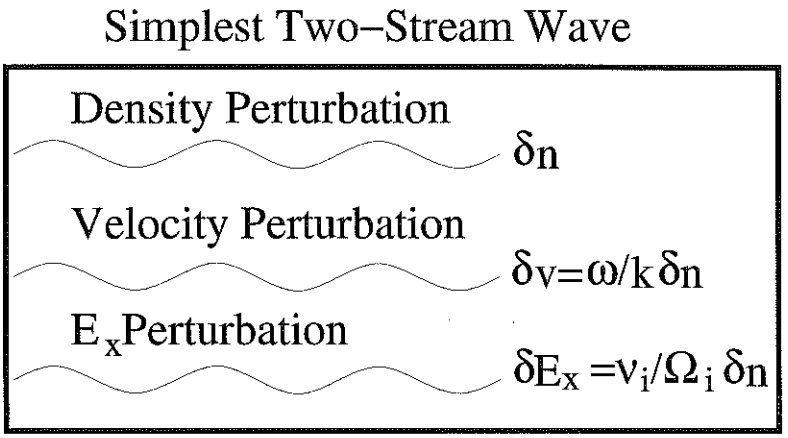


Figure 2.2: The simplest two-stream waves where $C_s = 0$ and $\nu_e \rightarrow 0$. When ω develops an imaginary component then two-stream waves may grow or decay, depending on the phase of ω.

the appearance of C_s in the dispersion relation does not imply that the principal mechanism governing the Farley-Buneman instabilityis similar to an acoustic instability. A number of simplifying assumptions make understanding this instability easier.

First, when $T_e = T_i = 0$, the acoustic speed, C_s, equals zero, but, as one can see from the dispersion equation (2.25), the wave still propagates. In fact, when $C_s = 0$, the instability grows for all wavelengths. This shows that temperature plays, primarily, a limiting role in the instability through diffusion of both electrons and ions.

Second, an important feature of the instability becomes apparent as $\nu_e \rightarrow 0$. Within this limit,the growth rate, ω_i, of equation (2.25) approaches zero, implying that electron-neutral collisions play an essential role in causing these waves to grow. However, even when $\nu_e = 0$, the waves propagate without growth or dissipation. In the frame of reference of the collisionless electrons, the ions stream past at the drift velocity, $V_d = E_0/B_0$. Any perturbation in the ion density simply propagates at this speed without modification. By assuming quasineutrality, we have prevented plasma oscillations. Figure (2.2) shows this simple Farley-Buneman system. The situation of $\nu_e \rightarrow 0$ is quite artificial since, in such a case, quasineutrality would break down.

Comparing the density perturbations to the electric field and velocity pertur-

bations in **k** space sheds some light on the underlying mechanism of the Farley-Buneman instability. For a cold system, the electric field magnitude derived from equation (2.19) becomes

$$\frac{E_x}{E_{z0}} = \frac{\nu_i \Omega_e}{\Omega_i \Omega_e + \nu_e \nu_i} \frac{\delta n}{n_0} \,. \tag{2.28}$$

As the electron-neutral collision rate, ν_e, increases from zero, the electric field magnitude decreases because the electrons now respond to the electric field by traveling in the direction of the electric field rather than by traveling only perpendicular to it via $\mathbf{E}\times\mathbf{B}$ motion. Because the electrons have the opposite charge of the ions, they respond by traveling in the opposite direction from the ions. As we see from (2.28), the electrons reduce the electric field. This, in turn, reduces the rate at which ions leave areas of enhanced ion density and enter areas of reduced ion density normally associated with wave propagation. This change in flow pattern causes an exponential growth in the density perturbation, an instability.

Comparing the ion velocities to the density perturbations makes the instability mechanism clearer. The linearized ion continuity equation gives us a relationship between the perturbed ion velocity and the perturbed density,

$$v_i = \frac{\omega}{k_x} \frac{\delta n}{n_0} = \left(\frac{V_d}{1 + \Psi_0} + i \frac{\Psi_0 k_x^2 V_d^2}{\nu_i (1 + \Psi_0)^3} \right) \frac{\delta n}{n_0} \,. \tag{2.29}$$

The imaginary term is zero when Ψ_0 is zero. This occurs when ν_e or m_i equals zero. When neither of these terms equals zero, a positive imaginary perturbation of the ion velocity exists. Adding an out-of-phase component to a simple propagating wave, either enhances or diminishes the wave, depending on the sign of the out-of-phase component. A number of intuitive arguments can be made that a positive Ψ_0 causes wave growth. We already made the first of these arguments; the electrons Pederson drift in a direction opposite to the ions and draw some ions with them into regions of already enhanced ion density.

A second, less intuitive, argument that the positive imaginary term of the ion velocity perturbations causes wave growth relies on the accepting that diffusive effects invariably cause wave damping. Then, if we can show that the out-of-phase component, caused by a positive Ψ_0, has the opposite sign as the diffusive components, we have a growing instability. Allowing the temperature to increase from zero adds diffusion to our system. Electron diffusion perturbs the electric

field making

$$\frac{E_x}{E_{z0}} = \left[\frac{\nu_i \Omega_e}{\Omega_i \Omega_e + \nu_e \nu_i} - ik_x \frac{T_e}{e} E_{z0}\right] \frac{\delta n}{n_0} \,. \tag{2.30}$$

This, plus the ion temperature effects, add an additional out-of-phase component to the perturbed velocity,

$$v_i = \frac{\omega}{k_x} s = \left[\frac{V_d}{1+\Psi_0} + ik_x^2 \frac{\Psi_0}{\nu_i(1+\Psi_0)} \left(\frac{V_d^2}{(1+\Psi_0)^2} - C_s^2\right)\right] \frac{\delta n}{n_0} \,. \tag{2.31}$$

The sign of the term containing C_s results from the combined electron and ion diffusion. Since C_s causes wave damping and has the opposite sign as the other out-of-phase component, $V_d^2/(1+\Psi_0)^2$, this term causes growth.

Two conditions must be met for growth, as seen in the relationship between v_i and δn, and as expressed by equation (2.31). The first condition is $Vd/(1+\Psi_0) > C_s$ and the second is that in order to have any out-of-phase component, either positive or negative, $\Psi_0 \neq 0$. Not only will setting the electron-neutral collision rate to zero cause $\Psi_0 = 0$, setting m_i to 0 also has this effect. This means that ion inertia plays a role similar to electron neutral-collisions, via an entirely different physical mechanism. A nonzero m_i also causes an out-of-phase relationship between the perturbed density and the perturbed velocity. Ion inertia achieves this effect by causing the ions to effectively overshoot the position one expects them to reach for simple wave propagation. They overshoot and head into regions of already enhanced ion density, causing wave growth. Both $m_i \neq 0$ and $\nu \neq 0$ are essential to enabling this instability to grow.

2.2.2 Three Dimensional Linear Theory

The full three dimensional dispersion relation derived by Fourier transforming equations (2.17) -(2.19) shows that two-stream waves propagate almost strictly perpendicular to $\mathbf{B_0}$. To a large degree, this strong selection of perpendicular waves justifies reducing this system to two dimensions for most analyses, including the simulations of chapters 3 and 4. There exists one interesting feature of the three dimensional dispersion relation. For slight angles off the magnetic field (called the aspect angle) the growth rate of the instability rises before rapidly decreasing [*Ossakow et al.*, 1975].

After some algebra, the dispersion relation can be written as

$$(k^2+\kappa^2k_z^2)\left(\omega^2+i\omega\nu_i-\frac{T_i}{m_i}k^2\right) - \frac{\gamma T_e}{m_i}k^4+k^2V_d\Omega_i(ik_y-i\kappa k_x) + \frac{m_e}{m_i}\nu_e(1+\kappa^2)k^2i\omega=0\ . \tag{2.32}$$

Again, assuming that the growth rate, ω_i is small compared to the frequency, ω_r, and the electron-neutral collision rate, ν_i, enables us to solve for ω_r and ω_i without resorting to a quadratic equation:

$$\omega_r = V_d(k_x-k_y/\kappa)/(\Psi_1+1) \tag{2.33}$$

$$\omega_i = \frac{\Psi_1/\nu_i}{\Psi_1+1}\left[\omega_r^2-k^2C_s^2+k^2\frac{\gamma T_e}{m_i}\left(\frac{\kappa^2k_z^2/k^2}{1+\kappa^2k_z^2/k^2}\right)\right] \tag{2.34}$$

$$\Psi_1 \equiv \frac{\Omega_e\Omega_i}{\nu_e\nu_i}\left(1+\kappa^2\frac{k_z^2}{k^2}\right)=\Psi_0\left(1+\kappa^2\frac{k_z^2}{k^2}\right) \tag{2.35}$$

When $k_z = 0$, the three dimensional dispersion relations simplify to the two dimensional equations. The growth rate, ω_i, increases initially with k_z because of the change in the last term of equation (2.34), then decreases rapidly because of the increase in Ψ_0. The maximum growth rate occurs at a very small aspect angle for large values of κ. Unfortunately, the equation for the angle at which the maximum growth rate occurs cannot be solved analytically but, instead, must be solved numerically.

2.3 Linear Kinetic Theory

The linear fluid theory predicts a growth rate which, for large k_x increases monotonically with wavenumber. This presents an almost intractable problem for simulations because a huge range of scale sizes would contain substantial amounts of energy. Also, such a system would undoubtably be strongly turbulent. Fortunately, when one applies kinetic theory to the ion motion, one sees that the very short wavelength modes are damped. *Farley* [1963] did the initial calculations of the Farley-Buneman instability using linear kinetic theory. *Schmidt and Gary* [1973] later built on that work to generate daytime growth rates for the two-stream instability. Because these calculations and the particular growth rates obtained are crucial to this investigation, we derive these equations.

The distribution of the ions evolves according to Boltzmann's equation. The fact that the ions collide almost exclusively with the neutral gas means that the BGK [*Bhatnagar et al.*, 1954; *Greene*, 1973] operator approximates the ion collision term quite well making

$$\frac{\partial f}{\partial t} + \mathbf{v}_i \cdot \frac{\partial f}{\partial \mathbf{x}} + \frac{e}{m_i}\mathbf{E} \cdot \frac{\partial f}{\partial \mathbf{v}_i} = -\nu_i \left[f - \frac{n_i}{n_0} f_0 \right] \tag{2.36}$$

where $f = f(\mathbf{v}_i, \mathbf{x}, t)$ is the time dependent distribution function while $f_0 = f_0(\mathbf{v}_i)$ is the initial homogeneous, pre-instability distribution function. We linearize this problem by decomposing f into $f = f_0(\mathbf{v}_i) + f_1(\mathbf{v}_i, \mathbf{x}, t) + f_{nl}$ where f_{nl} represents higher order, nonlinear terms which we shall ignore. Assuming f_1 varies as $\exp[i(\mathbf{k}\cdot x - \omega t)]$ we obtain

$$(-i\omega + i\mathbf{v}_i \cdot \mathbf{k} + \nu_i) f_1 + \frac{e}{m_i}\left(\mathbf{E}_0 \cdot \frac{\partial f_1}{\partial \mathbf{v}_i} + \mathbf{E}_1 \cdot \frac{\partial f_0}{\partial \mathbf{v}_i} \right) + \nu_i \frac{n_i}{n_0} f_0 = 0 \tag{2.37}$$

Dividing both sides by $(-i\omega + \mathbf{v}_i \cdot \mathbf{k} + \nu_i)$ and integrating over all velocities gives

$$-i \int_{-\infty}^{\infty} d^3v f_1 = \frac{e}{m_i} \int_{-\infty}^{\infty} d^3v \frac{(\mathbf{E}_0 \cdot \frac{\partial f_1}{\partial \mathbf{v}_i} + \mathbf{E}_1 \cdot \frac{\partial f_0}{\partial \mathbf{v}_i})}{(\mathbf{v}_i \cdot \mathbf{k} - \omega - i\nu_i)} + \frac{n_i}{n_0} \int_{-\infty}^{\infty} d^3v \frac{\nu_i f_0}{(\mathbf{v}_i \cdot \mathbf{k} - \omega - i\nu_i)} \tag{2.38}$$

Setting the zero order electric field to point only in the $\hat{y}$ direction and allowing only waves in the $\hat{x}$ direction makes this problem analytically tractable. When $k_y = 0$, the integral of the term containing $\mathbf{E}_0 \cdot \frac{\partial f_1}{\partial \mathbf{v}_i}$ goes to zero because

$$\int_{-\infty}^{\infty} \frac{\partial f_1}{\partial v_y} dv_y = f_1]_{-\infty}^{\infty} = 0 \ . \tag{2.39}$$

If $k_y \neq 0$ then the term containing $\mathbf{E}_0 \cdot \frac{\partial f_1}{\partial \mathbf{v}_i}$ need not equal zero because the $v_i \cdot \mathbf{k}$ component in the denominator now contains v_y. This makes this problem very difficult. However, once that term is gone the problem is soluble.

The initial distribution of the ions, f_0, is assumed to be Gaussian,

$$f_0 = \frac{n_0}{(2\pi)^{3/2} V_{th}^3} \exp\left[\frac{-(v - V_{id})^2}{2V_{th}^2} \right], \qquad T_i \equiv m_i V_{th}^2 \tag{2.40}$$

where $V_{id} \equiv (e/m) E_{0y}/\nu_i$, the ion drift rate. The two integrals of (2.38) can be cast in terms of the plasma dispersion function so that our ion density becomes

$$\frac{n_i}{n_0} = \frac{e}{m_i} \frac{E_{1x} Z'(\beta)}{2iV_{th}^2 k_x - \sqrt{2}\nu_i V_{th} Z(\beta)} \qquad \beta \equiv \frac{\omega + i\nu_i}{\sqrt{2} V_{th} k_x} \tag{2.41}$$

where the definition of Z and Z' [*Abramowitz and Stegun*, 1970; *Huba*, 1994] is

$$Z(\beta) = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} dt \frac{e^{-t^2}}{t-\beta} \qquad Z'(\beta) = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} dt \frac{e^{-t^2}}{(t-\beta)^2} = -2(1+\beta Z(\beta))$$

We need to combine this ion response function with an electron response function derived from the fluid equations. The electron density, n_e, derives from the electron continuity equation and the electron inertialess momentum equation, (2.7), to give an electron response function in 1-D of

$$\frac{n_e}{n_0} = \frac{-ik_x(e/m_e)E_{1x}}{-i\omega(1+\kappa^2)\nu_e + (\gamma T_e/m_e)k_x^2 + i\kappa\nu_e(e/m_e)E_{0y}k_x} \tag{2.42}$$

Since we evaluate the ion and electron density separately, we do not avail ourselves of quasineutrality. Instead, Gauss's Law determines the electric field. Fourier transforming Gauss's law gives

$$\nabla\cdot\mathbf{E} = (e/\epsilon)(n_i - n_e) \quad \Rightarrow \quad i\mathbf{k}\cdot\mathbf{E}_1 = (e/\epsilon)(n_i - n_e)\ . \tag{2.43}$$

Substituting in for n_i and n_e yields a dispersion equation

$$1 = \frac{\omega_{pe}^2}{-i\omega\nu_e(1+\kappa^2) + (\gamma T_e/m_e)k_x^2 + i\nu_e\kappa^2 V_d k_x} - \frac{\omega_{pi}^2 Z'(\beta)}{2V_{th}^2 k_x^2 + \sqrt{2}i\nu_i V_{th} k_x Z(\beta)} \tag{2.44}$$

An accurate solution to this problem requires numerical evaluation of the $Z(\beta)$ functions. However, in the limit of large, imaginary β, such as in the case when $|\omega| \ll i\nu_i$, the first term of an asymptotic expansion of Z gives

$$\begin{array}{rcl} Z(\beta) & \Rightarrow & -1/\beta \\ Z'(\beta) & \Rightarrow & +1/\beta^2 \end{array} \text{ when } \left\{ \begin{array}{rcl} \beta & \rightarrow & \infty \\ |arg(-i\beta)| & < & 3\pi/4 \end{array} \right. . \tag{2.45}$$

Substituting these approximations for Z into the kinetic dispersion relation, (2.44), causes this equation to revert to the two fluid dispersion relation of (2.23).

Figure 2.3 compares the solution of the dispersion relation for a two-fluid system to a hybrid system with kinetic ions and fluid electrons. Two important effects appear in this figure. First, the growth rate peaks at a specific, finite, wave number. The location of this peak depends on a number of parameters. Second, the frequency predicted for a specific wavenumber is reduced from the value predicted by fluid theory. This has the effect of reducing the phase velocity for all modes.

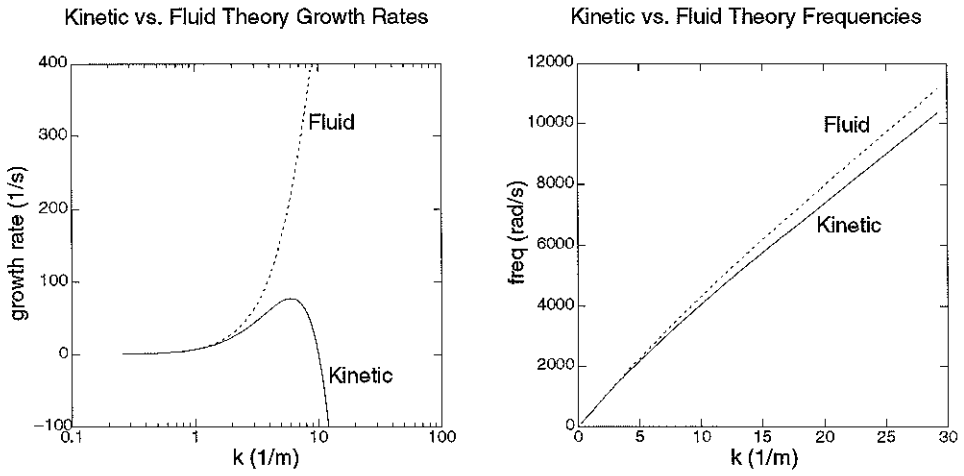


Figure 2.3: Linear solution of the one dimensional Farley-Buneman instability with fluid electrons and kinetic ions (solid line) and fluid electrons and ions (dashed line). Parameters: $B_0 = .25$ Gauss, $T_e = T_i = 208°K$, $\nu_i = 2.8 \times 10^3$, $\nu_e = 4 \times 10^4$ and $V_d = 600\text{m/s}$.

2.3.1 An Approximate Two Dimensional Kinetic Dispersion Relation

In order to compare our two dimensional simulations to linear theory, we need to extend the one dimensional kinetic theory to two dimensions. Equation (2.37) describes the linear distribution functions in three dimensions. However, it is very difficult to solve the integral-differential equation, (2.38), when a wave propagates in any direction but horizontal, $\mathbf{k} \neq k_x\hat{x}$. Both *Farley* [1963] and *Schmidt and Gary* [1973] evade this problem by assuming that, since the ion Pederson drift velocity is much less than the ion thermal velocity, the difficult $\mathbf{E}_0 \cdot \frac{\partial f_1}{\partial \mathbf{v}_i}$ term is negligible. Following their example enables us to generate a two dimensional linear kinetic dispersion relationship perpendicular to the plane of the magnetic field,

$$0 = 1 - \frac{\omega_{pe}^2}{-i\omega\nu_e(1+\kappa^2) + (\gamma T_e/m_e)k^2 + i\nu_e\kappa V_d(\kappa k_x - k_y)} + \frac{\omega_{pi}^2 Z'(\beta)}{2V_{th}^2 k^2 + \sqrt{2}i\nu_i V_{th}|k| Z(\beta)} \tag{2.46}$$

$$\beta \equiv \frac{\omega + i\nu_i}{\sqrt{2}V_{th}|k|} . \tag{2.47}$$

Figures 2.4 and 2.5 show the growth rate generated by a numerical solution of this dispersion relation. We use parameters identical to those used in simulations

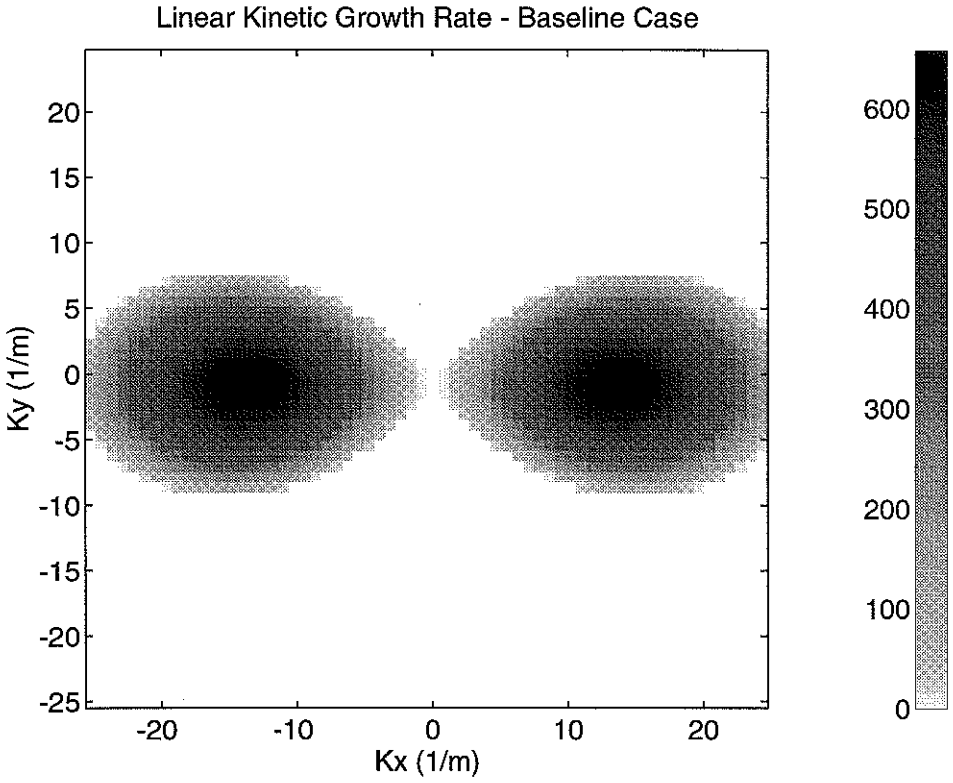


Figure 2.4: Growing components of the linear kinetic growth rate versus k_x and k_y for parameters: $B_0 = .25$ Gauss, $T_e = T_i = 208°K$, $\nu_i = 2.8 \times 10^3$, $\nu_e = 4 \times 10^4$ and $V_d = 800$m/s.

described in Chapter 4. These figures show the symmetry of the growth rate, a result not obvious from the dispersion equation (2.46). As in the one dimensional case, the imaginary part of ω, derived from the kinetic equations, matches the solution of the fluid dispersion relations for small k_x (long wavelengths). However, at shorter wavelengths waves are damped through the mechanism of linear Landau damping.

The real part of ω behaves analogously to the result from the linear two dimensional fluid equations except that the frequencies are somewhat reduced as shown in the one dimensional solution of figure 2.3. Also, the real part of ω tells us that waves with negative k_x have negative Re(ω) implying that the waves all propagate in the same direction as the electron drift direction.

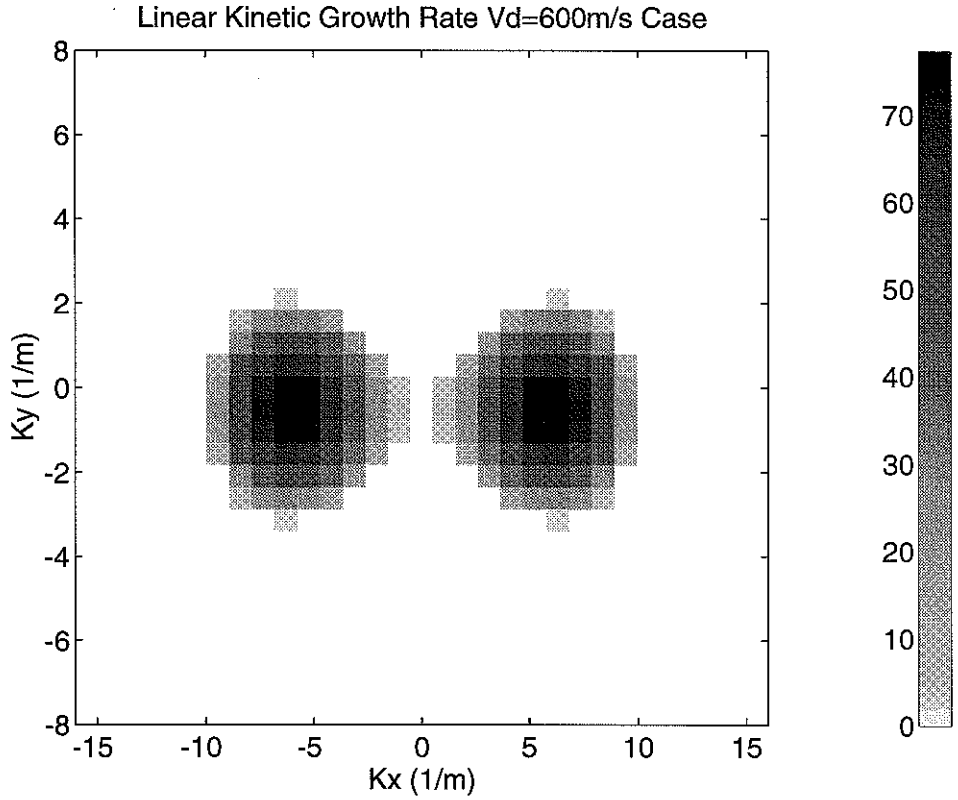


Figure 2.5: Growing components of the linear kinetic growth rate versus k_x and k_y for parameters where V_d has been reduced to $600m/s$.

Chapter 3

Simulation Methods

Computer Scientists are solving yesterdays problems using tomorrows tools while physicists are solving tomorrows problems using yesterdays tools.

– *Saul Teukolsky (paraphrased from talk)*

The linear theory of the FB instability predicts that ion Landau damping plays a crucial role in eliminating short wavelength waves, a feature neglected by a fluid model of ion behavior (see section 2.3). Any simulation of this system must, at a minimum, incorporate the short wavelength linear Landau damping because, otherwise, the shortest wavelengths have the largest growth rates. We approach this problem in two ways. First, we use a hybrid code approximating ion dynamics with a particle-in-cell (PIC) method and electron motion with a fluid method. Second, we use a two-fluid code which treats both ions and electrons as fluids, adding the necessary short wavelength damping through an enhanced, wavelength-dependent, ion-neutral collision rate.

Both codes use a fluid model of electron behavior described by equation (2.12) which neglects electron inertial and kinetic effects. The code further simplifies this equation by reducing it to 2 dimensions and limiting the electron temperature effects by assuming isothermal or adiabatic behavior. These assumptions enable us to write the equation defining the potential as:

$$n\nabla^2\phi + \nabla\phi\cdot\nabla n + \kappa\nabla\phi\times\nabla n\cdot\hat{z} =$$
$$\mathbf{E}_0\cdot\nabla n + \kappa\mathbf{E}_0\times\nabla n\cdot\hat{z} + \frac{\gamma T}{e}\nabla^2 n + (1+\kappa^2)\frac{m_e\nu_e}{e}\nabla\cdot(nv_i) \qquad (3.1)$$

The nonlinear term, $\hat{z}\cdot\nabla\phi\times\nabla n$ is multiplied by a factor of κ which, in the case of the E region, is a large number (≈ 100) making the nonlinear term the same order of magnitude as the linear terms, even for fairly small density variations. This term makes numerically solving equation (3.1) for ϕ proved very difficult. Also, the shortest timescale at which these equations respond is the electron-neutral collision rate.

We use finite difference equations to convert the partial differential equation (PDE) described by equation (3.1) into a matrix problem, $\mathsf{Ax} = \mathsf{b}$, where the matrix, A, and the vector, b, are known quantities and the vector, x, is the unknown. Appendix A briefly outlines how one converts differential equations into linear algebra equations. Periodic boundary conditions in both directions cause the potential, ϕ, to be determined only up to an arbitrary constant. In other words, if we know a potential, $\phi_1(x,y)$, which solves this equation, then any constant added to this potential gives us a new potential, $\phi_2(x,y) = \phi_1(x,y) + c$, which also solves this equation. The freedom represented by this undefined constant translates into a singularity in the matrix, A, and makes it very difficult to solve for x.

Newman and Ott [1981] attempted to eliminate this singularity by fixing the potential at a single point (node). This fails for complex reasons which are explained below, though a subtle modification of this approach succeeds. *Janhunen* [1992] attempted to solve equation (3.1) by using a successive-over-relaxation (SOR) technique (see Chapter 19, *Press et al.* [1988]). This method fails to converge when the nonlinear components of (3.3) become large, making the matrix highly non-diagonally dominant. We avoided these difficulties by carefully reformulating the finite difference equations to eliminate the singularity and by using a variant of the biconjugate gradient method, called the quasi-minimal residual method [*Freund and Nachtigal*, 1994] to handle the non-diagonal dominance of the matrix.

Since obtaining a numerical solution to equation (3.1) has been a substantial accomplishment of this project, we describe our methods in detail in the following section (3.1). This section also explains the flaw in the method used by *Newman and Ott* [1981]. Section (3.2) describes the other important numerical components of the hybrid code. Section (3.3) describes the methods used in the two fluid model.

3.1 Potential Solver

Simulating the Farley-Buneman instability with a fluid or hybrid code requires determining the potential from equation (3.1). Solving this problem numerically has been one of the principal difficulties in studying two-stream waves. At least one published result, *Newman and Ott* [1981], was done incorrectly. Nonetheless, we have found a technique which obtains accurate solutions for almost all density variations. The following sub-sections describe our methods and the error in the method used by *Newman and Ott* [1981].

The potential equation that we must solve is linear. While the complete set of fluid equations, (2.13) - (2.15), is nonlinear, the potential equation which must be solved for $\phi(x,y,t)$ is not. In other words, at any one time, t, $n(x,y)$ is independent of the value of $\phi(x,y)$, which we wish to calculate. Since the potential equation is linear, so is the resulting matrix problem.

3.1.1 Pseudo-Spectral Technique

When the maximum density perturbation in the simulation is small, $\delta n/n_0 \approx< 5$, a spectral method solves equation (3.1) very rapidly. To apply this method we write it in the form

$$\nabla^2\phi^{m+1} = (\nabla n\cdot\nabla\phi^m + \kappa\nabla n\times\nabla\phi^m + T(x,y))/n. \tag{3.2}$$

where $T(x,y)$ is the right-hand side (RHS) of equation (3.1) and m is an iteration counter. Initially, we guess a value of $\phi^0(x,y)$. $\phi^0(x,y) = 0$ is acceptable but inefficient. Using ϕ from the previous time-step would probably work better and simple extrapolation from the previous two time steps would yield a still better first guess. Using that, we approximate the right hand side of equation (3.2) by using finite difference methods as described in Appendix A. We then Fourier transform this approximation, divide each (k_x, k_y) mode by k^2 and take the inverse Fourier transform to give us a value we call $\phi^1(x,y)$. We then use $\phi^1(x,y)$ to generate a new value for the RHS of equation (3.2). We repeat this process until $|\phi^{m+1} - \phi^m| < \epsilon$ where ϵ is some acceptable error tolerance. Typically, we used $\epsilon < 10^{-6}$, though little difference appears if this criterion is varied by an order of magnitude. When $\kappa \sim 100$, this method works until the maximum density variation in the simulation exceeds approximately 10%. Beyond that it fails to

converge because $\kappa\nabla\phi\times\nabla n \sim \nabla^2\phi$ and this type of method only converges when the left-hand side of the modeled equation dominates the solution.

3.1.2 Technique for Solving the Potential Equation for All Densities

Solving the potential equation, (3.1), for arbitrary values of $n(x,y)$ has proven far more difficult then solving it in the case where the density is relatively uniform. Three difficulties arise. In order to understand these difficulties, it is necessary to understand the underlying method used to convert PDEs into sets of linear equations. Appendix A briefly explains the fundamentals of solving a PDE using finite difference methods. For a more thorough discussion see *Press et al.* [1988], chapter 19, or *Ames* [1977].

In our simulations the matrix, A, is initially nearly symmetric when $\kappa\nabla n \ll \nabla\phi$. However, it evolves into a very asymmetric matrix as the nonlinear electron behavior becomes important ($\kappa\nabla\phi\times\nabla n \gg \nabla^2\phi$). Most linear sparse matrix solvers require either symmetric, antisymmetric or Hermitian matrices. This is the first difficulty.

When we use finite difference techniques to convert the potential equation into a matrix, the dominant nonlinear term, $\kappa\nabla\phi\times\nabla n$, appears as off diagonal elements of this matrix. Both the iterative spectral method described above and the successive-over-relaxation (SOR) method (see *Press et al.* [1988]) require that the diagonal elements in the matrix dominate. Hence, neither of these methods converge when the nonlinear term, $\kappa\nabla\phi\times\nabla n$, has roughly the same magnitude as the linear term $\nabla^2\phi$, as will be the case when the nonlinear term causes saturation. This is the second difficulty.

The potential, ϕ, only appears in the potential equation, (3.1), as $\nabla\phi$. Therefore any function, $\phi(x,y)$, which solves potential equation (3.1) remains a solution even if we add an arbitrary constant to it. When such a PDE is converted into a matrix through finite difference equations, it is singular and much more difficult to solve than a nonsingular matrix. This is the third and most subtle difficulty to overcome.

To obtain a solution to a non-symmetric, non-diagonally dominant sparse matrix we use a variant of the biconjugate gradient method called the quasi-minimal residual approach, described in *Freund and Nachtigal* [1994]. This sophisticated solver functions despite the first two difficulties.

The singularity is eliminated using the following technique: Equation (3.1) is written in a flux-conservative form:

$$\nabla\cdot(n\epsilon\nabla\phi) = \nabla\cdot\left(\frac{\mathbf{J}_i}{\Psi_0} + \frac{T_e}{C_s^2 m_i}\nabla n\right) + \mathbf{E}_0\cdot\nabla n + \kappa\hat{z}\cdot\mathbf{E}_0\times\nabla n \qquad (3.3)$$

$$\epsilon \equiv \begin{pmatrix} 1 & -\kappa \\ \kappa & 1 \end{pmatrix}, \qquad \kappa \equiv \frac{\Omega_e}{\nu_e}\,. \qquad (3.4)$$

Again, periodic boundary conditions apply in both directions. An additional boundary condition is needed to make the PDE nonsingular. For this PDE, an appropriate condition requires that the D.C. component of the potential equals zero. This is a very difficult condition to apply in a non-spectral solver because it is a global property. An alternate, but often functionally equivalent boundary condition requires that we set $\phi(x,y)$ at one point in the simulation to zero. Then, after solving the matrix problem, we renormalizé the solution to make the D.C. component equal zero. On the mesh, this means dropping the finite difference equation of one node (at x_0, y_0) and using the equation $\phi(x_0, y_0) = 0$ instead. In matrix notation this means discarding a row and a column of the matrix.

This method will fail if one does not apply finite difference equations to the PDE, (3.3), with extreme caution. The node which has been set to zero has a discarded finite difference equation associated with it. After a solution for the remaining finite difference equations has been found, a test should be performed, the discarded equation should be evaluated to determine if it still applies despite the added restriction of $\phi(x_0, y_0) = 0$. If not, then the finite differencing has not been done properly. This is the problem with the approach used by *Newman and Ott* [1981]. If the differencing is done properly one can prove that the discarded equation is satisfied to within rounding error.

An alternative approach to solving the potential equation, discovered by my colleague, Dr. Corrado Ronchi, was to use a sparse matrix solver sophisticated enough to solve the problematic finite difference equations used by Newman and Ott. This routine, called MA28 [*Duff*, 1977; *Duff et al.*, 1992], recognizes and removes the singularities from matrix problems with a technique called "projecting out the null space." Unfortunately, this routine solved the matrix so slowly that it was an impractical solution. However, if one uses it to solve the non-singular matrix described in the following section, it works at about the same speed as the quasi-minimal residual approach.

Details of our Algorithm

Our algorithm starts by rewriting equation (3.1) as (3.3). We then replace all spatial derivatives with a half-step, centered, finite difference approximation:

$$\begin{aligned}&\nabla\cdot(n(x,y)\epsilon\nabla\phi)\approx\\&\frac{n_{i+1/2,j}(\partial_x\phi-\kappa\partial_y\phi)_{i+1/2,j}-n_{i-1/2,j}(\partial_x\phi-\kappa\partial_y\phi)_{i-1/2,j}}{\Delta x}\\&+\frac{n_{i,j+1/2}(\partial_x\phi-\kappa\partial_y\phi)_{i,j+1/2}-n_{i,j-1/2}(\partial_x\phi-\kappa\partial_y\phi)_{i,j-1/2}}{\Delta y}\end{aligned}\tag{3.5}$$

This transformation requires us to evaluate n between existing grid points. To avoid having to do this, the density at these intra-grid points is defined as the average of the density at the surrounding grid points. The resulting finite difference equation becomes

$$\begin{aligned}&\nabla\cdot(n(x,y)\epsilon\nabla\phi)\approx\\&\left(\frac{n_{i+1,j}+n_{i,j}}{2}\right)\left[\frac{\phi_{i+1,j}-\phi_{i,j}}{\Delta x^2}-\kappa\left(\frac{\phi_{i+1/2,j+1/2}-\phi_{i+1/2,j-1/2}}{\Delta x\Delta y}\right)\right]\\&-\left(\frac{n_{i,j}+n_{i-1,j}}{2}\right)\left[\frac{\phi_{i,j}-\phi_{i-1,j}}{\Delta x^2}-\kappa\left(\frac{\phi_{i-1/2,j+1/2}-\phi_{i-1/2,j-1/2}}{\Delta x\Delta y}\right)\right]\\&+\left(\frac{n_{i,j+1}+n_{i,j}}{2}\right)\left[\frac{\phi_{i,j+1}-\phi_{i,j}}{\Delta y^2}+\kappa\left(\frac{\phi_{i+1/2,j+1/2}-\phi_{i-1/2,j+1/2}}{\Delta x\Delta y}\right)\right]\\&-\left(\frac{n_{i,j}+n_{i,j-1}}{2}\right)\left[\frac{\phi_{i,j}-\phi_{i,j-1}}{\Delta y^2}+\kappa\left(\frac{\phi_{i+1/2,j-1/2}-\phi_{i-1/2,j-1/2}}{\Delta x\Delta y}\right)\right]\end{aligned}\tag{3.6}$$

This equation requires us to evaluate the potential, ϕ, between existing grid points. As we did for the density, we define the potential at these intra-grid points to equal the average of the potential at the surrounding grid points. We make this substitution, eliminate the terms that cancel and reorganize to yield

$$\begin{aligned}&\nabla\cdot(n(x,y)\epsilon\nabla\phi)\approx\\&\phi_{i,j}\left(\frac{-2n_{i,j}-n_{i+1,j}-n_{i-1,j}}{2\Delta x^2}+\frac{-2n_{i,j}-n_{i,j+1}-n_{i,j-1}}{2\Delta y^2}\right)+\\&\phi_{i+1,j}\left(\frac{n_{i+1,j}+n_{i,j}}{\Delta x^2}+\kappa\frac{n_{i,j+1}-n_{i,j-1}}{8\Delta x\Delta y}\right)+\\&\phi_{i-1,j}\left(\frac{n_{i-1,j}+n_{i,j}}{\Delta x^2}+\kappa\frac{-n_{i,j+1}+n_{i,j-1}}{8\Delta x\Delta y}\right)+\\&\phi_{i,j+1}\left(\frac{n_{i,j+1}+n_{i,j}}{\Delta y^2}+\kappa\frac{-n_{i+1,j}+n_{i-1,j}}{8\Delta x\Delta y}\right)+\end{aligned}$$

$$\phi_{i,j-1}\left(\frac{n_{i,j-1}+n_{i,j}}{\Delta y^2}+\kappa\frac{n_{i+1,j}-n_{i-1,j}}{8\Delta x\Delta y}\right)+$$
$$\phi_{i+1,j+1}\left(\kappa\frac{-n_{i+1,j}+n_{i,j+1}}{8\Delta x\Delta y}\right)+\phi_{i+1,j-1}\left(\kappa\frac{n_{i+1,j}-n_{i,j-1}}{8\Delta x\Delta y}\right)+$$
$$\phi_{i-1,j-1}\left(\kappa\frac{-n_{i-1,j}+n_{i,j-1}}{8\Delta x\Delta y}\right)+\phi_{i-1,j+1}\left(\kappa\frac{n_{i-1,j}-n_{i,j+1}}{8\Delta x\Delta y}\right) \qquad (3.7)$$

This algorithm approximates the left-hand side of equation (3.3). To use it to approximate a solution to that equation, we numerically evaluate the right-hand side, being careful to use a symmetrical finite difference operator so that the resulting vector has no D.C. component. Then, we move all terms to the right-hand side except the $\phi_{i,j}$ term. For a given mesh, this results in a set of linear algebraic equations which are written in matrix form. Looking at the structure of this matrix shows nine diagonal stripes, as does the matrix in equation (3.9). Also, this matrix is asymmetric and not diagonally dominant. In a diagonally dominant matrix, the term lying exactly on the diagonal is larger than the sum of the remaining terms in that row.

Numerous techniques can be used to solve linear algebra problems of the form $\mathsf{Ax} = \mathsf{b}$. The size of the matrix, A, precludes the use of a direct matrix solvers such as LU decomposition. The non-diagonal dominance prevents successive-over-relaxation methods from converging. Many more sophisticated solvers require the matrix be hermitian or anti-hermitian. We found that a variant of the biconjugate gradient method called the quasi-minimal residual approach will solve this difficult problem in almost every case. *Freund and Nachtigal* [1994] describe this method in detail and discuss using their algorithm. Fortunately, one need not have a detailed understand the internal workings of linear algebra solvers because, once a solution is found, it is a simple matter to test this solution by multiplying it by the matrix A and comparing the result with the expected b. In certain instances, this testing technique fails due to rounding error. However, this problem can be avoided by a method described in [*Press et al.*, 1988].

Explanation of why this algorithm removes the singularity

This finite difference algorithm successfully eliminates the singularity while many simpler approaches fail. In order to explain why this method succeeds, we first demonstrate why this, or almost any method, will successfully remove the sin-

gularity from Poisson's equation. Then, we demonstrate that our unusual finite difference representation of equation (3.7) has a removable singularity. Finally, we show why other algorithms may fail.

A typical finite difference equation used to approximate Poisson's equation, $\nabla^2\phi = d$, is

$$\frac{\phi_{i+1,j} - 2\phi_{i,j} + \phi_{i-1,j}}{\Delta x^2} + \frac{\phi_{i,j+1} - 2\phi_{i,j} + \phi_{i,j-1}}{\Delta y^2} = d_{i,j} \tag{3.8}$$

Applying these equations to solve Poisson's equation on a 4 by 4 mesh yields a 16 by 16 sparse matrix which must be solved:

$$\begin{bmatrix}
1 & a & & a & b & & & & & & b & & & \\
a & 1 & a & & & b & & & & & & b & & \\
 & a & 1 & a & & & b & & & & & & b & \\
a & & a & 1 & & & & b & & & & & & b \\
b & & & & 1 & a & & a & b & & & & & \\
 & b & & & a & 1 & a & & & b & & & & \\
 & & b & & & a & 1 & a & & & b & & & \\
 & & & b & a & & a & 1 & & & & b & & \\
 & & & & b & & & & 1 & a & & & b & \\
 & & & & & b & & & a & 1 & a & & & b \\
 & & & & & & b & & & a & 1 & a & & \\
 & & & & & & & b & & & a & 1 & a & \\
 & & & & & & & & b & a & & a & 1 &
\end{bmatrix}
\begin{bmatrix} \phi_{1,1} \\ \phi_{1,2} \\ \phi_{1,3} \\ \phi_{1,4} \\ \phi_{2,1} \\ \phi_{2,2} \\ \phi_{2,3} \\ \phi_{2,4} \\ \phi_{3,1} \\ \phi_{3,2} \\ \phi_{3,3} \\ \phi_{3,4} \\ \phi_{4,1} \\ \phi_{4,2} \\ \phi_{4,3} \\ \phi_{4,4} \end{bmatrix}
=
\begin{bmatrix} d_{1,1} \\ d_{1,2} \\ d_{1,3} \\ d_{1,4} \\ d_{2,1} \\ d_{2,2} \\ d_{2,3} \\ d_{2,4} \\ d_{3,1} \\ d_{3,2} \\ d_{3,3} \\ d_{3,4} \\ d_{4,1} \\ d_{4,2} \\ d_{4,3} \\ d_{4,4} \end{bmatrix} \tag{3.9}$$

$$a \equiv \frac{-\Delta x^2}{2(\Delta x^2 + \Delta y^2)} \qquad b \equiv \frac{-\Delta y^2}{2(\Delta x^2 + \Delta y^2)} \,. \tag{3.10}$$

We normalized the equations so that the main diagonal of the matrix contains only the value "1."

This matrix problem is singular and, therefore, difficult to solve. If we discard the final row and column of matrix (3.9) and require that $\phi_{4,4} = 0$, then the finite difference equation represented by that discarded row must remain valid.

Otherwise, one can expect some peculiar behavior at that point. To prove that the discarded equation remains valid, we must have

$$b(\phi_{1,4} + \phi_{3,4}) + a(\phi_{4,1} + \phi_{4,3}) = d_{4,4}. \tag{3.11}$$

Adding all the rows of equation (3.9) leaves only

$$(\phi_{1,4} + \phi_{3,4})(1 + 2a + b) + (\phi_{4,1} + \phi_{4,3})(1 + a + 2b) = \sum_{(i,j)\neq(4,4)} d_{i,j} \ . \tag{3.12}$$

This happens because the coefficients in front of all $\phi_{i,j}$, not adjacent to $\phi_{4,4}$, add to zero. The D.C. component of d must equal 0 in order for a solution to Poisson's equation with periodic boundary conditions to exist making

$$\sum_{i\neq 4,4} d_i = -d_{4,4} \ . \tag{3.13}$$

Also, from the definition of a and b, (3.10), we know $a + b = -1/2$. These two relations make equation (3.12) equivalent to equation (3.11). Therefore, the discarded equation remains valid and this finite difference technique successfully deals with the matrix singularity.

A similar analysis demonstrates the validity of equation (3.7). If we use the finite difference equation (3.7) to generate a matrix equation and sum all the rows together, most of the terms are zero as in the Poisson's equation example. Only those nodes neighboring the eliminated node are non-zero. The D.C. component of the right hand side of the potential equation, (3.3), must be zero for reasons identical to the reasons for which the D.C component of the right hand side of Poisson's equation must equal zero. The surviving terms from summing the rows of the matrix equation give an equation identical to the one dropped when we eliminated a node by setting the potential at that node to zero. The algebra necessary to demonstrate this is tedious and will not be presented here.

An algorithm that does not work

Many other finite difference schemes used to solve equation (3.1) fail to meet the requirement that the eliminated finite difference equation remains satisfied. For instance, the simple scheme used by *Newman* [1979], does not work. This scheme substitutes centered finite difference equations for equation (3.1) exactly as one

would expect, yielding:

$$
\begin{aligned}
& n\nabla^2\phi + \nabla\phi\cdot\nabla n - \kappa\hat{z}\cdot\nabla\phi\times\nabla n \Rightarrow \\
& \quad n_{i,j}\Big(\frac{\phi_{i+1,j} - 2\phi_{i,j} + \phi_{i-1,j}}{\Delta x^2} + \frac{\phi_{i,j+1} - 2\phi_{i,j} + \phi_{i,j-1}}{\Delta y^2} + \\
& \quad \left(\frac{n_{i+1,j} - n_{i-1,j}}{2\Delta x}\right)\left(\frac{\phi_{i+1,j} - \phi_{i-1,j}}{2\Delta x}\right) + \left(\frac{n_{i,j+1} - n_{i,j-1}}{2\Delta y}\right)\left(\frac{\phi_{i,j+1} - \phi_{i,j-1}}{2\Delta y}\right) + \\
& \quad \kappa\left[\left(\frac{\phi_{i+1,j} - \phi_{i-1,j}}{2\Delta x}\right)\left(\frac{n_{i,j+1} - n_{i,j-1}}{2\Delta y}\right) - \left(\frac{\phi_{i,j+1} - \phi_{i,j-1}}{2\Delta y}\right)\left(\frac{n_{i+1,j} - n_{i-1,j}}{2\Delta x}\right)\right] \\
& \quad = b_{i,j}
\end{aligned}
\tag{3.14}
$$

If one now adds up all rows of the resulting matrix one finds that each $\phi_{i,j}$ is multiplied by

$$
n_{i,j}\Big(-\frac{2}{\Delta x^2} - \frac{2}{\Delta y^2}\Big) + \frac{n_{i+1,j} - n_{i-1,j}}{\Delta x^2} - \frac{n_{i+2,j} - n_{i-2,j}}{4\Delta x^2} + \ldots \tag{3.15}
$$

Even the terms not adjoining an eliminated node do not add to zero and when one sums the rows of the matrix, the finite difference equation of the eliminated node is not satisfied.

However, one can easily think this is a viable algorithm. If $n(x,y)$ is nearly uniform, as is usually the case in the beginning of a numerical simulation, then the original equation, (3.3), reverts to Poisson's equation and the finite difference equation of the eliminated node will be satisfied. As the density, $n(x,y)$, develops structure, equation (3.3) no longer resembles Poisson's equation and the potential at the node dropped to eliminate the singularity fails to satisfy the dropped finite difference equation.

3.1.3 Validation of the Potential Solver

Accurately implementing a complex algorithm such as the one outlined above requires extreme care and a number of independent tests. It sounds terrible, but we have found that one of the most reliable methods of implementing such an algorithm is to repeat both the derivation and the coding twice and then to run the two codes, side by side, comparing the results. In this instance, that technique proved essential. Also, because of the complexity of this particular algorithm, we made other independent tests of its validity.

When density perturbations are small, one can compare the complex method against the pseudo-spectral method of section 3.1.1. These two methods apply

dramatically different methods to solve the same problem. For a given density, $n(x, y)$, one expects solutions generated by these two methods to agree for long wavelengths. However, for shorter wavelengths, the two methods yield differing solutions. The fundamental reason for this discrepancy lies in the inaccuracy with which finite difference techniques approximate differentials; the approximation is only valid when the wavelength of the variable being differenced is many times longer than the inter-node spacing of the mesh (see Chapter. 4, *Birdsall and Langdon* [1985]). A spectral method, on the other hand, accurately models the difference equations, though a pseudo-spectral method uses a mix of spectral and non-spectral methods.

A second test uses the algorithm of *Newman* [1979] described in section 3.1.2. While this method fails to obtain a solution, it can be used to test an existing solution. Again, since this method differs substantially from the complex method, we only compare longer wavelength modes. These two tests validate this code.

3.2 Hybrid Code

We have successfully incorporated the boundary value solver described above into a hybrid fluid-PIC code. This code is considered a hybrid because we assume the electrons behave as a fluid, while the ions are treated as PIC particles. Given the long time scales (milli-seconds) of the instability growth rate, this approach enables us to simulate this system well into saturation. Treating the ions as particles enables us to obtain the necessary short wavelength damping without the simplification of the two-fluid approach. By treating the electrons as a fluid we need not assume massive electrons as did *Janhunen* [1994*b*] ($m_i/m_e \approx 100$), nor do we have to work on the electron plasma or electron cyclotron timescale.

Our hybrid code applies standard PIC methods to the ions. This means that the ions are represented by a set of "macro-particles." Initially, we place these particles randomly into the two-dimensional space of our simulation. We also assign them a randomly generated Maxwellian distribution of velocities in three dimensions. From the particle positions and velocities, we approximate the density and current using PIC weighting (see *Birdsall and Langdon* [1985]) at every node of our mesh. We then use the algorithm of section 3.1.2 to solve for the electrostatic potential. The positions of the particles are advanced by numerically approximating

the Lorentz force equation using a Boris mover [*Boris*, 1970; *Boris and Roberts*, 1969]. Based on collisional cross sections, an appropriate fraction of particles are scattered using an ion-neutral collision algorithm [*Boyd*, 1991]. Finally, we reevaluate the densities and currents, repeating the entire process until the instability saturates and beyond.

The topic of particle simulation has many complexities and potential pitfalls. We will not discuss the components of our code which we consider standard. Nor will we analyze the more standard problems of particle codes. Readers interested in these topics should refer to the book by *Birdsall and Langdon* [1985]. They show how to implement a PIC code and discuss many of the issues relevant to these and any particle simulations.

Our particle simulator models ion-neutral collisions by elastically colliding a fraction of the ions with a thermal distribution of neutral particles [*Boyd*, 1991]. At each time step, we statistically determine the number of ions which will collide with neutral particles given a specific ion-neutral scattering cross-section, the densities, and the neutral gas temperature. Selecting an ion at random, we generate a random neutral particle by assuming the neutrals have a Maxwellian thermal distribution. If the relative velocity between the two particles exceeds a random number with a value between zero and the highest relative velocity encountered in the simulation thus far, the two particles collide. In the center-of-mass frame, we can calculate the velocities resulting from an elastic collision. This gives us a new velocity for the ion. We translate this back into the simulation frame of reference and forget about the individual neutral particle. Since we know the velocities in three dimensions, we calculate the collisional effects in three dimensions. This allows energy to scatter into and out of the plane of the simulation which lies perpendicular to the geomagnetic field lines.

Modeling ion behavior with a PIC method allows us to include all kinetic ion effects at the expense of slow computation and substantial particle noise. Even with nearly a million particles on a 64 by 64 mesh, random fluctuations in the particle locations lead to an expected standard deviation of 2% to 3% in the number of particles per cell. This level is comparable to the fully developed pure two stream waves when the driving electric field is near the minimum threshold for the instability. We make the driving electric field somewhat higher than ideal and use a short wavelength filter (see *Birdsall and Langdon* [1985], Appendix B) to further

reduce the particle noise.

Computing densities and currents using thousands or millions of test particles requires large amounts of computational power. Most supercomputers have the capability to subdivide a task and perform many parts of the task at once. Unfortunately, the gather and scatter routines of PIC codes can only be subdivided in this way with elaborate, usually machine dependent, algorithms. Because we run this code on a variety of workstations and supercomputers, such optimization has not been implemented.

3.2.1 The Problems with Pure PIC Codes

The disadvantage of using a PIC method to model both ion and electron behavior arises from three features of such methods [*Machida and Goertz*, 1988; *Janhunen*, 1994*b*]. First, these methods require vast amounts of computer time because they calculate the high speed behavior of hundreds of thousands to millions of electrons, while the instability being studied responds at the much slower timescales characteristic of the ion motion. Second, in order to make a solution possible while still using a reasonable amount of computer time, the electron mass must be artificially raised; both *Machida and Goertz* [1988] and *Janhunen* [1994*b*] set $m_i/m_e = 100$. The actual ratio m_i/m_e is actually closer to 50,000. The magnitude of this ratio affects a number of fundamental features of the simulation such as growth rate and phase velocity. Third, particle codes generate large amounts of numerical noise, because the densities and currents are determined by averaging over simulated particle positions and velocities. Even with nearly a million particles on a 64 by 64 mesh, random fluctuations in the particle locations lead to a variation of 2% to 3% in the number of particles per cell. This presents a problem for modeling E region plasma phenomena, where the difference between electron and ion densities is typically less than one part in a million. This means that statistical fluctuations cause differences between the ion and electron densities at least four orders of magnitude larger than those found in the electrojet. These differences invariably cause large electric fields, strongest at the scale of the mesh. Fortunately, these differences become smaller as one looks at longer wavelengths and both Machida and Janhunen see growth of waves at appropriate, long wavelengths. Since these are both nonlinear codes, long and short wavelengths couple together and, if one has a huge numerical error at the short wavelengths, this may very well affect the

longer, more important wavelengths.

3.3 The Two-Fluid Model

The two-fluid code simulates the behavior of the Farley-Buneman instability with far less numerical noise and far more rapidly than the hybrid code. We have implemented such a code, but a crucial problem arises. In the fluid theory for the two-stream instability, the growth rate goes to infinity as the wave number goes to infinity. Simulating such a system is almost impossible because the shortest wavelength allowed in the system will dominate the simulation. Fortunately, linear kinetic theory shows that below a certain wavelength these waves are damped (see section 2.3). A pure fluid code normally neglects such effects, but, in order to obtain a reasonable solution, it is necessary to incorporate at least some form of linear ion kinetic damping. The following section describes our two-fluid code with this damping.

We need to solve the time dependent components of the two-fluid system of equations, 2.13 and 2.14. Since the finite difference equations we use to approximate the spatial components of these equations are second order accurate, we decided to use the midpoint method, which is second order accurate with respect to time (see *Press et al.* [1988], chapter 16). This method requires that we solve the boundary value equation, (2.15), twice for each time step. We used the same boundary value equation solver as in the hybrid code, described in section 3.1.

To incorporate linear kinetic damping, we convert ν_i to $\nu_i^*(\mathrm{k})$ in the update equation for χ, (2.13). In order to determine an appropriate ν_i^*, we solve the linear kinetic dispersion relation of equation (2.46) given the parameters of the simulation desired. We then use the resulting growth rate, ω_i, in the linear two-fluid dispersion relation, (2.32) and solve for ν_i, which we call $\nu_i^*(\mathrm{k})$. By requiring that $\nu_i^*(\mathrm{k})$ be real, we do not match the two-fluid frequency, ω_r, perfectly with the kinetic frequency. Underlying this approach is the assumption that the only appreciable kinetic effect is linear ion Landau damping and that the details of this effect are unimportant except that it damps short wavelength modes. Of course, it would be difficult to assert the accuracy of this method had we not done the hybrid simulations.

Chapter 4

Simulation Results

There is something fascinating about science. One gets such wholesale returns of conjecture out of such a trifling investment of fact.

– Mark Twain

This chapter presents the results of the hybrid and fluid simulations. These simulations show (1) spectra similar to those of type I radar echoes, (2) wave growth at an angle offset from the drift direction, (3) nonlinear coupling to waves perpendicular to the electron drift direction, (4) a saturated wave phase velocity well below the velocity predicted by linear theory but significantly above the sound speed and (5) asymmetries in the spectra which match observed asymmetries. These simulations also demonstrate that the nonlinear electron $\mathbf{E} \times \mathbf{B}$ drift dominates the nonlinear behavior of the simulated Farley-Buneman waves and that other nonlinear, and even many linear, effects are smaller. This chapter illustrates these and other features with figures from and descriptions of the simulation results.

The majority of this chapter discusses results from the hybrid code. First, we give a description of the parameters used in the simulations then, we qualitatively describe the motion pictures and provide information on how to access them over the internet. This includes a description of those nonlinear effects not apparent from the spectral analyses, but visible in the motion pictures. After this, we discuss the effects of changing the driving electric field, certainly the most important parameter variation we make. Following this, we evaluate the effects of changing the electron temperature. Next, we present the results of a set of simulations in which we eliminate the crucial nonlinear $\delta\mathbf{E} \times \mathbf{B}$ electron drift term from the potential

Table 4.1: Parameters of Baseline Farley-Buneman Simulation

Physical Parameters		
Static magnetic field ($\mathbf{B_0}$)	.25	Gauss
Mean vertical $\mathbf{E}$ field (E_{0y})	-0.02	V/m
Mean Ion density (n_0)	10^{11}	m^{-3}
Neutral density (n_n)	6×10^{18}	m^{-3}
Initial Temperatures (T_e, T_i, T_n)	208	$^\circ K$
Effective ion mass (m_i for O_2^+)	4.6×10^{-26}	kg
e^--neutral collision frequency (ν_e)	4.0×10^4	s^{-1}
Ion-neutral collision frequency (ν_i)	2.8×10^3	s^{-1}
Simulation parameters		
Periodic Boundary Conditions		
Total Box Length	8 by 8	m
Grid resolution	64 by 64	
Time step	1×10^{-5}	s
Number of ions	921,600	

equation. This leads to a discussion of the role of ion kinetic effects, particularly ion Landau damping. Following this, we look closely at the spectra of some of these simulations and, using this spectral information, we evaluate the wave phase velocities of a number of simulations. We then compare the density perturbations to the potential perturbations.

While our hybrid simulations cost far more to complete than did our pure fluid simulations, they avoided a problem in the two-fluid code which we never fully overcame. When nonlinear terms dominate the solution (i.e., the waves are saturated), the two-fluid code develops features with scale sizes close to the minimum scale size of the mesh. Nonetheless, the two-fluid code gives similar saturation levels, nonlinear features and wave velocities as the hybrid code. Unfortunately, because of the small scale features, the two-fluid code could never stand on its own, requiring the hybrid results to eliminate doubts about its reliability. We consider this further in our discussion of the two-fluid code results.

4.1 Overview of the Hybrid Simulation Results

Our discussion of the hybrid simulation focuses on two sets of runs. First, we define a "baseline" simulation which attempts to represent typical conditions of the daytime equatorial electrojet at an altitude of approximately $105km$. Table 4.1

Table 4.2: Parameters of Improved Farley-Buneman Simulation

Physical Parameters		
Static magnetic field ($\mathbf{B_0}$)	.25	Gauss
Mean vertical $\mathbf{E}$ field (E_{0y})	-0.014	V/m
Mean Ion density (n_0)	2×10^{11}	m^{-3}
Neutral density (n_n)	2×10^{18}	m^{-3}
Initial Temperatures (T_e, T_i, T_n)	250	$^\circ K$
Effective ion mass (m_i for O_2^+)	5×10^{-26}	kg
e^--neutral collision frequency (ν_e)	1.7×10^4	s^{-1}
Ion-neutral collision frequency (ν_i)	9.5×10^2	s^{-1}

shows the parameters used for this simulation. These parameters have a number of flaws. The inconsequential flaws include a slightly low ion mass, a low plasma density, a high neutral density, a low temperature, and somewhat high collision rates. Table 4.2 shows a set of improved parameters used in a comparison simulation. The results of this run show small quantitative changes in the simulation results but no major qualitative ones and, because of the high cost of running these simulations, we believed it unnecessary to rerun and reanalyze all the data in light of the corrected parameters.

The baseline simulation also uses an artificially high driving electric field, E_{0y}, of $20mV/m$. Unlike the other flaws, this one serves a purpose: minimizing the effects of particle noise, enabling us to obtain rapid results (less than 48 hours per run). The other simulations in this set have a single parameter altered from the values used in the baseline case. This enables us to isolate the effects of the many components of the Farley-Buneman instability.

The second set of simulations mimic electrojet conditions more accurately than did the baseline set by using a lower driving force of $E_{0y} = 15mV/m$. However, if we change only this parameter in the baseline simulation, no Farley-Buneman waves develop because the large amount of particle noise prevents wave growth. Lowering the grid resolution to 32×32, halves the effective particle noise, enabling us to resolve waves generated by a $E_{0y} = 15mV/m \Rightarrow V_d = 600m/s$ driving electric field. With this more realistic driving field, we obtain qualitatively similar results to the baseline simulation. Unfortunately, the resolution of a 32×32 mesh is poor. We performed one run with $E_{0y} = 15mV/m$ on a 64 by 64 mesh using four times the number of particles as in the baseline case (3.6 million particles). This run is

called the "high resolution case" or the "$V_d = 600m/s$ case."

We made two other changes for this simulation. Linear kinetic theory predicts that, with a lower driving force, the dominant mode has a longer wavelength. So we expanded the $\hat{x}$ axis to 6 meters from the 4 meters of the baseline case. In order to better resolve the vertical modes, the high resolution simulation extended the vertical scale from 6 meters to 12.

This simulation devoured a huge amount of computer resources. First, 3.6 million particles required approximately 100MBytes of memory in single precision and twice that for double precision. Second, each update of the particle positions required many floating point operations. Third, the low driving force meant a low growth rate, causing the simulation to take longer to reach saturation than any of the baseline simulations. Fourth, after reaching saturation, we ran the simulation for a substantial duration in order to obtain reliable statistical data. As a result, this single simulation took almost a week of time on one of the worlds fastest and most expensive computers, a Cray YMP. Because of the computational expense of this high resolution simulation, we did only a limited number of comparison runs for this case.

4.1.1 A Qualitative Description of the Simulations

Figures 4.1 and 4.2 show the saturated state of Farley-Buneman waves generated by hybrid simulations for a few cases. The waves shown appear similar to those generated by all fully nonlinear simulations, with a few differences. The more V_d exceeds C_s, the larger the saturated wave magnitude and the more the dominant mode develops a vertical ($\hat{y}$ direction) component. The waves also become less coherent and show more complex, perhaps, turbulent motion. A comparison of the upper plot with the bottom one in figures 4.1 and 4.2 hints at these differences, but the motion pictures make them clear.

In the motion pictures, sequences of images of the same data shown in figures 4.1 and 4.2 are displayed as gray-scale or colormap plots, where the darkness or color of each pixel represents the density at that point. The reader is encouraged to view the motion pictures stored on the internet. They are available in the MPEG format and can be viewed using the MOSAIC tool at the internet address ftp://ee.cornell.edu/pub/jro/meers/fbi-movies.html.

The motion pictures show the growth of two-stream waves starting from particle

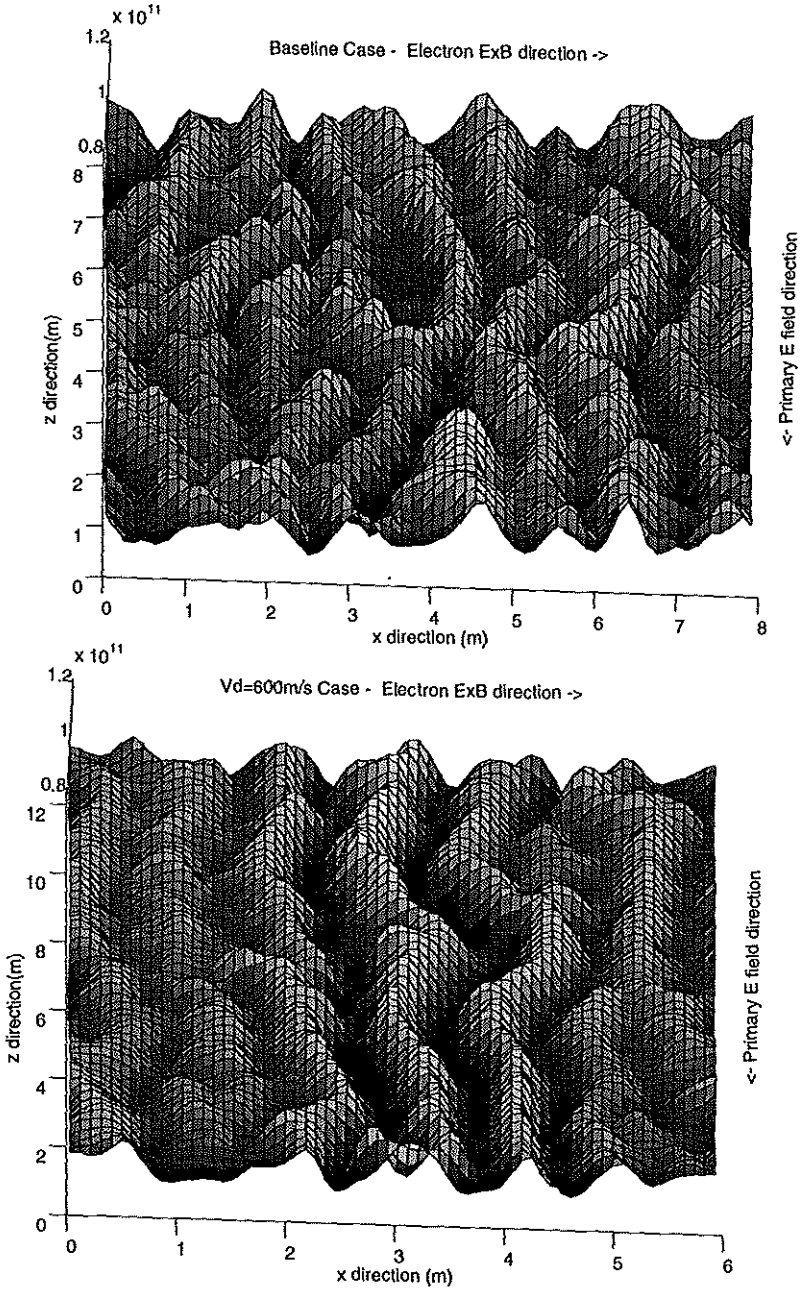


Figure 4.1: Density of waves in a saturated state for the baseline case (top) and the high-resolution, low-driving force case (bottom).

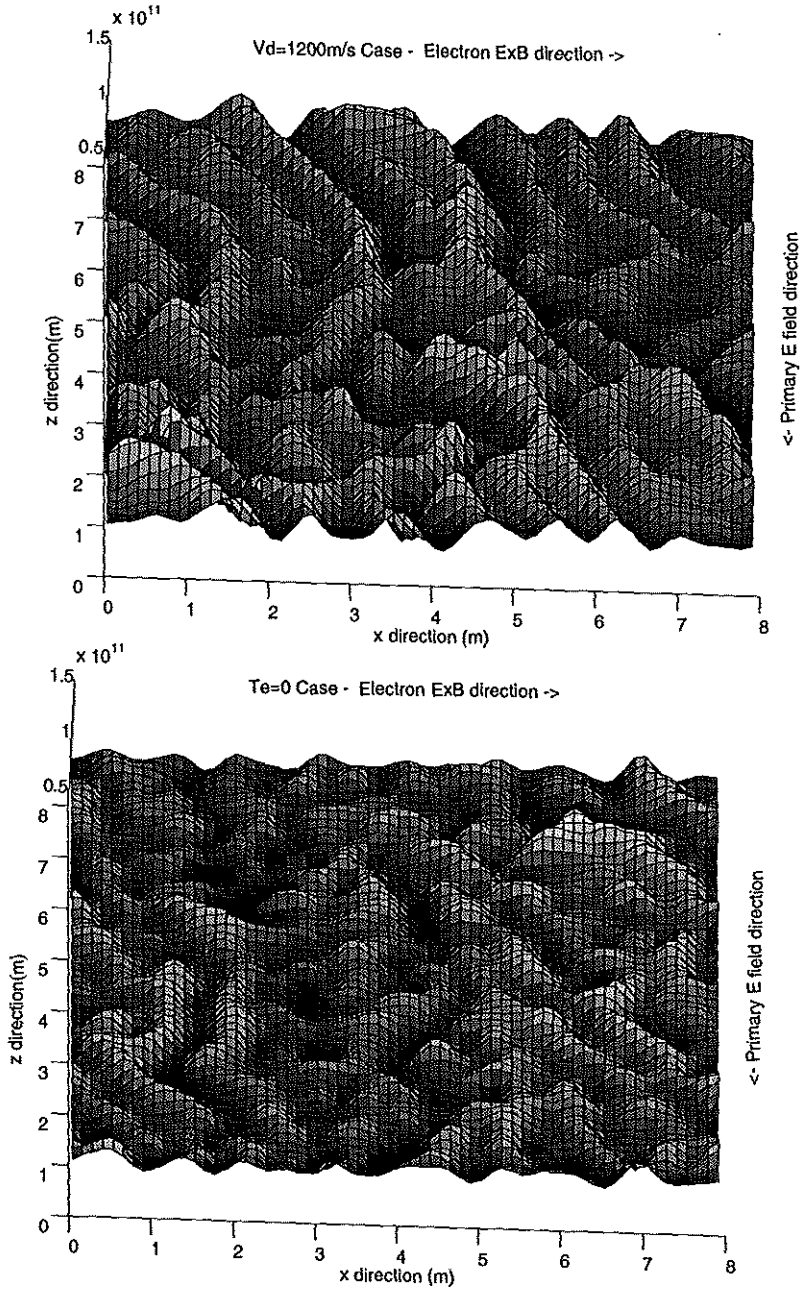


Figure 4.2: Density of waves in a saturated state for the baseline cases in which E_{0y} is changed to $30mV/m$ (top) and T_e is set to 0 in a simulation with $E_{0y} = -20mV/m$ (bottom).

noise and continuing until the waves reach saturation levels and beyond. As these waves develop and saturate, a number of interesting features become apparent. First, the waves grow roughly as described by linear kinetic theory, with the waves having the highest linear growth rate growing fastest. Motion pictures of the low driving force (high resolution) simulation show simple, coherent waves spanning the height of the simulation. While these waves travel toward the right, in the direction of the electron drift, they also develop a small upward, vertical tilt. A close inspection reveals a smaller numbers of waves with a downward tilt. As the driving force, E_{oy}, increases, the waves show more complex behavior in which the tilting angle becomes more pronounced and the waves lose much of their coherence.

Qualitative description of the apparent saturation mechanism

The first nonlinear effect apparent in the motion pictures is the development of secondary waves traveling upward on the crests and downward in the valleys of the principal waves. In one particular low resolution run, with a 32 by 32 mesh and a low driving force, we observe a sequence of events leading to saturation. Initially, the principal waves grow and travel to the right. Then, secondary waves become apparent traveling upward on the crests and downward in the valleys. The principal waves bend, developing vertical components. However, in one instance, one of the primary waves fails to bend and continues growing while the other primary waves in this simulation appear to stop growing. Then, the secondary waves, on this large wave, become very pronounced. After a short while, this wave quickly bends and, almost immediately, its magnitude diminishes to a level comparable with the other waves. As time continues, these waves grow, bend and shrink again, though never again as dramatically as that first wave.

If the secondary waves play an important role in saturation, then an explanation for the failure of any other waves in this simulation to reach the magnitude achieved by that first large wave probably lies in the amount of energy distributed throughout the spectrum. At startup, the entire simulation spectrum contains only the energy caused by particle noise and linear effects. This means that the secondary waves grow from a very small initial perturbation, allowing the primary waves to reach a large magnitude before the secondary waves have grown to a sufficient amplitude to inhibit primary wave growth. Later in the simulation, more energy is distributed throughout the spectrum, seeding the secondary waves which,

in turn, saturate the primary waves.

The sequence of events seen in this simulation leads us to believe that when the secondary waves reach some critical magnitude they cause the primary waves to bend and drop in magnitude. These bent primary waves have a lower linear growth rate than do horizontally propagating primary waves. This reduced growth rate, combined with the interaction between primary and secondary waves, appears to lead to a saturated state.

Only in one simulation do we see this dramatic overshoot, where the wave does not couple to secondary waves until it has reached a relatively large magnitude. If we rerun the same simulation with a different random number generator seed, we obtain similar results but no waves reach the magnitude of that single early wave. In simulations with a higher driving force, E_{0y}, the processes of wave growth, development of secondary waves, bending of the primary wave and saturation all seem to occur nearly simultaneously. These simulations do not make it apparent that each step leads to the next as do the simulations with the low driving force.

The vertically traveling secondary waves probably account for the vertically oriented density irregularities associated with type I radar echoes [*Cohen and Bowles*, 1967]. A simple parametric analysis of the fluid equations, (2.17) -(2.19), shows that when the primary, horizontal waves attain a certain amplitude, these secondary waves must grow. This will be discussed in detail in chapter 5.

Other features seen in the motion pictures

When these same density waves are viewed as a grey-scale movie, the secondary waves are still seen but they appear less dramatic, probably because the color motion pictures tend to draw attention to the extremes of the density enhancements and depletions while the gray-scale motion pictures give more linear weighting to the densities. The gray-scale motion pictures, however, show a feature that does not stand out in the colormap version. At the edges, or nulls, of the principal waves (i.e., along the $n \approx n_0$ lines), a rippling develops. Sometimes these ripples develop into eddies which break off from the principal wave. These caviton-like density depressions tend to move much more slowly than the Farley-Buneman waves and usually disintegrate after the next wave front collides with them. In the process, though, they interact with this oncoming wave, and lose some of their coherence.

The following subsections examine the effects of changing two parameters, the

driving electric field and the electron temperature. Changing the strength of the driving electric field, E_{0y}, leads to some qualitative changes in the behavior of the Farley-Buneman instability. Varying the electron temperatures has a lesser effect on the wave behavior but we discuss it below because of the importance of the question of electron temperature effects.

4.1.2 Drift Velocity Effects

An important parameter in these simulations is the strength of the driving electric field E_{0y}. This field supplies free energy to the system and, according to linear theory, determines a number of important behaviors. First, as E_{0y} increases, the zero-order electron drift speed, $V_d = E_{0y}/B_0$, increases. According to linear fluid theory, an increased electron drift speed causes the phase velocity of the wave, V_{ph}, to increase proportionally and the growth rate, γ, to increase with increasing V_d^2 as,

$$V_{ph} = V_d/(1+\Psi_0) \tag{4.1}$$

$$\gamma = \frac{\Psi_0}{\nu_i(1+\Psi_0)}(V_d^2 - C_s^2)k^2(\cos\Theta - \sin\Theta/\kappa) \tag{4.2}$$

$$\Psi_0 \equiv \frac{\nu_e\nu_i}{\Omega_e\Omega_i}$$

where Θ is the elevation angle. Linear kinetic theory reduces the phase velocity a little, while reducing the growth rate, especially for short wavelengths, considerably. This is described in detail in section 2.3. Kinetic theory also predicts that the wavelength of maximum growth moves to shorter wavelength (higher k) as one increases the driving electric field. We expect the wavelength with the largest growth rate to dominate the simulation, though nonlinear effects may change this somewhat.

The Farley-Buneman waves are still expected to have a vertical component, as discussed in the work of *Newman and Ott* [1981] and *Janhunen* [1994*b*]. However, neither studied the dependence of this vertical component on the electron drift speed or on other electrojet parameters.

We also expect that, as the driving electric field increases, nonlinear effects should become more pronounced. The wave magnitude at saturation should increase, though the relationship between E_{0y} and the saturation magnitude need

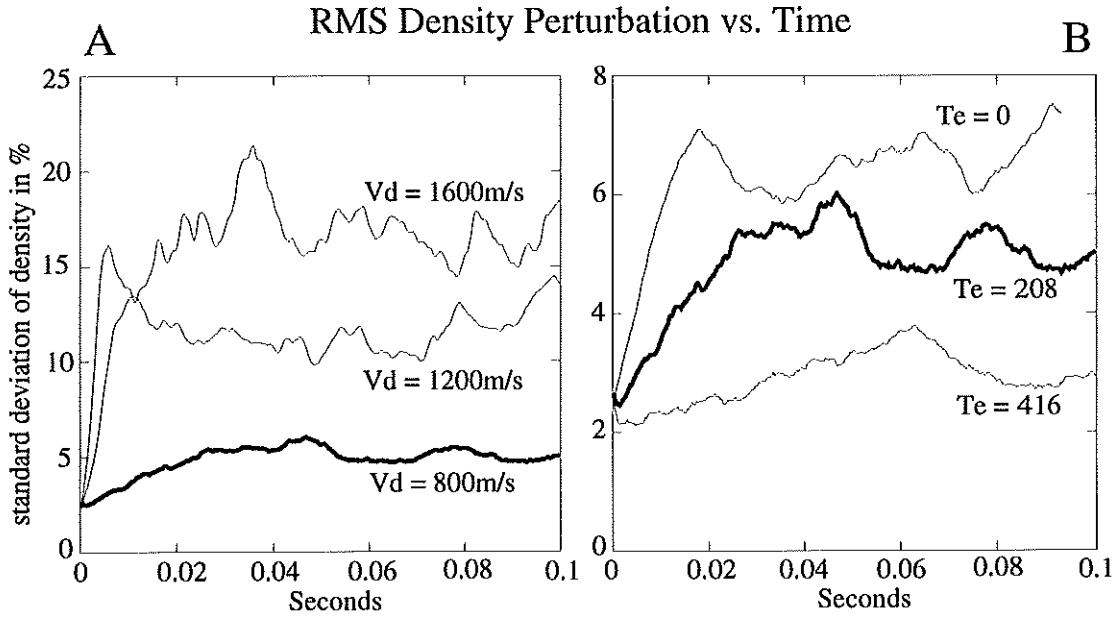


Figure 4.3: These figures represent time versus the standard deviation of the spatial variation of the density, $\langle|\delta n/n_0|^2\rangle$, for (A) three drift velocities with $T_e = T_i = T_n = 208K$ and (B) three electron temperatures with constant $V_d = 800m/s$ and $T_i = T_n = 208K$. The baseline case is the bold line. The relative growth rates between simulations behave as predicted by linear theory. Hence, in figure (A), the growth rate scales proportionally to the square of the drift velocity and, in (B), the growth rate scales with the negative of the square of the acoustic speed.

not be simple. As nonlinear effects increase, we expect spectral broadening, more coupling between modes and, perhaps, turbulent behavior.

Some of our predictions were validated by the simulations, while others failed. Figure 4.3A shows the time evolution of a set of runs with differing driving electric fields, E_{0y}. Each curve follows the time history of the standard deviation of the density taken for the 64 by 64 nodes of the simulation,

$$\langle|\delta n/n_0|^2\rangle^{1/2} = \left[\sum_{i}^{all\ nodes} (\frac{n_i}{n_0} - 1)^2\right]^{1/2} . \tag{4.3}$$

Figure 4.4 compares the baseline case to the high resolution, low driving force simulation. It plots both the standard deviation of the density and its maximum value in the simulation at 25 milli-second intervals. From these figures, one sees that the growth rate scales proportionately to V_d^2 less a constant factor, C_s^2, as

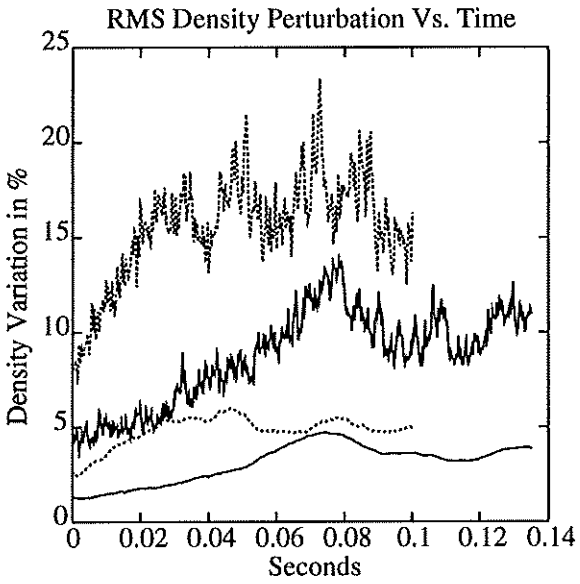


Figure 4.4: Time vs. $\langle|\delta n/n_0|^2\rangle$ of the simulation with $V_d = 600m/s$ (solid lines) compared to the baseline case where $V_d = 800m/s$ (dotted lines). The lower, smooth, curves show the standard deviation of the density over the entire simulation area, $\langle|\delta n/n_0|^2\rangle$, while the upper, jagged, curves show the maximum density on the mesh at a given time.

predicted by equation (4.3). One also sees that the saturation amplitude increases with increasing V_d.

In the high resolution ($V_d = 600m/s$) simulation, the dominant mode lies close to the mode predicted by linear kinetic theory except for a small vertical component. As V_d increases, the wave number, k, of the dominant mode increases but by less than kinetic theory predicts. Whatever causes this reduced dominant wavenumber may also be responsible for the increased vertical component.

The harder driven systems ($V_d \geq 800m/s$) run into the limitations of our hybrid simulation. First, linear kinetic theory predicts growth down to a size considerably smaller than the dominant mode allowed in the simulation. Further, the evidence from the polar Farley-Buneman instability indicates that, for the case where the system is driven by a large E_{0y}, electron heating occurs. These simulations are incapable of modeling that phenomenon because our fluid model assumes isothermal electron behavior with no heating mechanism allowed. Incorporating such a mechanism should be possible, but has not yet been done.

As expected, our motion pictures show that an increasing driving force causes

nonlinear effects to become more pronounced. The secondary waves traveling on the crests and in the valleys of the principal waves become more obvious. Also, the eddies at the edges of these waves seem larger and interact more substantially with the principal waves. In the low drift-speed simulation ($V_d = 600m/s$) the secondary waves and eddies clearly affect the primary waves, but remain relatively distinct from them. In the highest drift-speed simulations ($V_d \geq 1200m/s$) the eddies and secondary waves so disrupt the primary waves that distinguishing between primary and secondary waves becomes difficult.

The spectral analysis of section 4.4 reveals a number of unpredicted features. One sees that the dominant mode does not travel exactly parallel to the electron drift direction, developing a substantial vertical component instead. This vertical component increases with increasing E_{0y} and seems to peak with waves traveling upward at an angle of 45 degrees. These waves do not travel at the speeds predicted by linear theory, but instead move with a phase velocity between the linear theory velocity and the acoustic velocity. This effect is especially pronounced in cases with large driving electric fields, where V_d and C_s are markedly different. We also see the development of long wavelength modes that travel substantially slower than the sound speed. These do not develop until the simulation saturates and seem to result between coupling from the stronger, short wavelength, modes.

4.1.3 Electron Temperature Effects

The baseline case assumes isothermal electrons with $T_e = T_n$. This is not always a valid assumption and a more sophisticated model of electron kinetic effects may be necessary, particularly for extrapolating our results to the auroral two-stream instability. However, we find that drastically changing the electron temperature in the simulations has a minimal effect on the behavior of the saturated waves. The pure particle simulations of *Janhunen* [1994*b*] and *Machida and Goertz* [1988] show that, when one drives a Farley-Buneman system with a large E_{0y}, the electron temperature rises above the neutral temperature but does not display complex wave-dependent electron kinetic effects. Figure 4.3A shows the density variations versus time for three electron temperatures. Even for these extreme cases, the effects are not dramatic.

Lowering the electron temperature has a similar effect to raising the drift velocity. The waves saturate at a higher magnitude, the largest mode has a greater

vertical component (i.e., its direction of travel has a greater upward component), the waves become less coherent and show more turbulent motion. Additionally, all these features travel more slowly as a result of the lower acoustic velocity of this reduced temperature simulation.

The simulation with twice the electron temperature shows all the features of a simulation with a lower drift velocity: coherent waves, lower saturation amplitude and less vertical component to the dominant waves. One would expect this in a system where the difference between V_d and C_s has been reduced. Also, the waves travel more rapidly because of the increased acoustic speed.

4.2 Linearized Electron Behavior

Linearizing the electron behavior drastically changes the results of the simulations and reveals the importance of the nonlinear $\kappa\hat{\mathbf{z}}\cdot\nabla\phi\times\nabla n$ term in equation (3.3). In order to eliminate this term from equation (3.3), our numerical method requires that we also eliminate the $\nabla\phi\cdot\nabla n$ term as explained in section 3.1.2. Since this term is not multiplied by a factor of $\kappa \sim 100$ as is the $\hat{\mathbf{z}}\cdot\nabla\phi\times\nabla n$ term, we assume that it has a relatively small effect. Dropping these two terms from the potential equation reduces it to

$$n\nabla^2\phi = \mathbf{E}_0\cdot\nabla n + \kappa\hat{\mathbf{z}}\cdot\mathbf{E}_0\times\nabla n + \frac{T_e}{e}\nabla^2 n + \Gamma\nabla\cdot\mathbf{J_i} \tag{4.4}$$

which we solve with the pseudo-spectral Poisson solver described in section 3.1.1. In this case, the waves saturate at amplitudes almost an order of magnitude larger than those obtained in the baseline simulation (Figure 4.5). Figure 4.6 shows a typical density profile of the saturated waves obtained from a simulation employing linearized electron dynamics. These waves are far more monochromatic and uniform than those from the fully nonlinear simulation. They also propagate in a direction parallel to the horizontal drift velocity direction, which contrasts with the dominant mode of the fully nonlinear simulation.

In the simulation with linearized electron behavior, nonlinear ion effects saturate the wave because we have eliminated all other potential mechanisms. We have good evidence that ion kinetic effects and not ion fluid effects saturate the wave. First, the velocity distribution of the ions shows a stretching out of the distribution around the electron drift speed, a characteristic signature of Landau damping

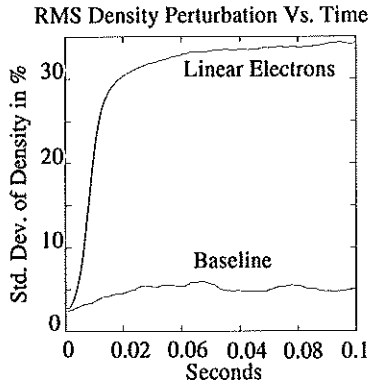


Figure 4.5: Time history of the standard deviation of $\delta n/n_0$ for the simulation with linearized electron behavior compared with the baseline case.

(discussed in section 4.3). Second, when we watch the particle motion in phase space, by using an animation tool (HDF), we see evidence of particle trapping and wave breaking.

The pronounced differences between the fully nonlinear simulations and those with linearized electron dynamics make a strong case for the importance of the $\kappa\hat{z}\cdot\nabla\phi\times\nabla n$ term in the potential equation. The fully nonlinear simulation saturates at a much lower density perturbation level, $\delta n/n_0$, than does the linearized one. Also, the linearized simulation shows no wave turning, secondary waves or mode coupling. It also shows little or no evidence of ion kinetic effects, as we discuss below.

4.3 Ion Kinetic Effects

The importance of ion kinetic effects on the resulting wave behavior is of also interest. Since our hybrid simulations use PIC particles to model ion dynamics, our results may be affected by ion kinetic effects.

Figure 4.7 shows the saturated phase space distribution of ions in the high resolution case compared with a calculated Gaussian distribution having the same mean and the same assumed temperature of $208°K$. The distribution of ions in velocity space differs from the calculated Gaussian profile only by an offset of its mean velocity, due to the ion Pederson drift velocity,

$$V_{id} = \frac{eE_{0y}}{m\nu_i}\hat{y} \ . \tag{4.5}$$

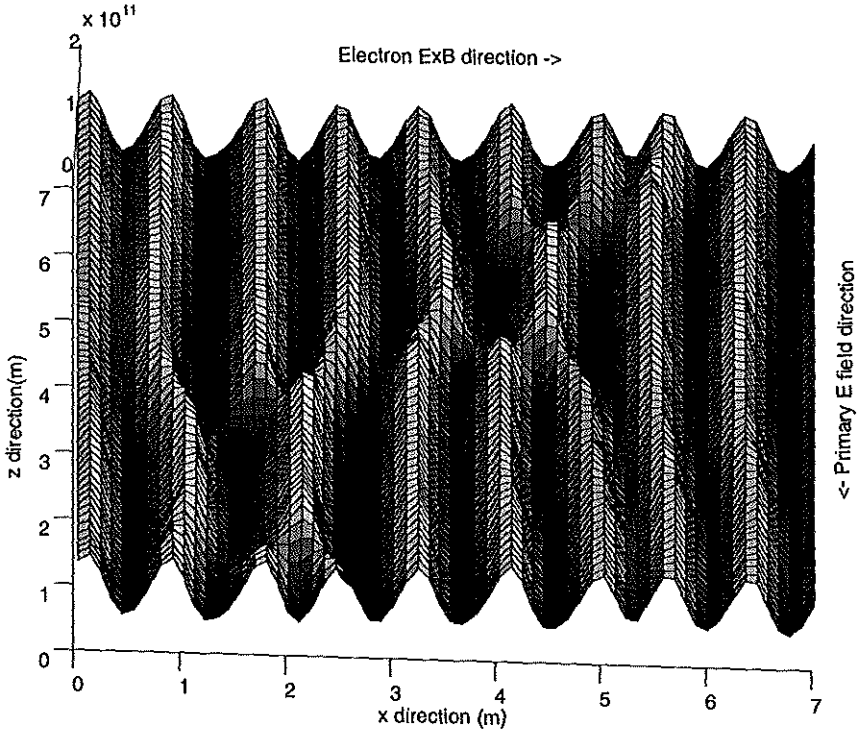


Figure 4.6: A typical snapshot of the density profile in the case where the electron behavior has been linearized. This wave travels horizontally.

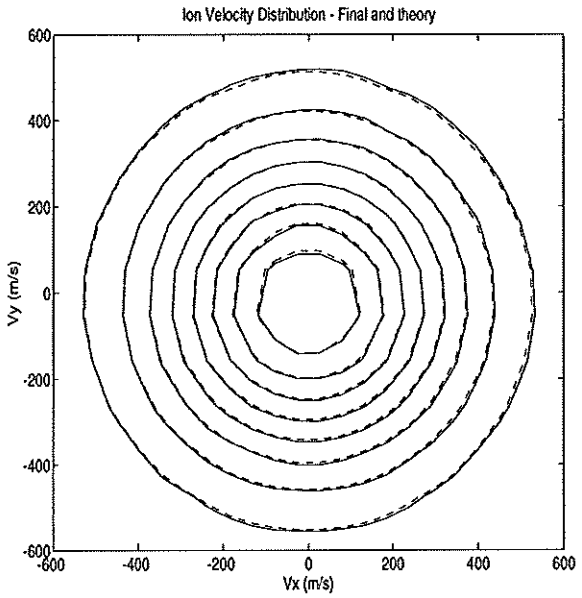


Figure 4.7: The velocity distribution of ions for the high resolution case. The solid lines show contours of the ion velocity distribution recorded at the end of the simulation. The dashed lines show contours for a theoretical Gaussian distribution with a temperature, $T_i = 208K$, and a vertical offset due to the ion Pederson drift.

If ion Landau damping were important, one would expect to see a stretching of the distribution around the resonant frequency, as described in Chapter 6 of *Nicholson* [1983] and Chapter 7 of *Chen* [1984]. The distribution of the low driving force simulation shows little or no evidence of ion kinetic effects while a simulation driven harder shows a very small effect. Figure 4.8 compares the ion velocity distribution of the baseline case to a calculated Gaussian distribution with the same ion Pederson drift speed and a theoretically imposed temperature of $208°K$. In this case, one sees an almost imperceptible stretching of the distribution for particles having velocities approaching the electron drift speed, V_d.

Figure 4.9 shows a case where the electron behavior has been linearized as described in section 4.2. Here we see a strong signature of ion Landau damping. Also, when we watch the particle motion in phase space, evidence of particle trapping and wave breaking is seen. However, when we incorporate the nonlinear terms back into the simulation these effects do not appear. Instead, two dimensional, nonlinear $\delta\mathbf{E}\times\mathbf{B}$ effects seem to prevent the primary wave from becoming energetic enough

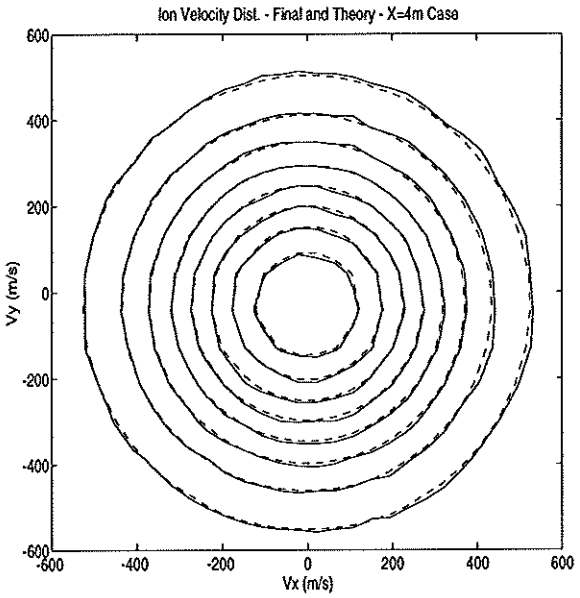


Figure 4.8: The velocity distribution of ions for the 4m by 8m baseline simulation. The solid lines show contours of the ion velocity distribution recorded at the end of the simulation. The dashed lines show contours for a theoretical Gaussian distribution with a temperature, $T_i = 208K$, and a vertical offset due to the ion Pederson drift.

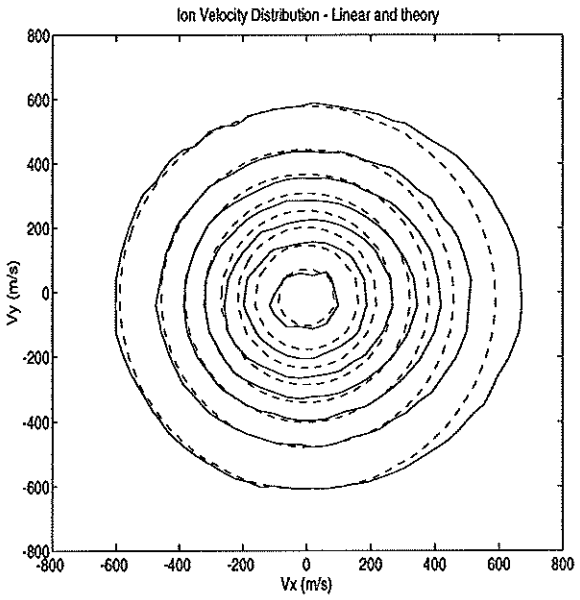


Figure 4.9: The velocity distribution of ions for the $E_{0y} = 15mV/m$ case with linearized electron behavior. The solid lines show contours of the ion velocity distribution recorded at the end of the simulation. The dashed lines show contours for a theoretical Gaussian distribution with a temperature, $T_i = 208K$, and a vertical offset due to the ion Pederson drift. The stretching of the distribution for large positive values of v_x indicates the importance of Landau damping.

to drive these ion kinetic effects. It is not surprising that the simulations with the linearized electron behavior show evidence of ion kinetic effects. In a particle code, as in reality, nonlinear phenomena always saturate the waves. By eliminating nonlinear electron effects, all that remains are nonlinear ion effects. Theoretically, nonlinear ion fluid effects could saturate the system, but these terms appear relatively small, while, as we know from linear kinetic theory, ion kinetic effects show their importance by selecting the dominant mode. What is surprising is how small this effect appears when the nonlinear electron effects are present. This method of looking for kinetic effects by comparing the particle distributions with theoretical distributions may be relatively insensitive, even though, for the simulations with linearized electron behavior, this method proved effective.

The apparent absence of strong ion kinetic effects suggests that a fluid model describing ion behavior may successfully model the Farley-Buneman instability. Our experience shows that a pure two-fluid code, which entirely ignores kinetic effects, develops numerical instabilities due to growth at the shortest wavelength allowed by the simulation. However, one can roughly imitate the linear component of Landau damping with a wavelength dependent viscosity or diffusion. If the details of the mechanism causing short wavelength damping are important, then a two-fluid code will fail to give accurate results. However, if the only important ion kinetics effect is this damping, then a two-fluid code using an enhanced k-dependent damping, as described in section 3.3, should give reasonable solutions at a far lower computational cost than the hybrid code.

4.4 Spectral Analysis

When radars probe the E region, they measure one wavelength at a time. Therefore, comparisons of radar observations with simulation results should be performed in Fourier space. Radars also clearly record Doppler shifts which correspond to phase velocities. Since phase velocities are defined as the ratio of two spectral quantities, frequency and wavenumber, they too should be discussed as a part of spectral analysis. Because of the importance and controversy surrounding the phase velocities of two-stream waves, they are discussed in their own section (4.5).

Spectral analysis in both time and space reveals much about the simulation not obvious from the original images. Some of the nonlinear features discussed

earlier, such as the secondary, nonlinear electron $\mathbf{E}\times\mathbf{B}$ drifts waves, appear highly localized in space and therefore spread out in k-space. These features appear in the spectra because no information is lost when transforming data from real space to k-space. On the other hand, obvious features seen in the motion pictures are difficult to discern when viewed in the spectra.

Simulation spectra for all modes, including vertically oriented modes, show a remarkable similarity to radar spectra. This occurs despite the differences between what radars and our simulations actually measure. Radars average a much larger area of space for a much longer period of time than a realistic simulation can currently model.

We generate simulation spectra as follows. The hybrid code saves periodic snapshots of the density, potential and currents at fixed intervals (typically every $0.25ms$). We calculate a spatial spectrum at each timestep by Fourier transforming each snapshot with a two dimensional fast Fourier transform (FFT). Then, for each wavenumber, k, we Fourier transform the time series into frequency, ω, space. Unfortunately, we cannot display the resulting three dimensional information as such, so we limit ourselves to examining cross-sections, averages and individual spectra.

4.4.1 Time Averaged Spectra

Averaging over the temporal or frequency variations enables us to easily see the dominant modes of the simulation. Figure 4.10 shows the spectrum of the high resolution, low driving force, simulation. From this we see that the dominant mode, having $k_y = 0$, travels horizontally. The $k_y = 0$ modes for positive and negative k_x are identical because of the symmetry resulting from Fourier transforming real data, $f(\mathrm{k}) = f^*(-\mathrm{k})$.

For both positive and negative k_x, substantial energy propagates with a non-zero k_y. The difference between positive and negative k_x ($k_y \neq 0$) means that an asymmetry exists between these modes. Modes with $k_x > 0$ and $k_y > 0$ have only positive frequencies and, therefore, travel up and to the right while modes with $k_x < 0$ and $k_y > 0$ have only negative frequencies and travel down and to the right. This becomes clearer in the spectral cross-sections discussed in the following subsection. We shall call this asymmetry a "$\pm k_x$ asymmetry." Motion pictures of this particular simulation do not make this asymmetry clear, but the spectrum

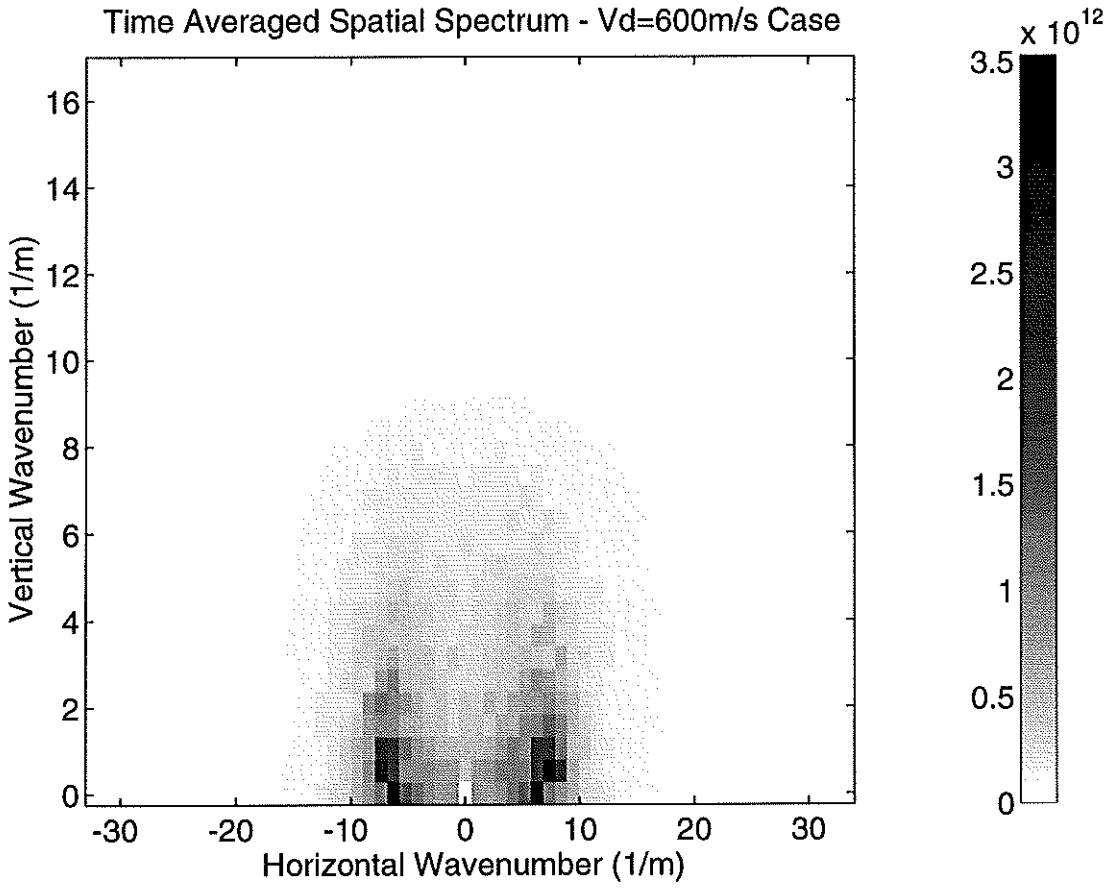


Figure 4.10: Magnitude of the spatial spectrum of the density perturbations averaged over 200 spectra spanning $50ms$ of simulation results. Each averaged spectrum derives from the saturated component of the high resolution simulation where $E_{0y} = 15mV/m$. Approximately one half the area of the spectrum is completely quiet (white) due to the particle smoothing algorithms. This figure shows only spectral data in the upper half plane because when computing a Fourier transform of real data, the transformed data has the symmetry $f(\mathrm{k}) = f^*(-\mathrm{k})$. Hence, the negative wavenumbers in the k_y direction are just the complex conjugate of the positive k_y wavenumbers shown.

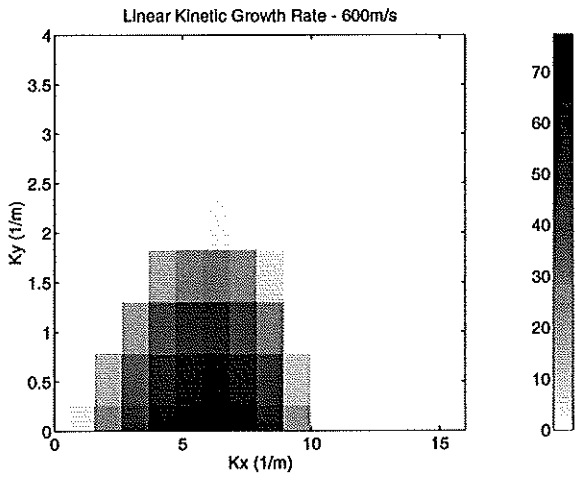


Figure 4.11: The positive growth rates predicted by linear kinetic theory for the low driving force simulation with $E_{0y} = 15mV/m$.

of figure 4.10 shows a small $\pm k_x$ asymmetry. In simulations with larger driving forces, this $\pm k_x$ asymmetry is clear in both the motion pictures and spectra.

We argue that three modes dominate the simulation: a horizontal mode, a mode with a slight upward tilt and a smaller mode with a slight downward tilt. One could also interpret the spectrum as showing a single mode with a substantial spectral width which causes it to extend beyond the $k_y = 0$ line.

Comparing this time averaged spectrum with the growth rates predicted by linear kinetic theory, shown in Figure 4.11, reveals that the dominant mode lies on the point of peak growth while the other active modes lie on points of lower growth. One also sees from this figure that linearly damped modes contain substantial amounts of energy.

The time averaged spectrum from the baseline case shows more complex features than those exhibited by the $E_{0y} = 15mV/m$ case. Figure 4.12 shows the time averaged spatial spectrum for the saturated baseline case. Here, the dominant mode lies at $k_x \approx 6$, $k_y \approx 3$. Unlike simulation using a low driving force, the dominant mode has a substantial vertical component for positive k_x but relatively little vertical activity for modes with a negative k_x. This shows a dramatic $\pm k_x$ asymmetry, visible in the motion pictures as well as in the spectra. This asymmetry agrees with the results of *Janhunen* [1994*b*] who saw this feature in his pure particle simulations where waves with phase velocities satisfying $\mathbf{V}_{ph} \cdot \mathbf{E}_0 < 0$ contained significantly more energy than did waves satisfying $\mathbf{V}_{ph} \cdot \mathbf{E}_0 \geq 0$.

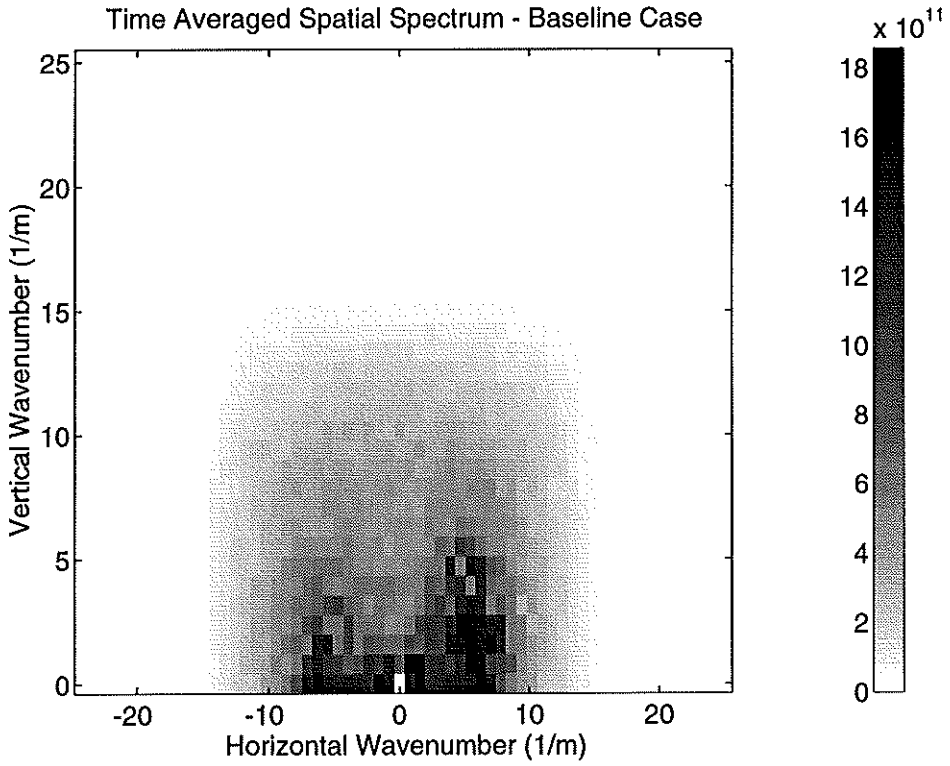


Figure 4.12: Magnitude of the spatial spectrum of the density perturbations averaged over 200 spectra spanning $50ms$ of simulation results. Each averaged spectrum derives from the saturated component of the baseline simulation. Approximately one half the area of the spectrum is completely quiet (white) due to the particle smoothing algorithms. This figure shows only spectral data in the upper half plane because when computing a Fourier transform of real data the transformed data has the following symmetry, $f(\mathrm{k}) = f^*(-\mathrm{k})$. Hence, the negative wavenumbers in the k_y direction are just the complex conjugate of the positive k_y wavenumbers shown.

Figure 4.12 reveals a second, smaller, pair of modes near $k_x \approx \pm 1$. These modes show an upward tilt similar to that seen in the dominant, $k_x \approx 7$, mode. The modes between these long wavelength modes and the dominant mode show a substantial decrease in activity, in spite of the fact that they have a larger linear growth rate than does the long wavelength, $k_x \approx 1$, mode. We have two possible explanations. First, the long wavelength mode exists because it steals energy from the dominant mode or, second, the intervening modes damp out due to nonlinear processes. The first explanation seems more plausible, especially when we take into account that these long wavelength modes travel more slowly than the phase velocity predicted by linear theory. If this theory is correct, then these modes probably do not appear in the low driving force simulation because (1) the coupling effect is not as strong or (2) the simulation does not allow an appropriate mode and, so, no coupling occurs. The motion pictures do not clearly show the longer wavelengths, probably because we perceive small scale features more readily than large scale features.

Not only does the baseline simulation develop long wavelength modes absent from the low driving force simulation, the dominant mode is also considerably broader than the comparable mode in the low driving force simulation. One might expect that increasing the driving force would increase the importance of nonlinear behaviors which often lead to mode broadening. As seen in figure 4.12, the baseline simulation also scatters much more energy throughout k-space than does the low driving force simulation.

The linear growth rate shown in Figure 2.4 indicates that the fastest growing mode lies almost half way between the minimum and maximum wavenumbers allowed by the simulation. Both finite differencing errors and our particle noise filtering algorithms damp these high **k** modes. As a check to see if we have artificially suppressed the dominant mode we repeat this simulation with the horizontal dimension spanning only $4m$ instead of $8m$. Figure 4.13 shows this spectrum, which differs from the baseline case spectrum in a number of ways. It shows broader and less distinct modes, and the dominant modes have slightly higher k_x values and less vertical spread, though not as high as predicted by linear theory.

Why do neither of these simulations show peak growth anywhere near the $k_x \approx 14$, $k_y = 0$ point predicted by linear kinetic theory? Figures 4.14 and 4.15 show the spectra of the 8 by $8m$ and 4 by $8m$ simulations with linearized electron behavior. We see the same dominant wavelength as in the nonlinear case. Note

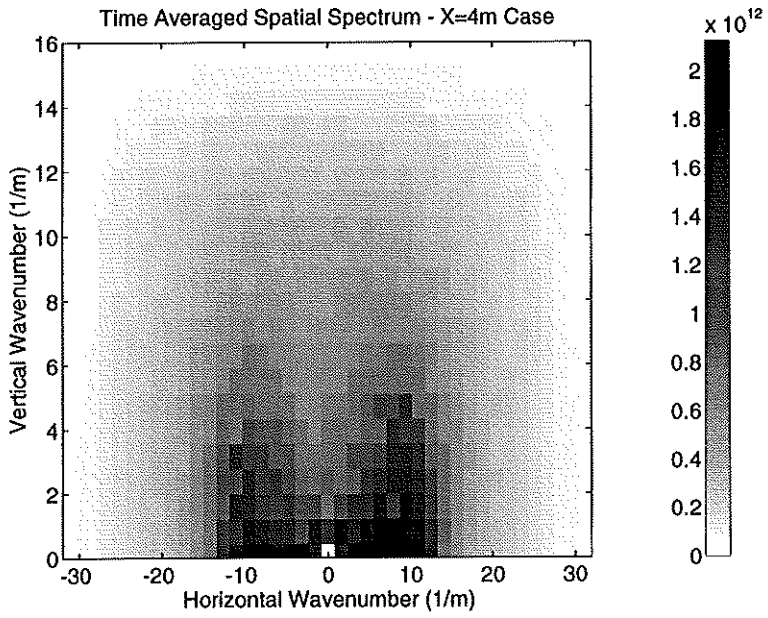


Figure 4.13: Magnitude of the spatial spectrum of the density perturbations averaged over 200 spectra spanning $50ms$ of simulation results. Each averaged spectrum derives from the saturated component of the baseline simulation where the horizontal axis spans only four meters. This gray-scale map does not show the entire spectrum, only those modes with appreciable density perturbations.

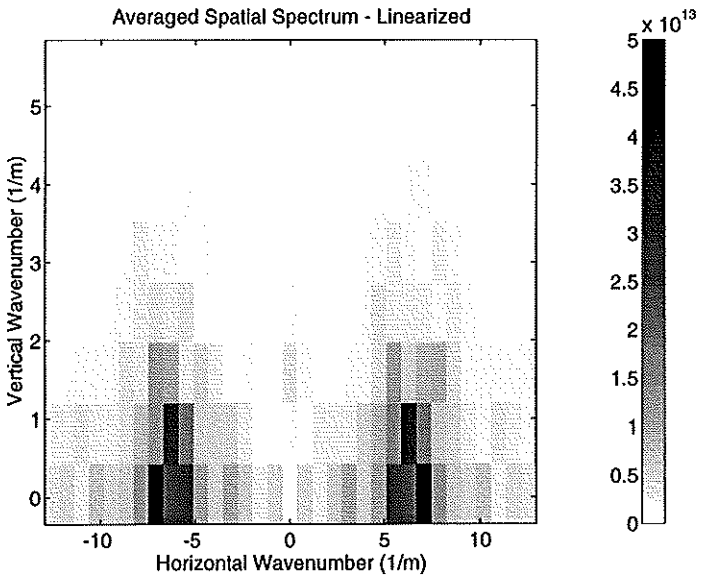


Figure 4.14: Magnitude of the spatial spectrum of the density perturbations averaged over 200 spectra spanning $50ms$ of simulation results. Each averaged spectrum derives from the saturated component of the simulation with linearized electron dynamics. The parameters used in this simulation match the baseline case except that the horizontal system length spans only four meters. This colormap does not show the entire spectrum, only those modes with appreciable density perturbations

that the waves of these linearized simulations saturate at much higher density perturbations and the ion velocity distribution differs from the nonlinear case as discussed in section 4.3. Even so, it seems unlikely that these simulations develop almost identical dominant wavelengths by coincidence. We conclude that nonlinear kinetic effects such as nonlinear Landau damping play an important role in selecting the dominant frequencies, particularly when the electron drift velocity exceeds the sound speed by a considerable amount.

In-situ measurements made by rockets can compare wavenumber to wave amplitude. For example *Pfaff et al.* [1987*a*], shows a series of power versus frequency measurements where the frequency was measured in the rocket's frame of reference. Unfortunately, the rocket accelerates, spins and, sometimes, precesses. None of these effects have been removed from the data shown in the figures, making them difficult to interpret. Hence, *Farley* [1985] asserts only that $\langle |\delta N(k)|^2 \rangle \sim k^{-n}$ where $n \sim 2 - 4$.

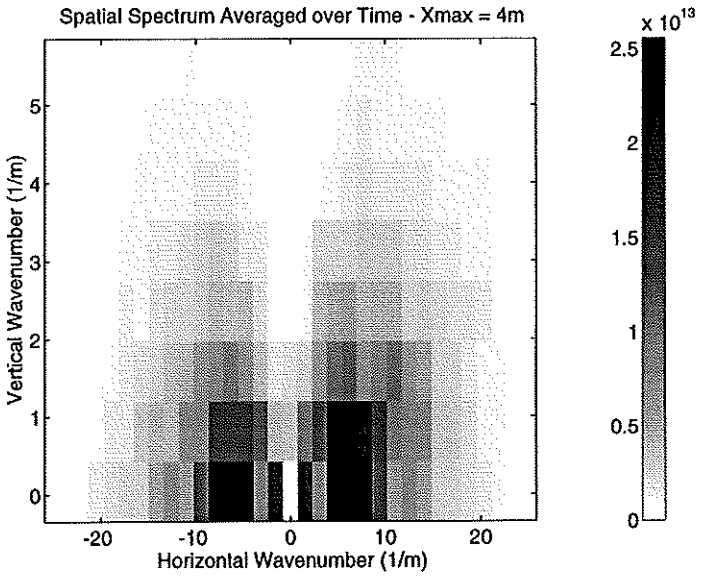


Figure 4.15: Spatial spectrum of the density perturbations averaged over 100 spectra from the saturated component of the linearized electron simulation where the horizontal axis spans only four meters. This grey-scale map does not show the entire spectrum, only those modes with appreciable density perturbations

We can compare the simulations wavenumber to its amplitude, though only for a small range of wavenumbers. Figure 4.16 shows this information for strictly horizontal waves from the $V_d = 600m/s$ simulation. To avoid the potentially deceptive appearance of this data, we show it in both a log-log and a log-linear plot. Beyond $k_x \approx 15$, numerical damping dominates. We see that the initial decrease in magnitude with k_x goes just slightly faster than k_x^{-4}. Because of the poor quality of both the rocket and simulation data, a more useful comparison of experiment to simulation awaits both larger scale simulations and more refined *in-situ* measurements.

4.4.2 Cross-sections of Spectra

Another method of examining the three dimensional spectra with our two dimensional visualization tools is to inspect individual planar cross-sections of the spectra. Using sets of these we can obtain a relatively detailed picture of the full spectra. Also, any one cross-section of $n(k_x, k_y, \omega)$ looks a lot like a dispersion relationship. These spectra tell us precisely where in (ω, k) space the dominant modes lie, their

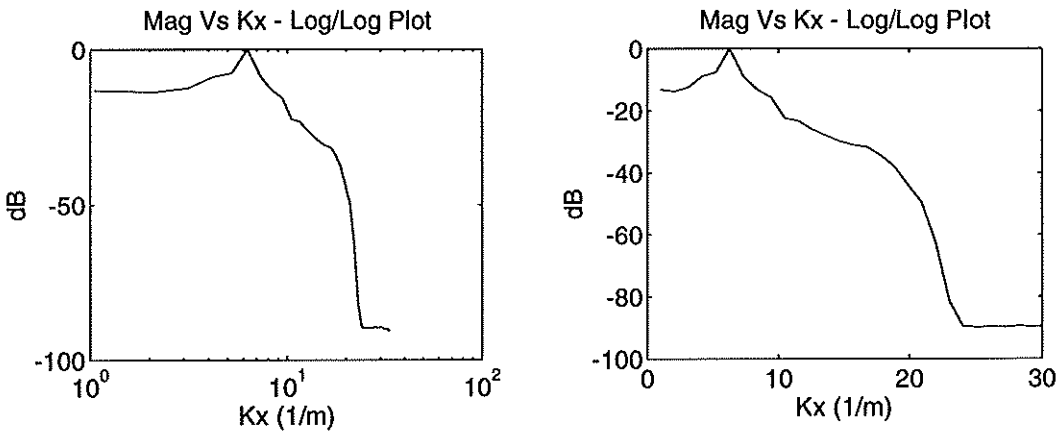


Figure 4.16: Relative Magnitude of the density perturbation in dB versus k_x for $k_y = 0$ from the $V_d = 600m/s$ simulation. Both a log/log and a log/linear plot of the same information is shown.

spectral widths and the phase velocity of each mode. This subsection shows and discusses a number of these spectral cross-sections. The information derived from these spectra is partially redundant with the averaged spectra. We will concentrate on those features not apparent in the averaged spectra.

Figures 4.17 and 4.18 show spectral cross-sections for the baseline case and the high resolution ($V_d = 600m/s$ case). The negative frequencies in these figures can be confusing, so we'll discuss their implications. To generate these figures, we Fourier transform real data. The resulting spectra contain the symmetry, $f(k_x, k_y, \omega) = f^*(-k_x, -k_y, -\omega)$ from which we plot cross-sections of the magnitude. If we were to plot the cross-sections of $\pm k_x$ versus $\pm\omega$ for all $\pm k_y$ then half our images would be redundant. We need to discard the redundant images. Perhaps the most intuitive way to do this would be to plot only $\pm k_x$ versus $+\omega$ for all $\pm k_y$. This eliminates the counter-intuitive negative frequencies. However, to obtain the same information as shown in figure 4.17, we would have to generate seven images. We chose instead to plot $\pm k_x$ versus $\pm\omega$ for only $+k_y$ because we obtain double the information on each figure with only a small amount of increased confusion. One result of this choice is that the right hand side of the $k_y = 0$ image exactly mirrors the left hand side. Also, note that $(k_x, k_y) = (0, 0)$ on this image because this reflects the D.C. component of the spectrum.

In both these figures, three features stand out. First, the definition of a multi-dimensional phase velocity is $\mathbf{V}_{ph} \equiv \omega \mathrm{k}/k^2$. If $\mathbf{V}_{ph} > 0$, then all waves propagate

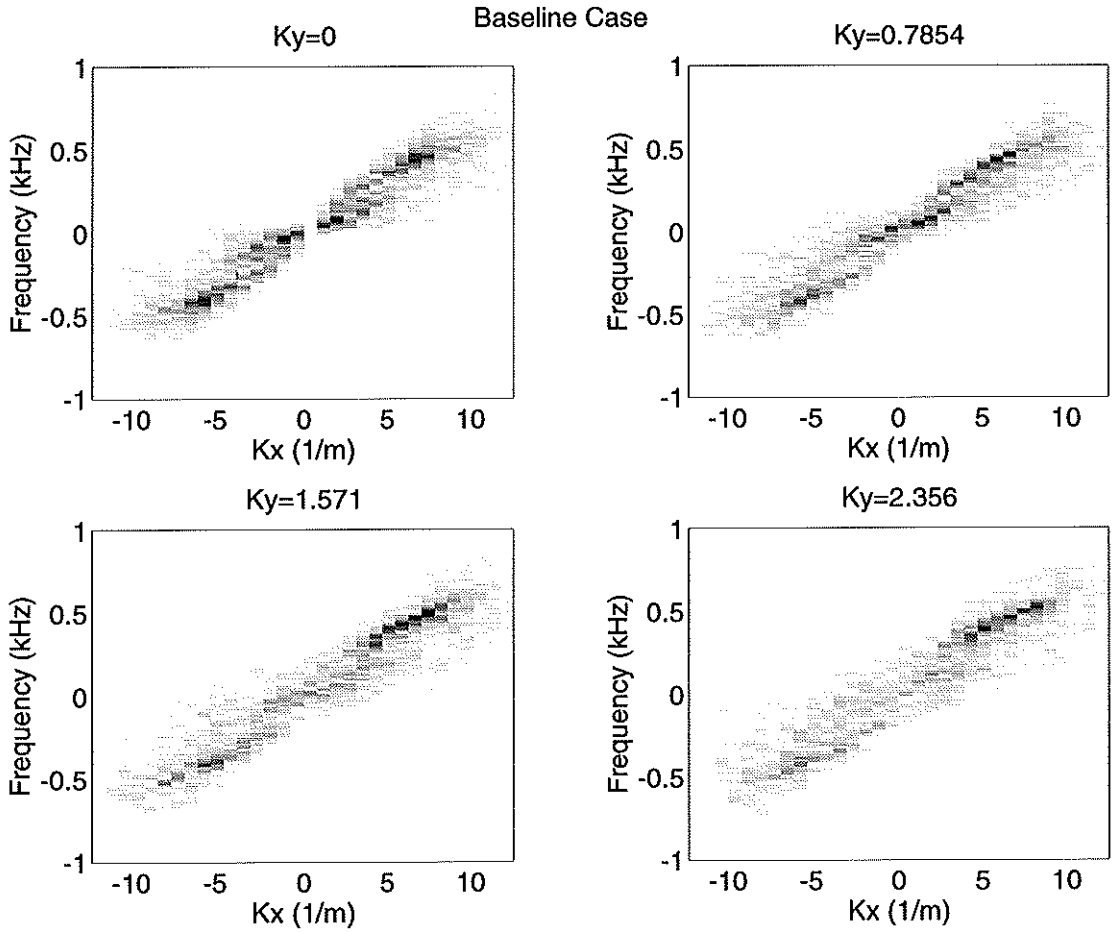


Figure 4.17: Dispersion relationship for four k_y in the baseline case. The lower half of the $k_y \neq 0$ plots represents downward propagation. See text for a more detailed explanation.

exclusively to the right. Second, the $\omega > 0$, $k_x > 0$ peaks have magnitudes which exceed the $\omega < 0$, $k_x < 0$ peaks. Hence waves propagating upward and to the right have larger amplitudes than waves propagating downward and to the right. Third, the curve which follows the largest amplitude of the density variations through (ω, k_x) space is not a straight line. This means that the phase velocity is wavelength dependent.

The hybrid simulation run most recently has provided us with some conflicting information. We modified the baseline case so that the x-axis spans 4 instead of 8 meters. Since this simulation still uses 64 nodes in the x-direction, it now has twice the horizontal resolution at the expense of simulating only half the area.

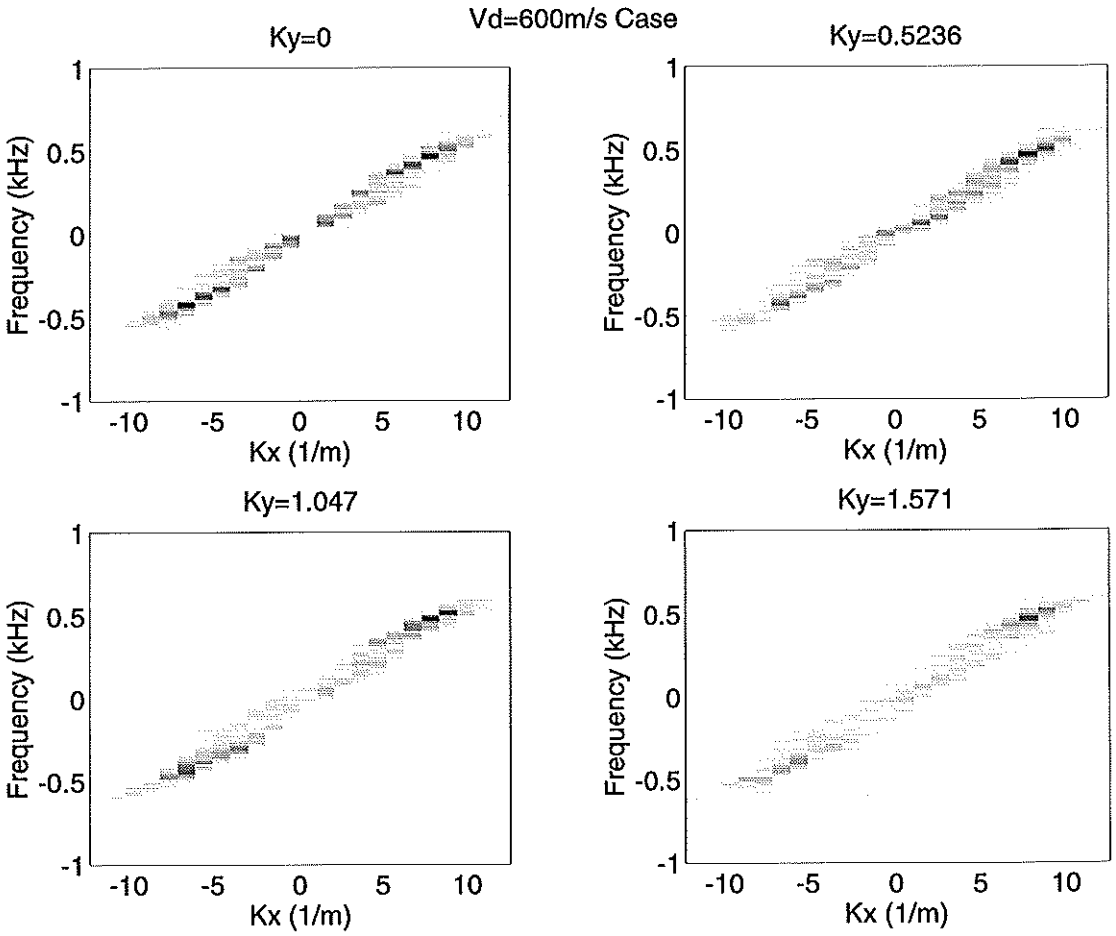


Figure 4.18: Dispersion relationship for four k_y in the $V_d = 600$ case. All the same features appear as in the baseline case, but with a simpler spectrum.

More importantly, it moves the wavelength of the fastest growing mode away from the upper half of the simulation spectrum where numerical damping is significant.

Figure 4.19 shows four cross-sections of the spectrum generated by this $x = 4m$ case. They differ from the spectral cross sections of the baseline case (Figure 4.17) in several ways. First, the dominant modes of the $x = 4m$ simulation lie at slightly higher values of k_x as discussed in the preceeding subsection. Second, proportionally less energy appears in the $k_y \neq 0$ modes of the $x = 4m$ case, though this is not a large discrepancy. Third, the long wavelength modes ($\lambda \geq 2m$) appearing in the $x = 4m$ case are less distinct, though still visible in the $k_y = 0.7854$ cross-section. Fourth, the $k_y = 0.7854$ spectra shows more energy in the downward propagating modes, though the $k_y = 1.571$ spectrum shows still more energy in the upward propagating modes. The motion pictures of this simulation show upward tilting of the dominant modes similar to those seen in the baseline case.

During the growth stage of a simulation, the spectra reveal different behavior than from the waves in a saturated state. During the early part of the high resolution, $V_d = 600m/s$, simulation, between $20ms$ and $60ms$, the waves grow roughly exponentially, as seen in figure 4.4. This indicates a linearly growing wave. Figure 4.20 shows four cross sections of the 3D spectrum from the growing stage of the high resolution simulation. In this case, the dominant mode lies on the $k_y = 0$ axis, essentially at the point predicted by linear kinetic theory. In this growing stage, linear kinetic theory also predicts the phase velocity of the wave quite well, as discussed in the next section.

The spectrum of a simulation where the electron behavior has been linearized, described in section 4.2, shows that the dominant mode may not lie exactly on the $k_y = 0$ line. However, as seen in Figure 4.21, the spectrum is far less complex than in the case with nonlinear electron behavior, with sharper peaks and far less scattering into other modes. Another important difference between the spectra of the linearized cases and the fully nonlinear ones is that the magnitude of the density perturbations in the linearized case exceeds that of the fully nonlinear case a factor of by more than eight. Further, the linearized electron cases do not show the perpendicularly traveling secondary waves, eddies or coupling to long wavelength modes.

One possible explanation for why the dominant mode of this linearized simulation does not travel exactly horizontally is numerical. Periodic boundary conditions

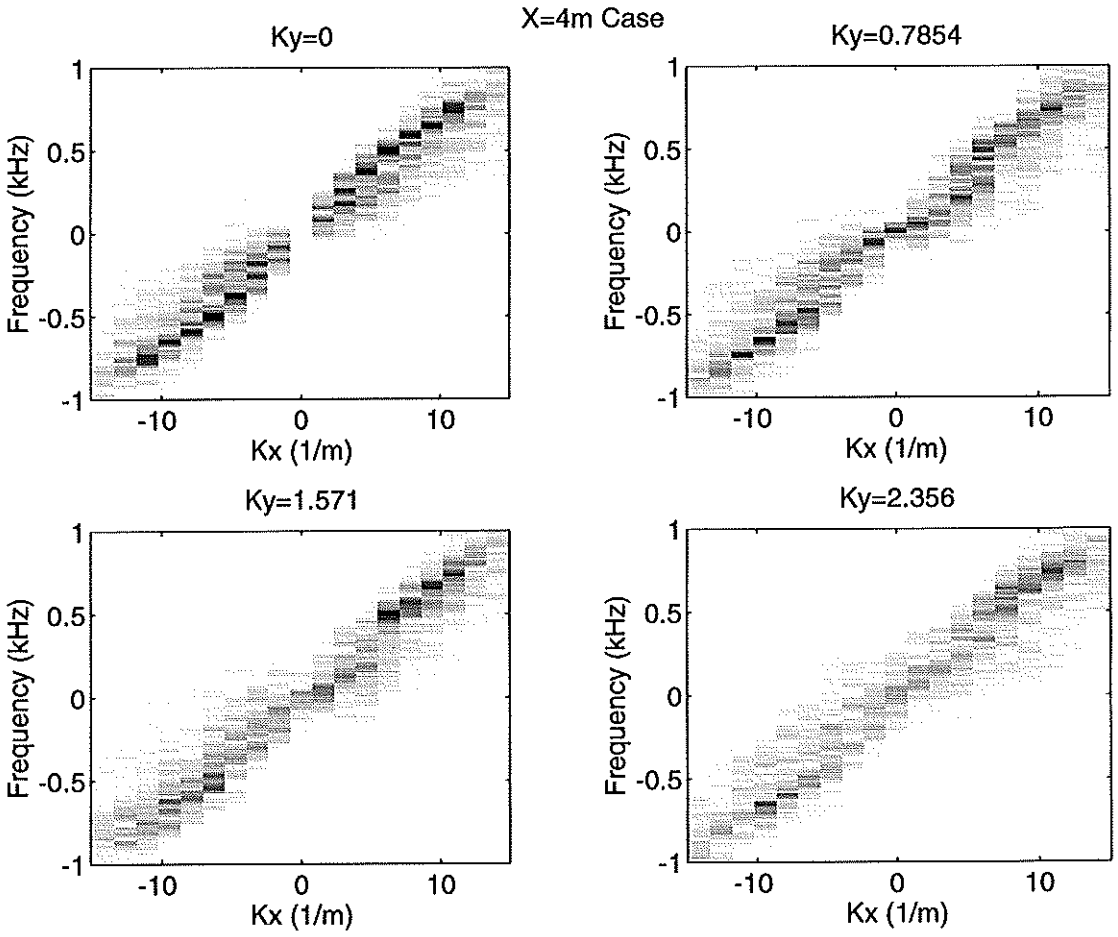


Figure 4.19: Dispersion relationship for four k_y in the $x = 4m$ case.

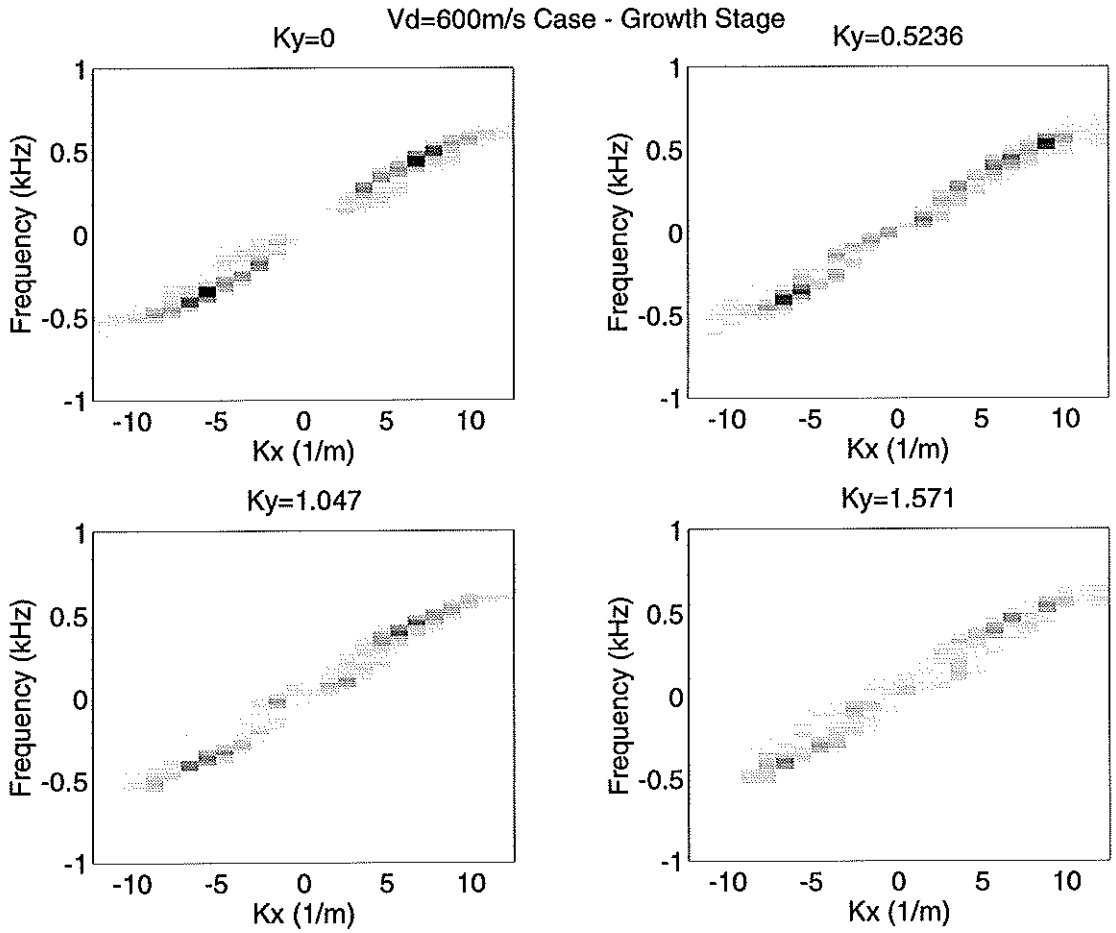


Figure 4.20: Dispersion relationship for four k_y during the linear growth phase of the $V_d = 600$ case. All the same features appear as in the baseline case, but with a simpler spectrum.

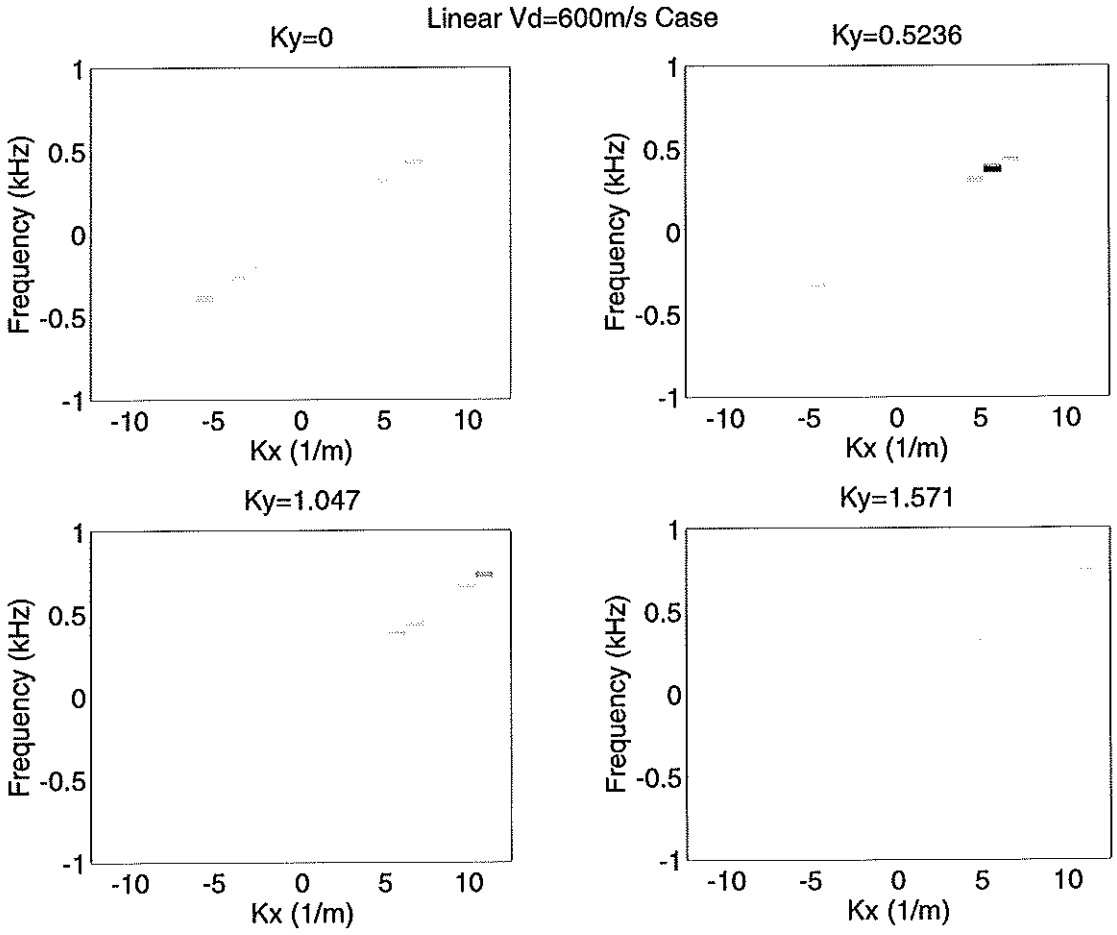


Figure 4.21: Dispersion relationship for four k_y in the linearized electron, $V_d = 600$ case.

quantize the angles allowed for a monochromatic wave. Since the top of one wave has to connect with the bottom of the next, monochromatic waves with wavelength of 1m in a 6m by 12m box can only have angles of [0, $\sin^{-1}(1/12)$, $\sin^{-1}(2/12)$, ...]. Further, the simulation cannot adiabatically switch between angles. If the linear growth rate between a mode with a 0° angle differs only slightly from one with a small angle, then random variations can cause the small angle mode to dominate early on. Because no mechanism allows the simulation to switch to the zero angle mode, this mode then dominates.

4.4.3 Individual Spectra

In spite of the differences between what a radar measures and our simulation records, spectra generated by these differing methods appear quite similar. Radars probe the E region ionosphere one wavelength at a time, measuring the temporal spectrum for a single value of k and, usually, showing normalized power versus Doppler shift [*Kudeki et al.*, 1987; *Cohen and Bowles*, 1967]. Radars also average together large volumes of space. The huge Jicamarca radar has a full-width at half-max beam width of one half a degree while the much smaller Cornell University Portable Radar (CUPRI) spans five degrees. This means that at 100km Jicamarca averages the waves contained in a box roughly ten kilometer on a side while CUPRI averages ten times that. Additionally, these radars take measurements over long periods of time and, to reduce noise, average together many spectra. Simulations cannot model the Farley-Buneman instability over anything approaching these vast spans of space and time. Nonetheless, the spectra generated by the simulation look quite similar to those of the radar observations.

Figures 4.22, 4.23, 4.24 and 4.25 show the spectra of the lowest (k_x, k_y) values generated by the high resolution simulation. To make these spectra similar to the radar spectra, we have normalized the power displayed in each plot and use as a horizontal axis the phase velocity, $\omega/|k|$. This corresponds to the Doppler shift displayed in *Kudeki et al.* [1987] and *Cohen and Bowles* [1967]. Compared with the radar data, the simulation spectra contain substantial amounts of noise. Due to the expense of running this high resolution simulation for long times, we only obtained 256 snapshots of saturated wave behavior. Since the snapshots occurred at intervals of $0.25ms$, the total interval lasted 64ms. A typical radar takes 128 samples with an inter-pulse period of 2ms and then takes ten of these spectra and

averages them together to make a single spectrum. This spectrum contains over a second worth of data. Radars also average echoes from a huge volume of space making their spectra less noisy than those from the simulations.

From these spectra, we can determine the magnitude of the amplitude change between upward and downward tilted modes. For example, the mode with the largest amplitude lies at $k_x = 7.33$, $k_y = 0.5236$. This mode travels up and to the right. The comparable mode traveling downward and to the right at $k_x = -7.33$, $k_y = 0.5236$ has a 7dB smaller amplitude. A quick inspection of neighboring spectra shows that that, typically, an upward tilted mode has a 3dB to 10dB amplitude increase over comparable downward tilted modes.

Individual spectra enable us compare spectral widths with Doppler shifts. One sees that for the dominant modes, around $k_x = 7.33$, $k_y = 0.5236$, these spectra are quite narrow and well defined, far narrower than spectra seen by radar. This may result from the fact that radar generally cannot inspect the dominant modes. Further, because of the averaging done by radars, we expect radars to have artificially broad signals.

Probably, the best comparison between radar and simulation spectra can be made for vertical modes. For these modes, the large array at Jicamarca has made numerous measurements of 3m irregularities. Neither the Jicamarca measurements nor the simulations show net Doppler shifts for these modes. Also, the spectral widths of these vertically oriented modes appear similar, though a more detailed comparison needs to be made.

Noise in the simulation spectra tends toward a zero Doppler shift. When we calculate the first moment of a noisy spectrum in order to obtain the phase velocity, this noise generally reduces the calculated phase velocity to a number below what we would expect based on a visual inspection of the spectrum. This noise may have a physical origin, but probably results from particle noise. This problem is discussed further in the next section.

4.5 Phase Velocities

When radars observe type I echoes, they measure both reflected power and Doppler shift. Numerous papers have reported that two-stream waves travel at close to the acoustic speed and not the velocity predicted by linear theory (see section 1.6.1).

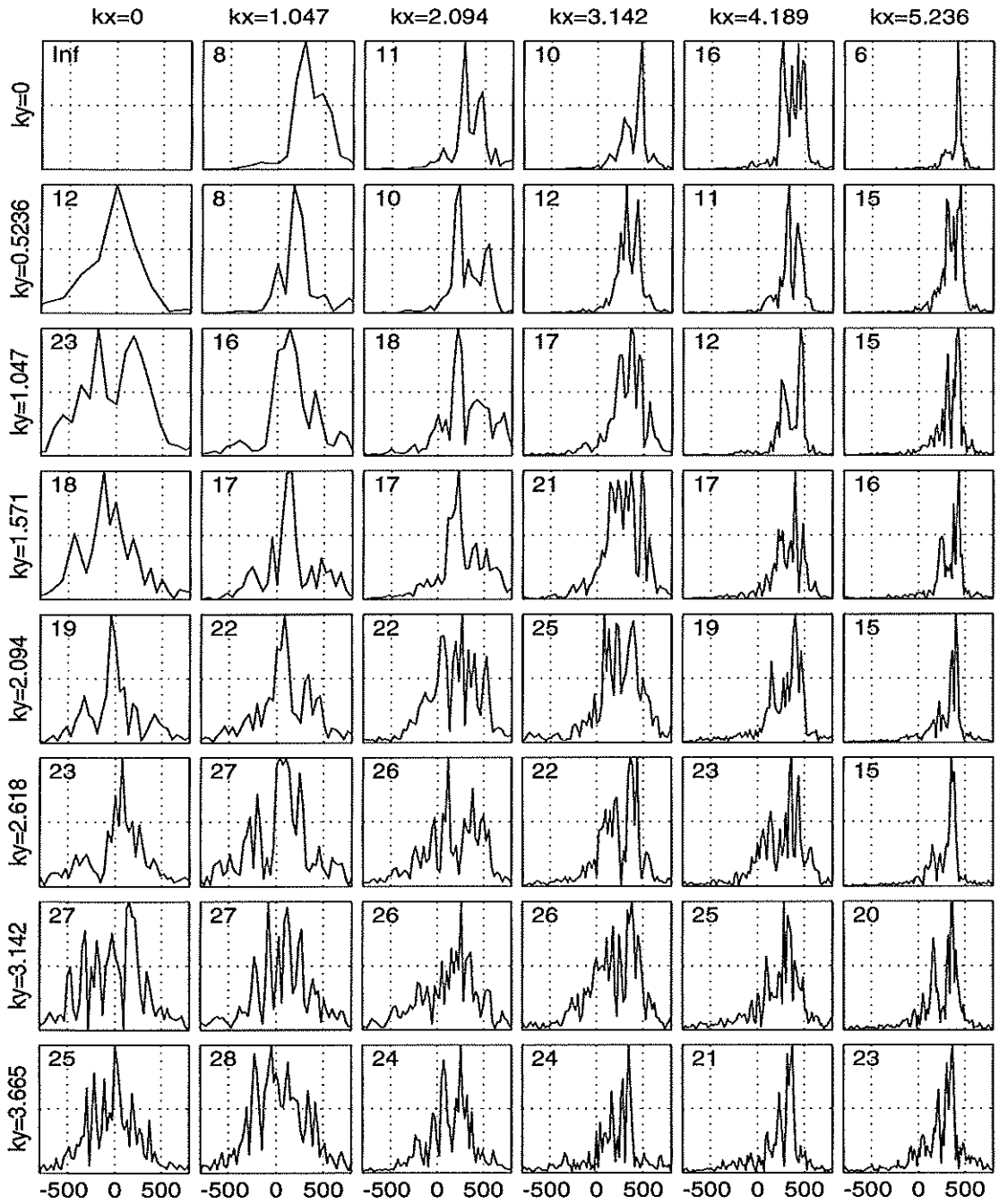


Figure 4.22: Self-normalized spectra of the $V_d = 600m/s$ simulation for $k_x = 0m^{-1} \Rightarrow 5.236m^{-1}$. The x-axis shows the phase velocity, $\mathbf{V}_{ph} = \omega\mathbf{k}/|k|^2$, in units of m/s, while the y-axis shows the power in arbitrary units. The number in the upper left corner of each plot shows the relative drop in the density perturbations in dB ($dB = -20\log(\delta n/\delta n_{max})$). Larger numbers indicate smaller density irregularities.

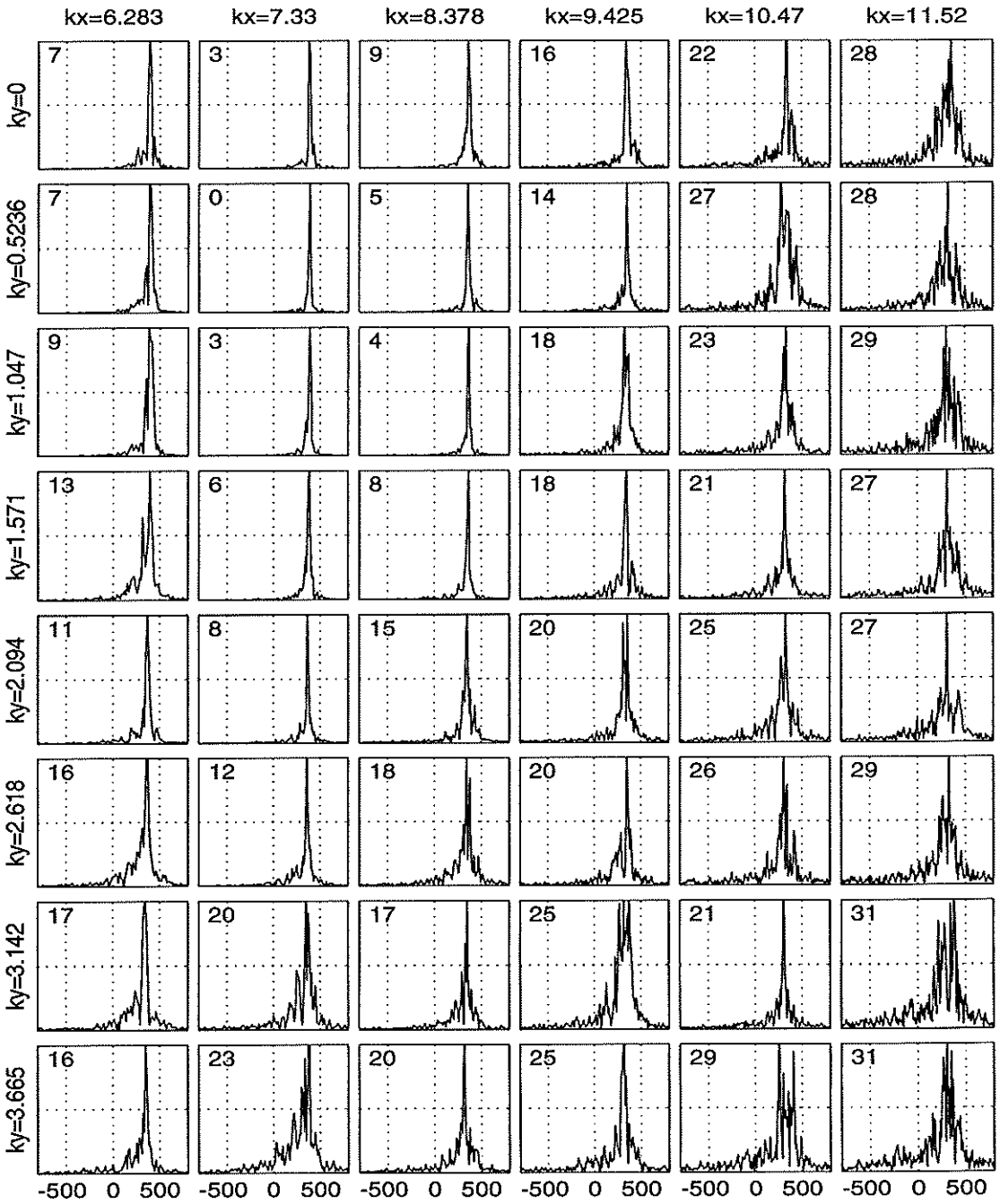


Figure 4.23: Self-normalized spectra of the $V_d = 600m/s$ simulation for $k_x = 6.283m^{-1} \Rightarrow 11.52m^{-1}$.

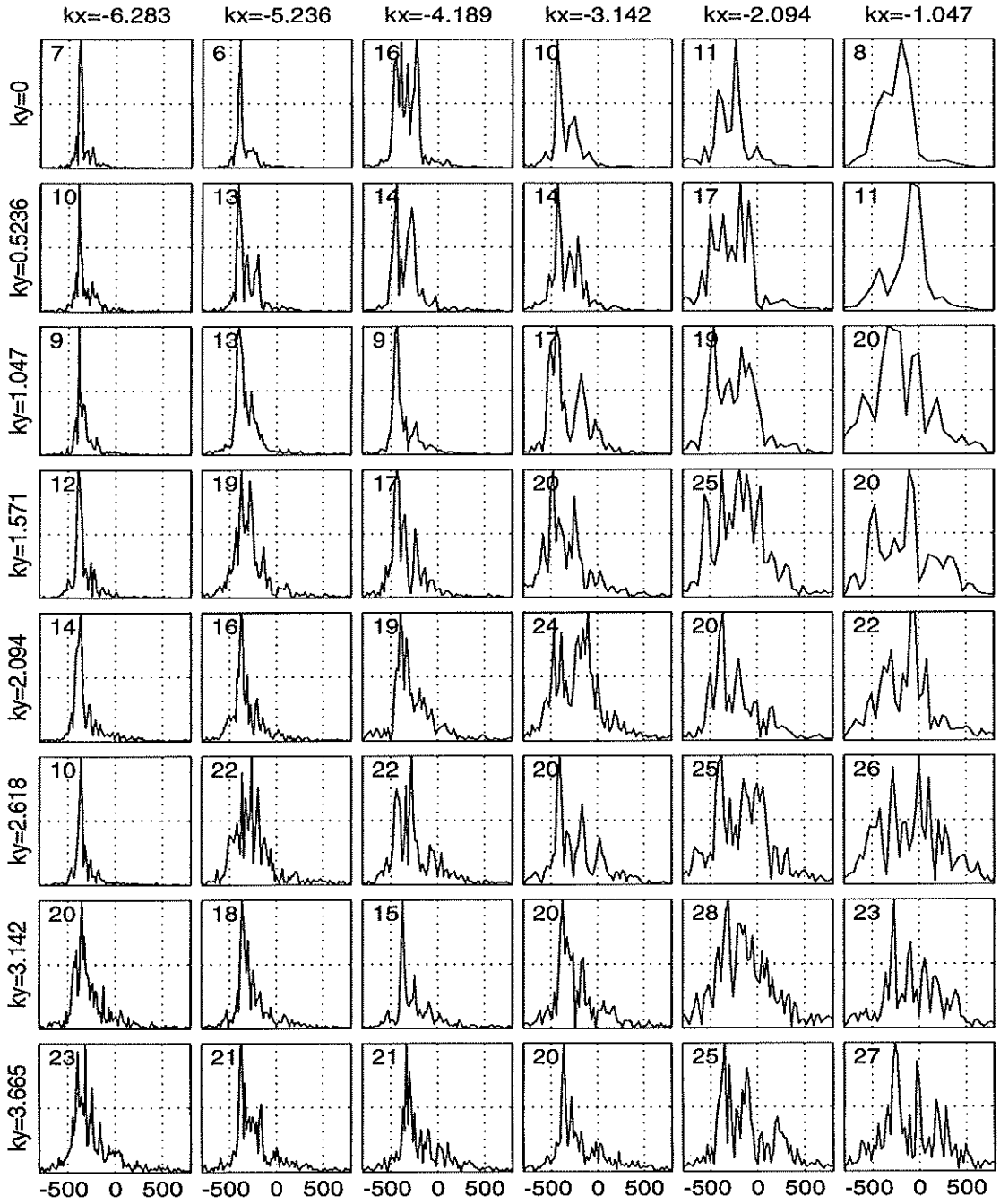


Figure 4.24: Self-normalized spectra of the $V_d = 600m/s$ simulation for $k_x = -6.283m^{-1} \Rightarrow -1.047m^{-1}$

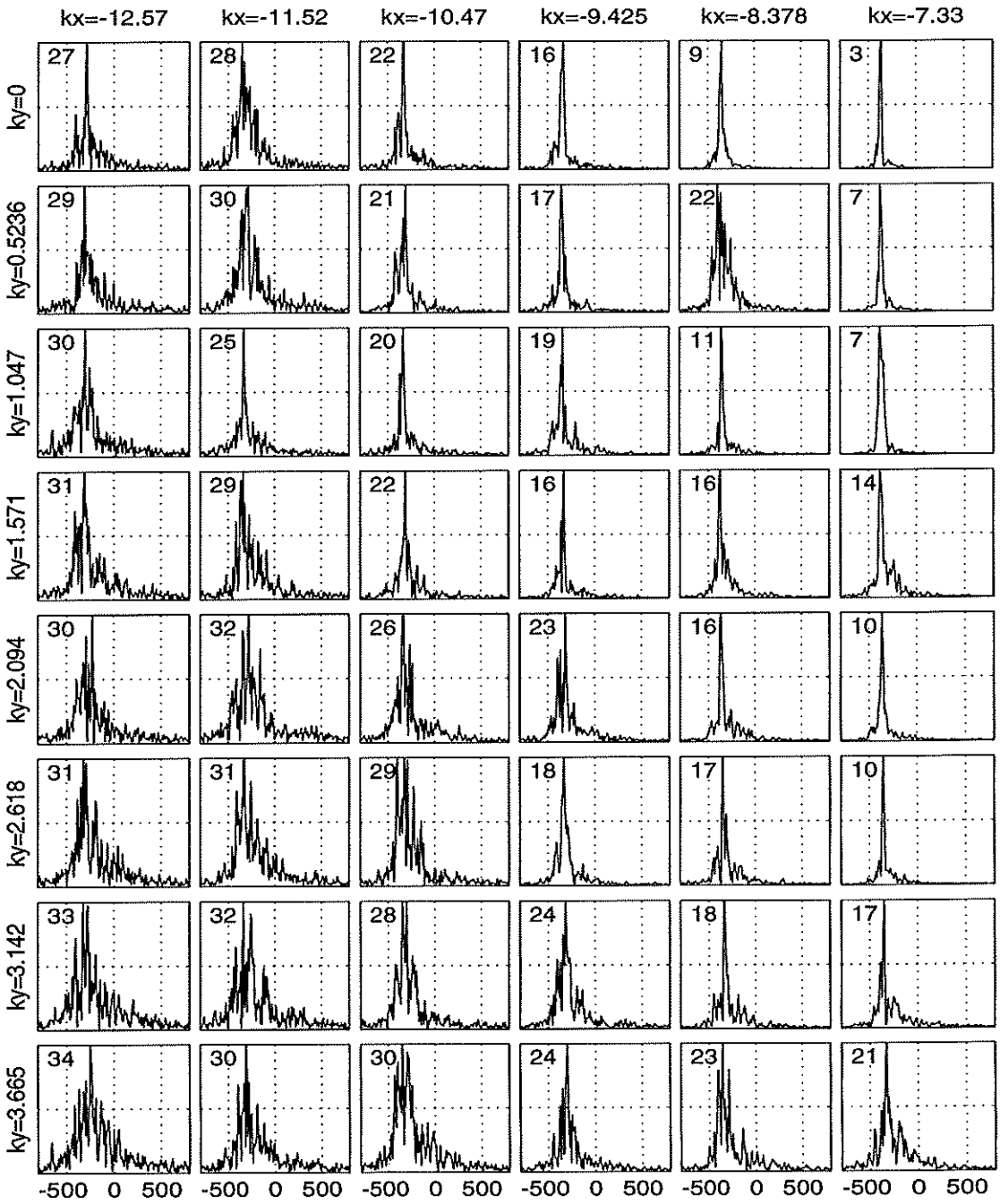


Figure 4.25: Self-normalized spectra of the $V_d = 600m/s$ simulation for $k_x = -12.57m^{-1} \Rightarrow -7.33m^{-1}$

On the other hand, the rocket measurements do not corroborate this. This section discusses the phase velocities generated by our simulations in detail.

In general, the waves seen in the hybrid simulations travel at phase velocities between the velocity predicted by linear theory and the acoustic velocity. Initially, while the waves grow from noise, but before reaching saturation (i.e., during the mostly linear stage), the dominant modes travel at a speed predicted by linear kinetic theory. This speed differs substantially from that of linear fluid theory, as described in section 2.3. After saturation, we observe slower velocities. The dominant mode now travels slower than the velocity predicted by linear theory but above the acoustic velocity. Also, we observe some long wavelength modes traveling slower than the acoustic velocity.

The spectral cross-sections, Figures 4.17 and 4.18, show frequency versus wavenumber. Since $\mathbf{V}_{ph}$ is defined as $\omega \mathrm{k}/k^2$, the slope of the distribution in these figure is the phase velocity. Figure 4.26 compares two cross-sections of the full three dimensional spectrum with the acoustic speed, the linear growth rate, the linear fluid phase velocity, and the linear kinetic phase velocity. These figures show that the velocity of these waves lies between the linear kinetic velocity and the acoustic speed, but nowhere near the linear fluid velocity.

When the system is driven harder than in the $V_d = 600m/s$ case, then the waves still travel at a speed between the acoustic speed and the linear kinetic phase velocity. Figure 4.27 shows that, in the baseline case, the phase velocity remains closer to the acoustic speed than in the $V_d = 600m/s$ case. Additionally, the long wavelength mode seen at $k_x \approx \pm 2$ travels slower than the sound speed. Since 50MHz radars measure relatively long wavelength modes, they may pick up these slower waves. The simulation with the lower driving force, $E_{0y} = 15mV/m$, does not develop such clear long wavelength modes.

Finally, when the drift velocity is increased dramatically, the phase velocity of the waves seen in the simulations remain roughly at the same speed while linear theory predicts they should increase in proportion to the electron drift speed. In fact, the highest phase velocity we have seen in the hybrid simulations is $C_s(1+\Psi_0)$. We do not have an explanation for this behavior.

During the early growth stage of the instability, linear kinetic theory predicts the velocity quite well, as seen in figure 4.28. As the wave saturates, displaying nonlinear behavior not obvious on spectral plots, the average phase velocities of

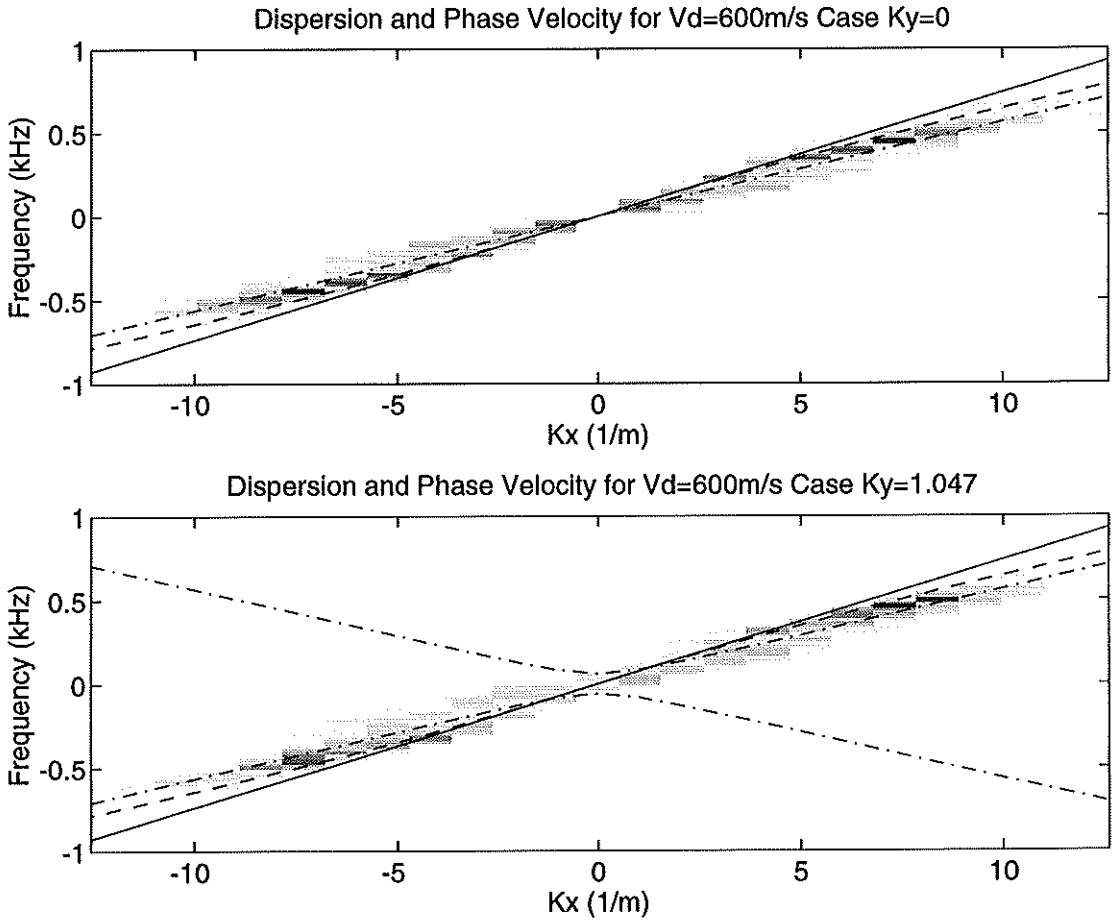


Figure 4.26: Dispersion relationship for $V_d = 600$ case where $k_y = 0$ (top) and $k_y = 1.571$ (bottom). The solid lines show where the spectrum would lie if the waves traveled at the velocity predicted by linear fluid theory. The dashed lines track waves traveling at the linear kinetic velocity while the dash-dot curves track waves traveling at the acoustic speed. In the lower plot there are two acoustic speed curves which derive from the relation $\omega = \pm|k|C_s$ while in the top plot this relation yields two straight lines, of which we show only the important one.

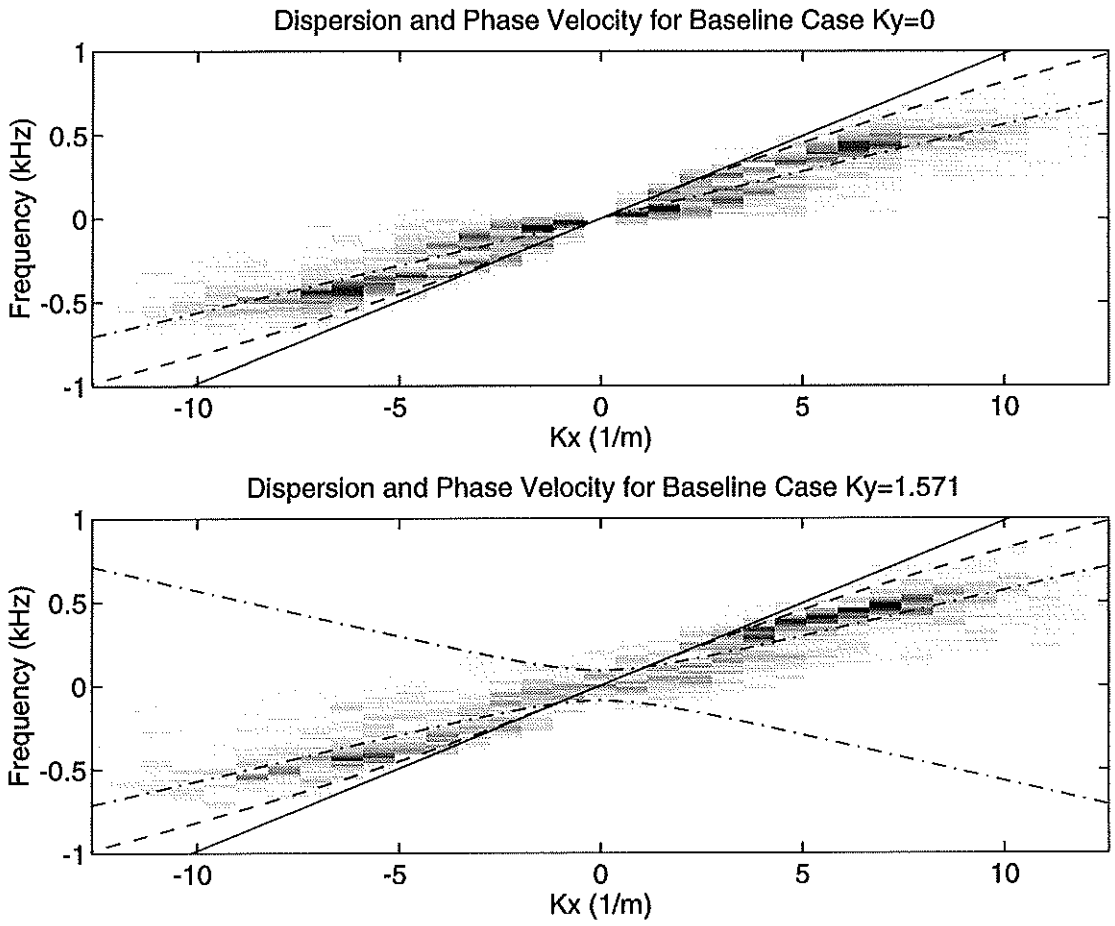


Figure 4.27: Dispersion relations for the saturated baseline case where $k_y = 0$ (top) and $k_y = 1.571$ (bottom). The solid lines show where the spectrum would lie if the waves traveled at the phase velocity predicted by linear fluid theory. The dashed lines tracks waves traveling at the linear kinetic velocity while the dash-dot curves track waves traveling at the acoustic speed. In the lower plot there are two acoustic speed curves which derive from the relation $\omega = \pm|k|C_s$ while in the top plot this relation yields two straight lines, of which we show only the important one.

the dominant modes slows. Even in this saturated state, some waves travel at the velocity predicted by linear kinetic velocity. However, these modes no longer dominate the simulation spectra. For the case of linearized electron behavior, the linear kinetic theory does a better job predicting the phase velocities, as seen in figure 4.29. Comparing this to a simulation where the electron behavior has not been linearized shows that the reduced phase velocity is largely due to the nonlinear electron behavior.

Figure 4.30 shows phase velocities from the $x = 4m$ case. The dominant modes, indicated by the darkest patches, travel at just below the phase velocity predicted by linear kinetic theory, while the shorter wavelength modes travel at a reduced velocity closer to the sound speed. A close inspection of this figure reveals that the longest modes ($\lambda \geq 2m$) may also be traveling quite slowly.

The dominant modes travel more rapidly in this $x = 4m$ case than in any other case. The baseline case differs from the $x = 4m$ case only in the length of its horizontal axis (8 meters instead of 4). In the baseline case, however, dominant modes travel at velocities just above the acoustic velocity. A possible explanation for this discrepancy is that grid effects reduce the prominent short-wavelength phase velocities in the baseline case. This type of error is well understood in the case of a PIC code model of plasma oscillations. There the phase velocities decrease as the wave length approaches the grid size (see [*Birdsall and Langdon*, 1985]). However, there are important differences between plasma waves and two-stream waves. The plasma wave frequency depends on density while the two-stream wave frequency depends primarily on the constant drift velocity and, to a lesser degree, on the slightly density dependent constant Ψ_0.

A second possible explanation for the slowing of the dominant modes lies with the interaction between long and short wavelengths. The baseline case shows strong, slowly traveling 4 and 8 meter wavelength modes which appear coupled to the shorter wavelength modes. The $x = 4m$ case cannot develop 8m modes, and so the interaction between long and short wavelengths should diminish, possibly preventing the slowing effect. One method of distinguishing between this mechanism and the numerical effect described above is to run the baseline case with a 128 by 128 mesh. This simulation would require increased particle noise filtering, but would still allow the same spectral resolution as the $x = 4m$ case while reducing numerical errors associated with the grid size.

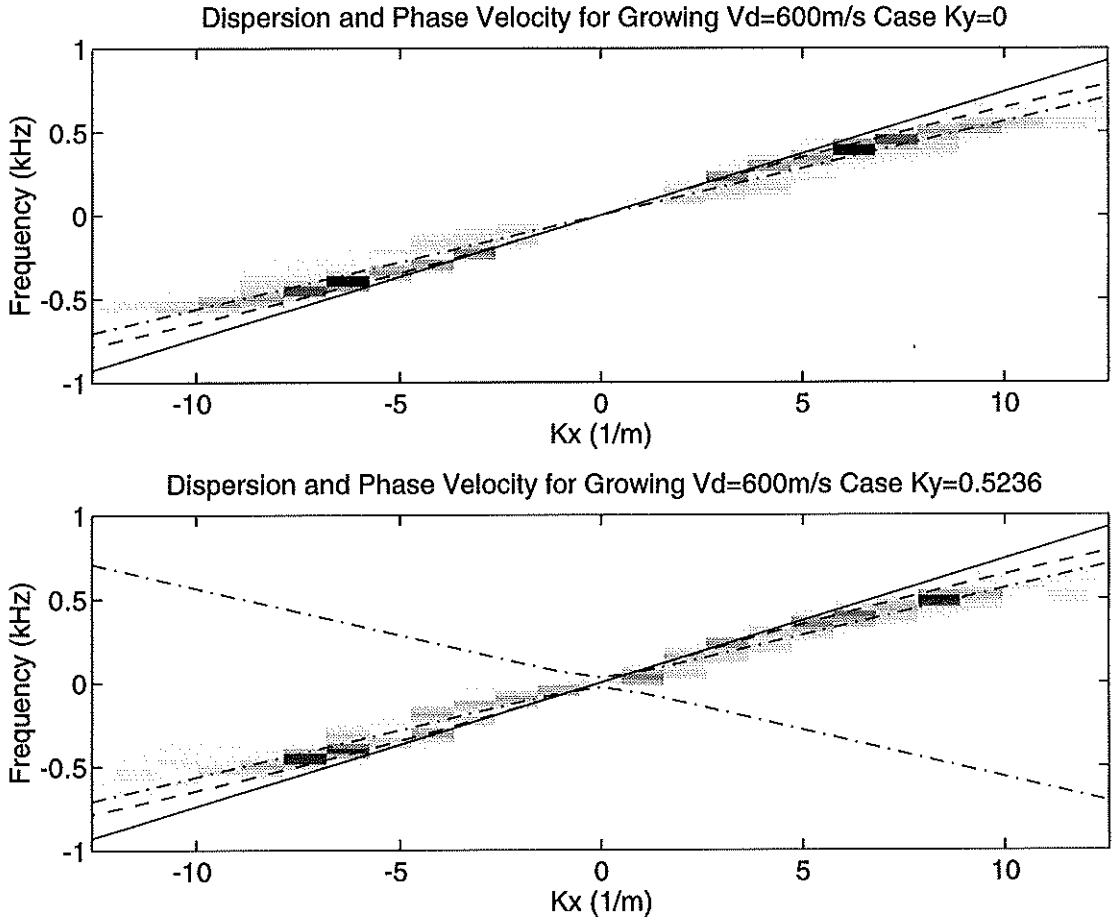


Figure 4.28: Dispersion relationship for the growth stage of the $V_d = 600$ case where $k_y = 0m^{-1}$ (top) and $k_y = 0.5236m^{-1}$ (bottom). The solid line shows where the spectrum would lie if waves traveled only at the velocity predicted by linear fluid theory. The dashed lines track waves traveling at the linear kinetic velocity while the dash-dot lines track waves traveling at the acoustic speed.

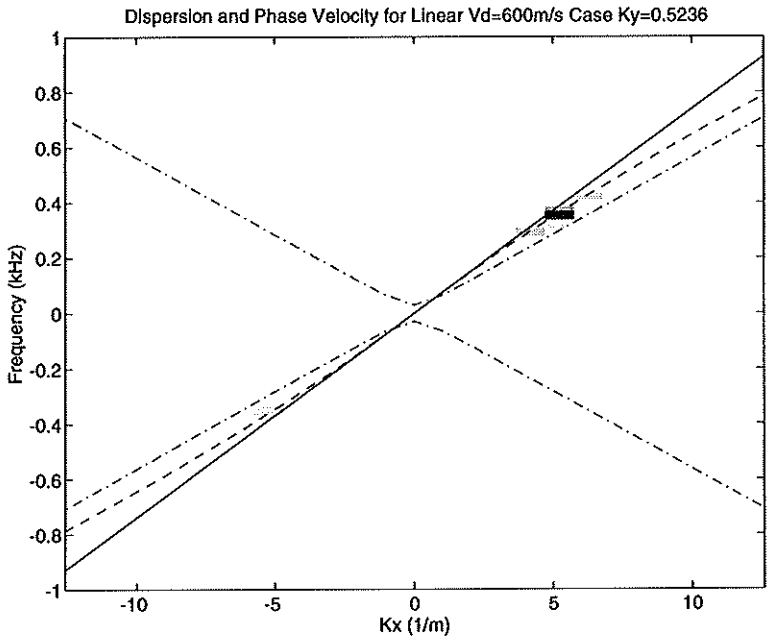


Figure 4.29: Dispersion relationship for the low driving force simulation with linear electron dynamics in a saturated state. The solid line shows where the spectrum would lie if waves traveled at the velocity predicted by linear fluid theory. The dashed lines track waves traveling at the linear kinetic velocity while the dash-dot lines track waves traveling at the acoustic speed.

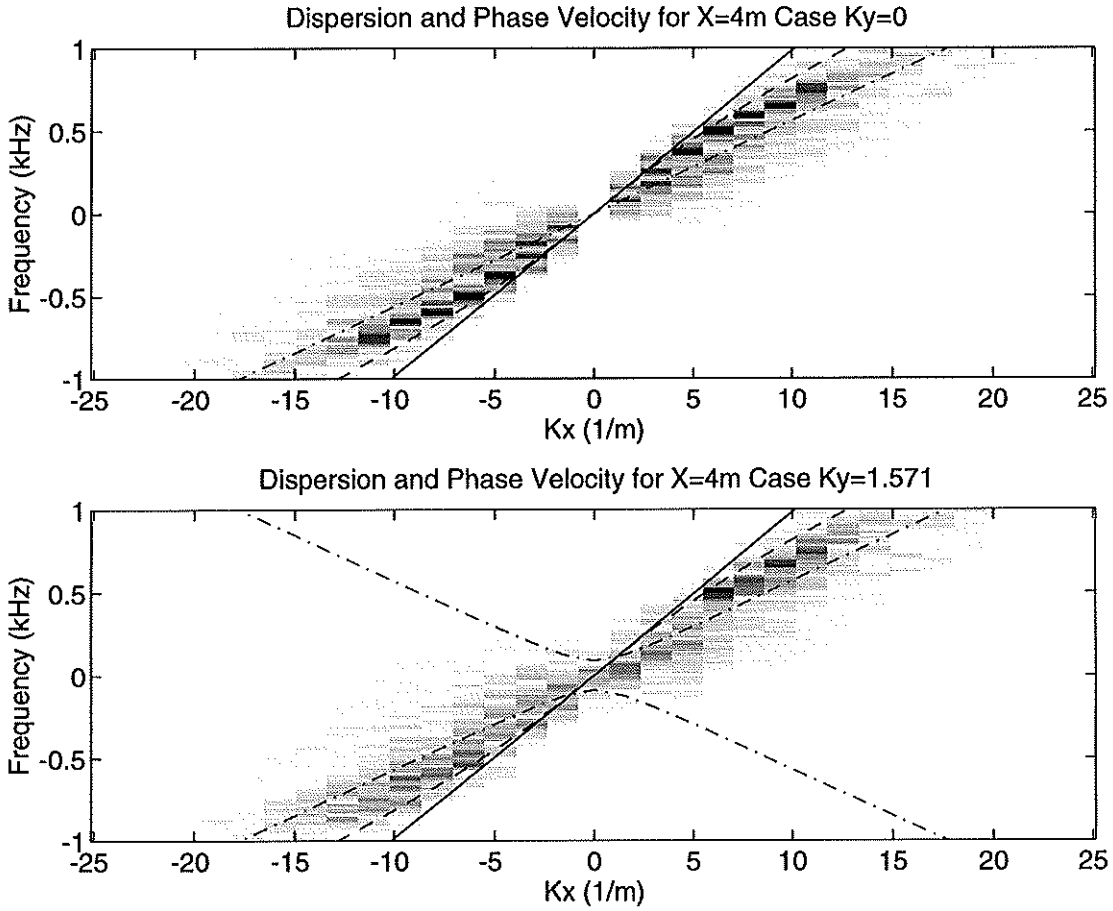


Figure 4.30: Dispersion relationship for the $4m$ by $8m$ baseline simulation in a saturated state for $k_y = 0$ (top) and $k_y = 0.7854$ (bottom). The solid line shows where the spectrum would lie if waves traveled at the phase velocity predicted by linear fluid theory. The dashed line denotes the linear kinetic phase velocity while the dash-dot line tracks waves traveling at the acoustic speed.

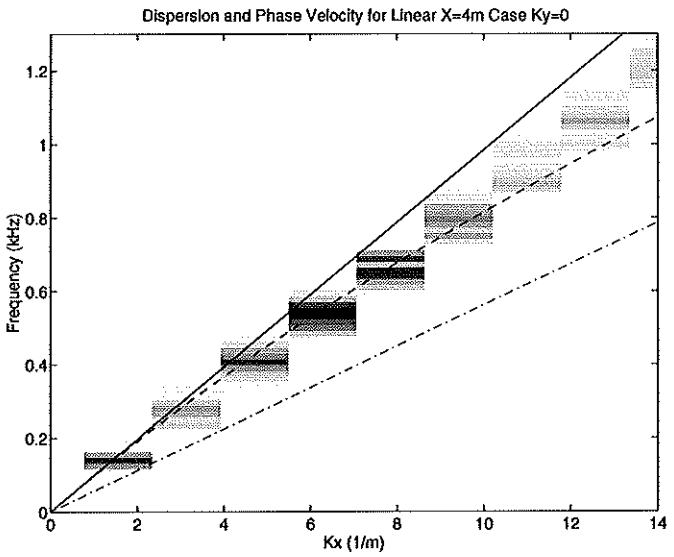


Figure 4.31: Dispersion relationship for the $4m$ by $8m$ baseline simulation with linear electron behavior in a saturated state. The solid line shows where the spectrum would lie if waves traveled only at the velocity predicted by linear fluid theory. The dashed line tracks waves traveling at the linear kinetic velocity while the dash-dot line tracks waves traveling at the acoustic speed.

Though the phase velocities of the dominant modes in the $x = 4m$ case exceed those seen in the baseline case, they still lie below the velocities predicted by linear kinetic theory. Figure 4.31 shows the phase velocity resulting from an identical simulation in which the electron dynamics have been linearized. Here, linear kinetic theory predicts the phase velocity quite well. This demonstrates that the slowing of the dominant modes seen in the baseline case cannot be attributed solely to numerical grid effects.

In order to compare our phase velocities to those obtained from radar echoes, we analyze our data with a method similar to that used on the radar data. Radar jocks [*Sahr*, 1990] obtain the Doppler shift from an echo or averaged set of echoes by calculating the first moment of the spectra,

$$V_{ph} \equiv \frac{\int_{-\infty}^{\infty} d\omega \; n^2(\omega)\omega/k}{\int_{-\infty}^{\infty} d\omega \; n^2(\omega)} \approx \frac{\sum_{\omega_{max}}^{\omega_{min}} n^2(\omega)\omega/k}{\sum_{\omega_{max}}^{\omega_{min}} n^2(\omega)} \tag{4.6}$$

We perform the same analysis on our spectra. Figure 4.32 shows the phase velocity of the lowest and most accurately simulated modes for the high resolution simulation. One caveat: the noise of the spectra tends to pull the moment toward zero

Doppler shift, as seen in the plots of the individual spectra, Figures 4.22 through 4.25. For example, the spectrum for $k_x = 5.236$, $k_y = 1.047$ shows a peak at approximately $440m/s$ but the moment of this spectrum is only $370m/s$.

4.5.1 Phase Velocity Vs. Elevation Angle

While individual radars can generally probe only a single wavelength, some of them can look at multiple elevation angles. A number of studies have looked at the change in phase velocity with elevation angle [*Cohen and Bowles*, 1967; *Farley*, 1985] and have concluded that the phase velocity of two-stream waves is largely independent of elevation angle. Some controversy exists about this conclusion. The radars capable of looking mostly horizontally are invariably quite small and therefore have large beam widths. When a radar with a 5° degree beam width probes the electrojet at an elevation angle of 70° degrees (20° degrees off vertical) , it looks at an area covering 25km by 25km. The gradient-free portion of the electrojet is usually smaller than this, so it may be that these radars simultaneously observe both vertically oriented two-stream waves driven by the perturbed electric field of gradient drift waves and primary (pure) two-stream waves.

Figure (4.32) plots the V_{ph} versus k, where we obtain V_{ph} by taking frequency moments of the spectra. The simulation noise reduces the estimated Doppler shift as described in the previous section, making this approach inaccurate. As an improved method of comparing phase velocity to elevation angle, we show the power spectra at each angle for a given |k|. In other words, we plot a cylindrical cross-section of the three dimensional spectrum for which |k| equals a constant. Figure 4.33 shows this information for the low driving force simulation and Figure 4.34 shows it for the baseline case. If one follows the phase velocity at each angle, indicated by the dashed line, then it appears that the phase velocity falls off with a mostly cosine like dependence. However, if one follows the peak of the spectra indicated by the darkest patches versus elevation angle, the phase velocity remains constant somewhat longer. At large elevation angles, the phase velocity is reduced, but the power is also reduced making the error in moment calculation large. When the system is driven harder, as in the baseline case, Figure 4.34 shows a further exaggeration of this flattening of the phase velocity with respect to the elevation angle.

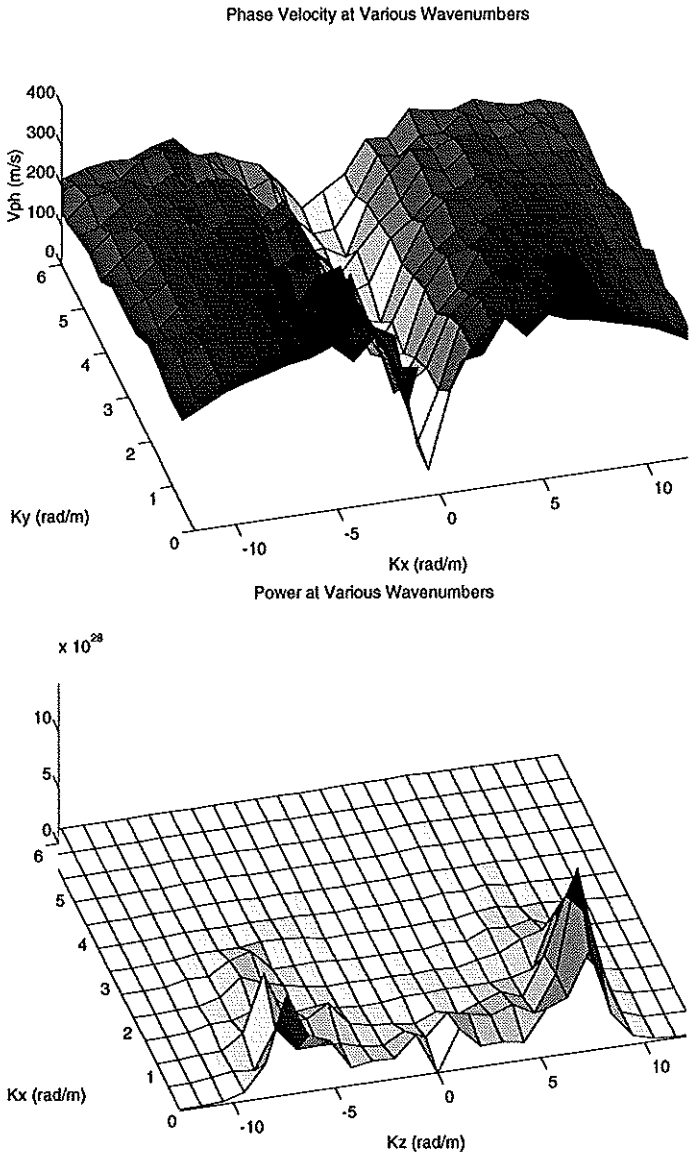


Figure 4.32: The upper plot shows the magnitude of the phase velocity at each wave number while the lower plot shows the relative power in that mode from the $V_d = 600m/s$ simulation.

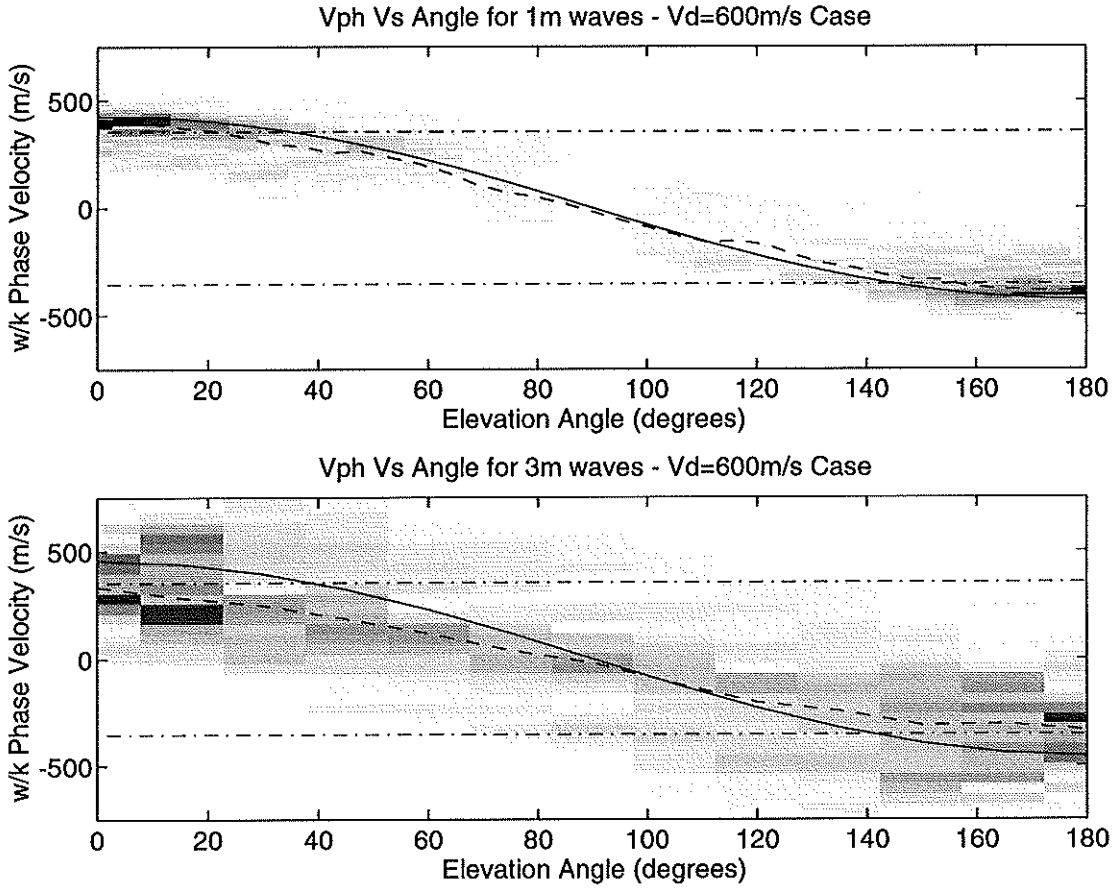


Figure 4.33: Spectra of $1m$ (top) and $3m$ (bottom) waves from the saturated $V_d = 600m/s$ case. These are interpolated from the full 3D spectra. The 3m wavelength plot interpolates from a very limited number of spectra and may be highly inaccurate. The dashed line shows the first moment of the power spectra (i.e., the estimated phase velocity). Notice the difference between the moment (dashed line) and the peak. The solid line shows the linear fluid theory phase velocity prediction while the dash-dotted line show the acoustic speed.

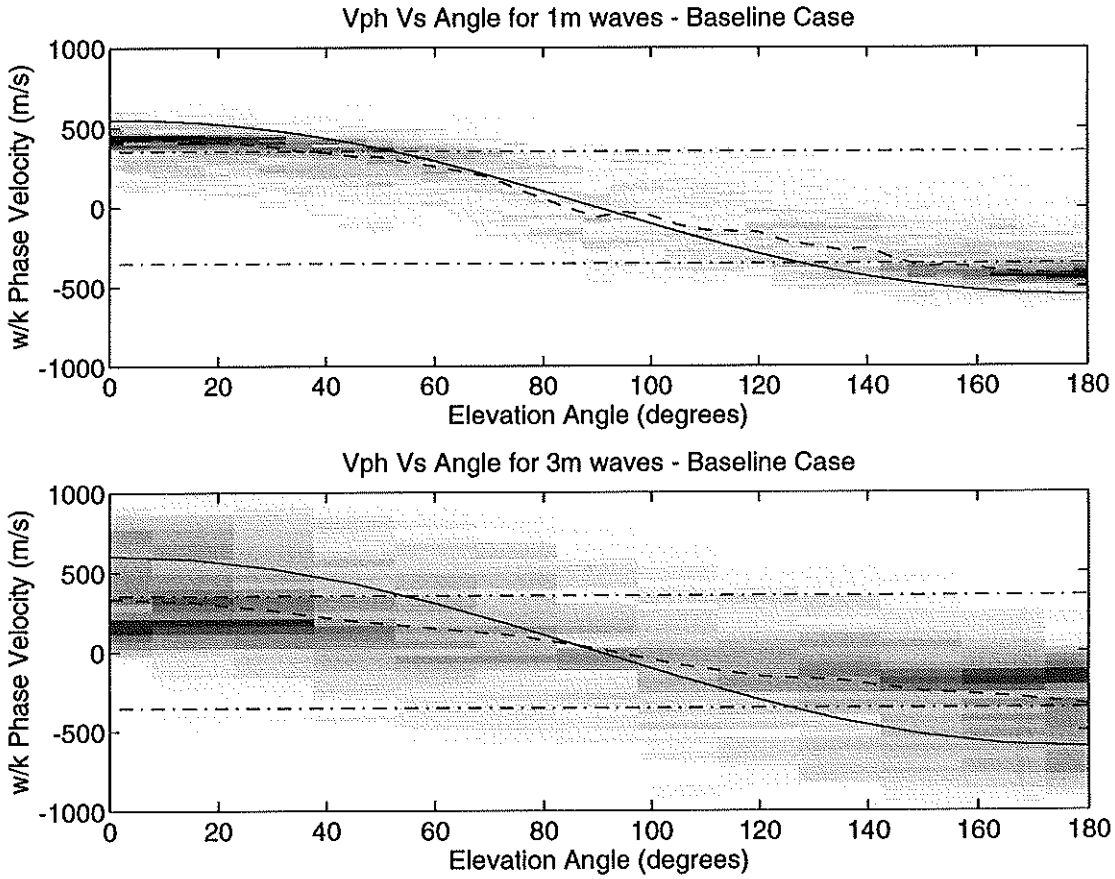


Figure 4.34: Spectra of $1m$ (top) and $3m$ (bottom) waves from the saturated baseline case. These are interpolated from the full 3D spectra. The $3m$ wavelength plot interpolates from a very limited number of spectra and may be highly inaccurate. The dashed line shows the first moment of the power spectra (i.e., the estimated phase velocity). Notice the difference between the moment (dashed line) and the peak. The solid line shows the linear fluid theory phase velocity prediction while the dash-dotted line show the acoustic speed.

4.5.2 Power Vs. Elevation Angle

Radars can measure echo strength as a function of elevation angle. We can derive similar quantities from the simulation results. Figure 4.35 shows the relative density perturbation in dB versus elevation angle. For one meter waves we see a decrease of roughly 0.35 dB/degree. For three meter waves, this information becomes less clear because we approach the scale-size of the simulation. In spite of the poor quality, we estimate that the magnitude diminishes at, very roughly, 0.25 dB/degree.

Ierkic et al. [1980] observed a 0.3 dB/degree rate of change in magnitude with elevation angle for 3m waves. This matches the simulation's result of 0.25 dB/degree to well within the uncertainty of the measurement. This agreement may be less meaningful than it appears. First, the simulation's resolution for 3m waves is very rough. Second, this number probably depends on the difference between C_s and V_d. The larger the difference, the broader the modes and, therefore, the slower the fall-off rate. Third, radars capable of comparing elevation angle to magnitude generally have such large beam widths that they may mix measurements from pure two-stream waves from the upper electrojet with coupled gradient drift/two-stream simulation generates a number roughly comparable to the observed rate.

4.6 Potential and Current Perturbations

Until now, we have exclusively evaluated perturbations in the densities. The set of variables describing the state of the simulation also includes the potential and the ion currents. Figures 4.36 and 4.37 show a snapshot of these quantities for the high resolution simulation at $t = 75ms$. The density, potential and horizontal ion current density all show similar spatial structure. The vertical ion current density shows only a hint of this structure overlayed on a noisy background. These features are discussed further in section 5.1.1.

The secondary, nonlinear $\mathbf{E} \times \mathbf{B}$ drift waves discussed in section 4.1.1 require an appreciable perturbed electric field. Figure 4.38 shows the perturbed electric field derived from the potential of Figure 4.36. From the horizontal electric field, E_x, we see that the perturbed electric field is roughly double the ambient field of $15mV/m$ with occasional peaks of three times this value. The vertical electric field, E_y, shows mostly noise with some structure matching that of the density

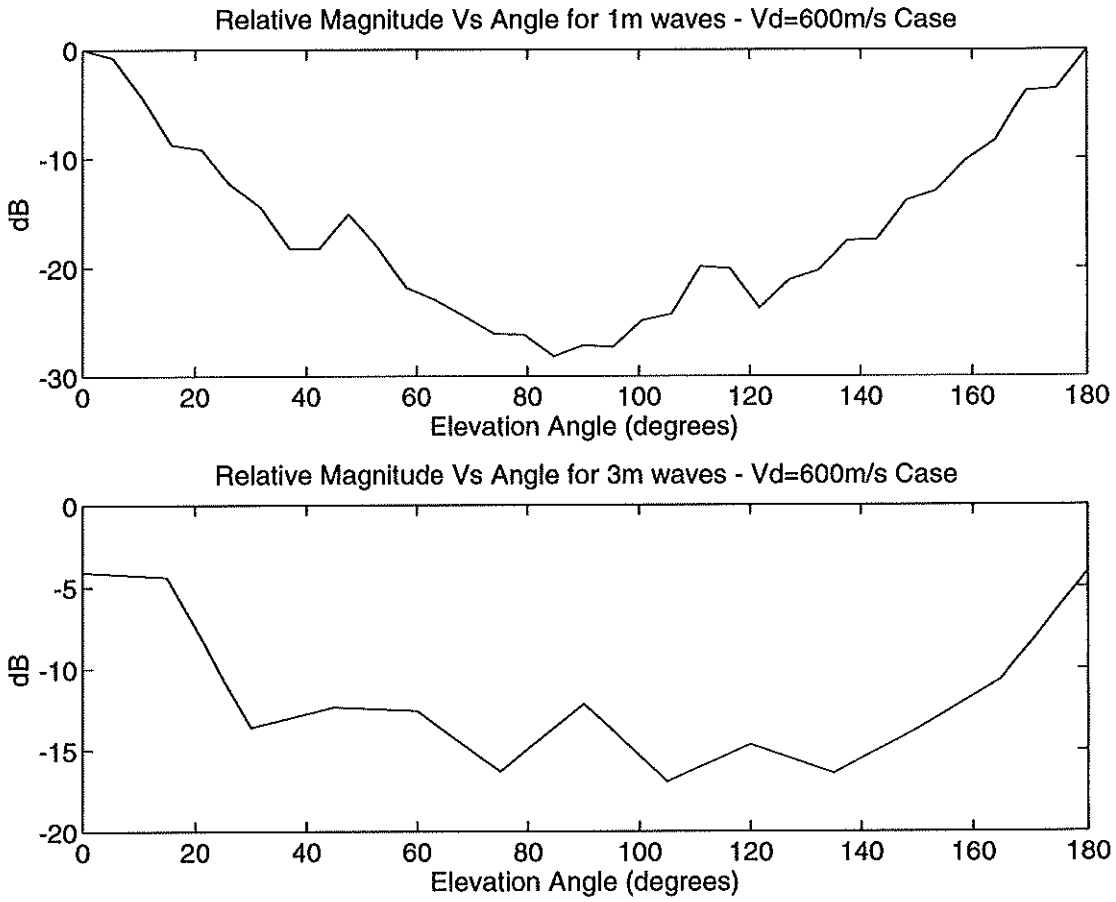


Figure 4.35: Relative Magnitude of the density perturbation in dB versus the elevation angle for $1m$ (top) and $3m$ (bottom) waves from the saturated $V_d = 600m/s$ case. These are interpolated from the full 3D spectra. The 3m wavelength plot interpolates from a very limited number of spectra and may be highly inaccurate.

perturbations. The largest values of E_y appear at the points of maximum density variation.

Rockets are capable of comparing density variations to electric field variations in pure two-stream waves. *Pfaff et al.* [1987*a*] published one of the most extensive data sets on this topic from the CONDOR equatorial electrojet campaign. They observe density variations on the order of 10% and horizontal electric fields on the order of 10mV/m. Our high resolution simulation shows both higher density perturbations and higher fields than this. This may result from the relatively high ambient electric field which we use to overcome particle noise. Other possible explanations of this discrepancy include how the measurements are made by the rockets or, possibly, the electron kinetic effects neglected by the simulation.

In the future, we intend to fly a simulated rocket through our simulation results. This should include simulated probes with their various limitations. Two serious problems make a meaningful comparison with the CONDOR data difficult. First, rocket experiments have a very hard time determining temperatures and so don't yield a good measurement of the crucial acoustic speed. Second, the pure two-stream waves measured in the CONDOR campaign were probably driven with a lower driving electric field than used in the low driving force hybrid simulation as described above. It may be that the hybrid code is unable to resolve the Farley-Buneman instability for cases similar to those seen by the CONDOR rockets. However, a pure two-fluid code may succeed.

The latest rocket study of the equatorial electrojet from the 1994 Guara campaign is just becoming available. *Pfaff et al.* [1995] showed preliminary results of a rocket flying through a pure two-stream instability with an approximately $600 m/s$ electron drift speed. However, they inferred a dominant wavelength of roughly 9 meters. This does not agree with the results of linear theory and our current simulations do not explore wavelengths this long. Further, they estimated that these two-stream waves travel at velocities just under the drift speed. This agrees with linear theory but may be at odds with the simulations and the radar data, depending on the temperature. We expect to receive a more complete analysis of this new data in the near future.

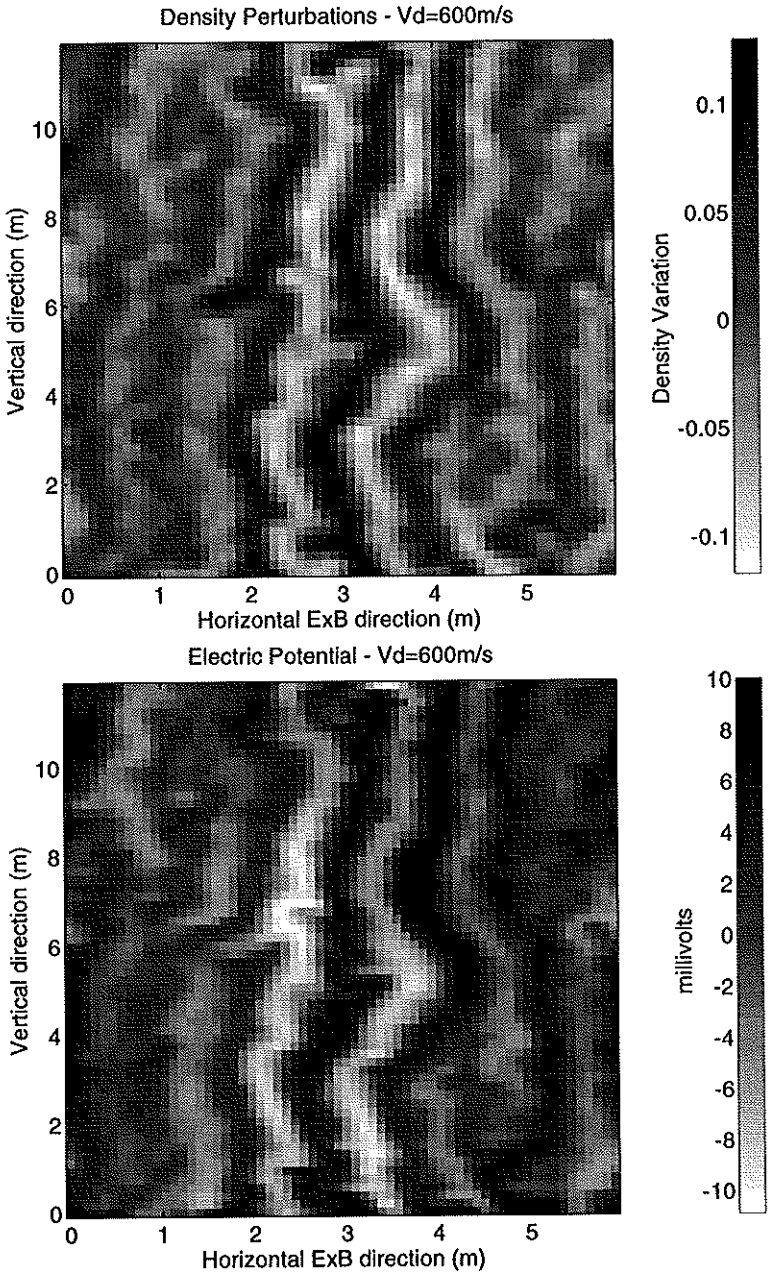


Figure 4.36: The top plot shows the percent variation in the density of saturated two-stream waves from the high-resolution simulation while the bottom shows the potential.

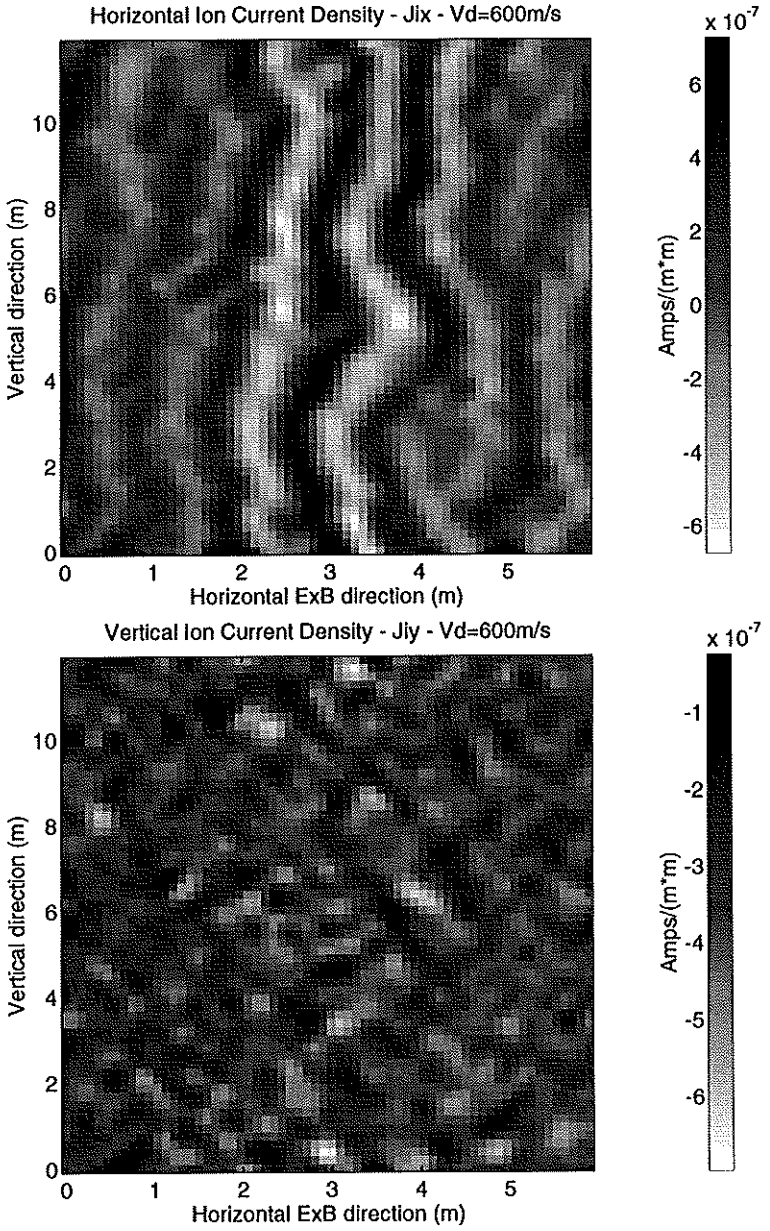


Figure 4.37: The top plot shows the horizontally traveling ion current of saturated two-stream from the high-resolution simulation while the bottom show the vertically traveling ion current.

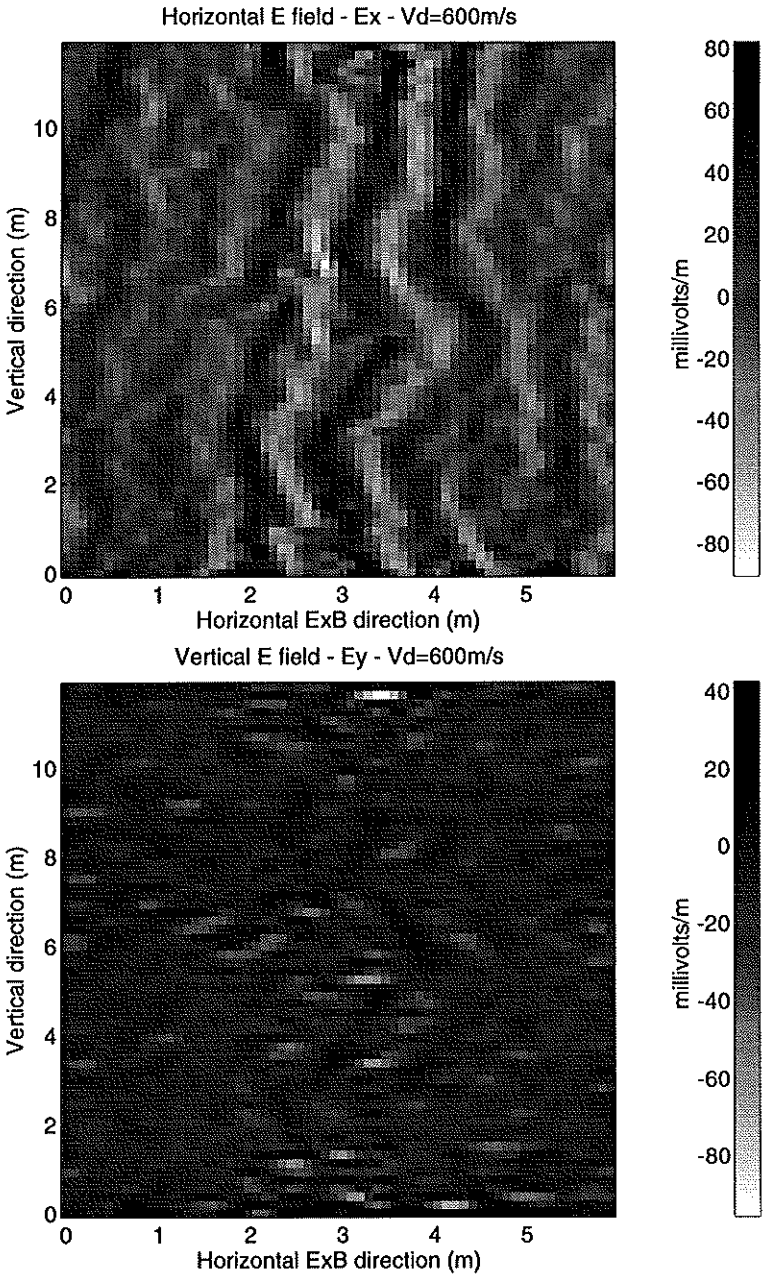


Figure 4.38: The top plot shows the horizontal component of the electric field, E_x, of saturated two-stream waves from the high-resolution simulation while the bottom show the vertical component of the electric field, E_y

4.7 Two-Fluid Code Results

The two-fluid code gives results similar to those of the hybrid code. However, the two-fluid code has a problem. When nonlinear terms dominate the solution, this code develops features with scale sizes close to the mesh size. Nonetheless, the two-fluid code produces waves with similar saturation levels, nonlinear features, wave velocities and peak wavelengths as the hybrid code. Unfortunately, because of the small scale features, the two-fluid code, in its current form, cannot stand on its own. It requires the hybrid results to eliminate serious doubts about its reliability. We will present the results of a single two-fluid simulation and compare these results to similar hybrid simulations.

Figure 4.39 shows the time evolution of the density perturbation of a two-fluid simulation with parameters similar to those of the low-resolution ($V_d = 600m/s$) hybrid simulation. The only parameter difference is that the two-fluid code simulates waves in a 4m by 4m box while the low resolution hybrid code simulates waves in a 3m by 6m box. Also, a small difference in the effective ion-neutral viscous damping rates may result from the different methods used by these two codes to model this effect. Since the linear physics of the two-fluid code with k-dependent damping matches the linear physics of the hybrid code, we expect the growth rates and wavelength of maximum growth to match. On the other hand, saturation levels depend on the details of the nonlinear physics, so the fact that the saturation level of these codes matches so well implies that the same nonlinear physics is probably responsible for the saturation levels in both these codes. This match also implies that ion kinetic effects primarily damp short wavelength mode and that nonlinear ion kinetic effects, which are absent from the two-fluid code, do not appear to play an important role in saturation beyond their linear Landau damping effect.

Comparing the spectrum from this two-fluid simulation (Figure 4.40) to the high resolution simulation's spectrum (Figure 4.18) shows both similarities and differences. First, the two-fluid spectrum originates from a 32×32 grid simulation with only 128 samples taken over time. Nonetheless, the phase velocities and the points of peak instability match reasonably well. Both these features result from the details of choosing the k-dependent damping. The spectral widths are difficult to judge but appear similar. This spectrum does not show an asymmetry in

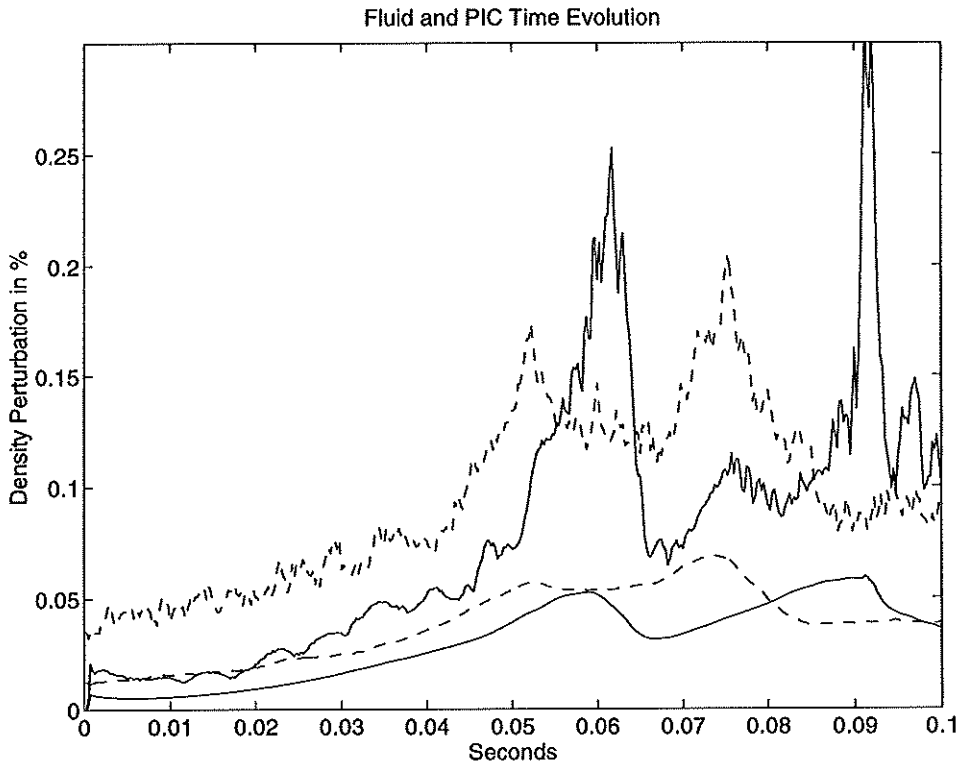


Figure 4.39: Density perturbation versus time from the two-fluid simulation with $V_d = 600m/s$ (solid lines) compared to the 32×32 hybrid simulation with matching parameters (dotted lines). The lower, smooth curves show the standard deviation of the density over the entire simulation area, $\langle |\delta n/n_0|^2 \rangle$, while the upper, jagged, curves show the maximum density on the mesh at a given time.

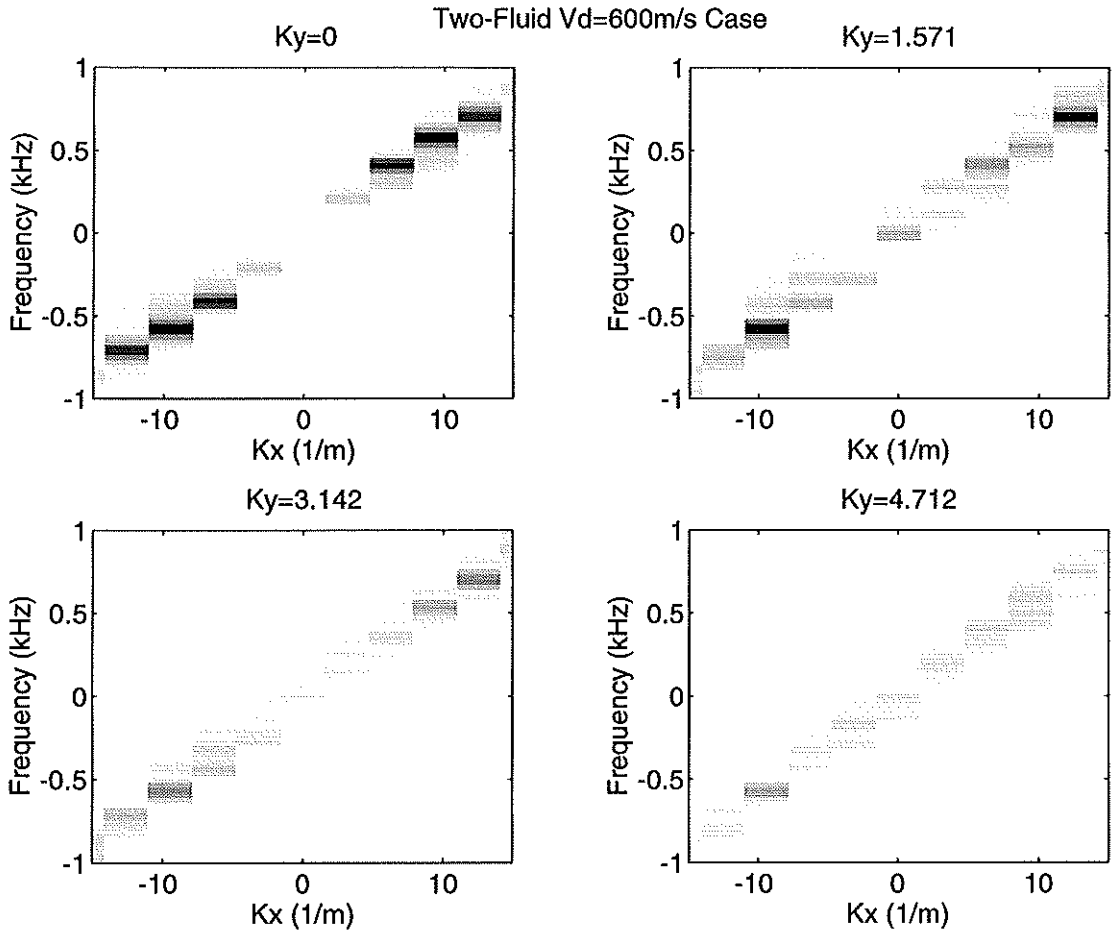


Figure 4.40: Dispersion relationship for four k_y from the Fluid simulation.

magnitude between the downward and upward components of the waves. However, the $k_y = 1.571$ plot shows a small asymmetry in the location of the peaks, with the upward traveling component having a shorter wavelength.

Motion pictures of this fluid simulation show many of the same fundamental features as do motion pictures of the comparable hybrid simulation. We see similar wavelength primary waves grow and propagate rightward. The secondary waves traveling upward along the crests and downward in the valleys of the primary waves appear, but are less pronounced. The viewer is distracted by the short wavelength peaks which develop due to the poor modeling of highly nonlinear behavior. Whenever a localized extremum develops, a large magnitude mesh-scale wave develops. These appear numerical in origin. This behavior undoubtably affects the wave behavior. Nonetheless, as figure 4.39 shows, the saturation amplitude matches that

of the hybrid code implying that the exact details of short wavelength behavior do not influence the saturation level of the two-stream waves. However, to make the two-fluid code a reliable tool, this high k behavior needs to better reflect the physics of the electrojet and show less numerical error.

Chapter 5

Nonlinear Theory

There is something fascinating about science. One gets such wholesale returns of conjecture out of such a trifling investment of fact.

– *Mark Twain*

Simulations and theory should complement one another. In studying a complex phenomenon, the computational physicist applies a theoretical model to develop a simulation technique, then compares the simulation results with the physical phenomena. The simulation has two advantages over the experiment. First, it has excellent diagnostic capabilities. Second, it enables one to isolate physical phenomena by turning on and off various components of the theoretical model. Using the insight gained from simulation, the theoretical model may be modified. Then the process of simulating, comparing with observations and modifying the theory continues. By this process, one hopes to obtain a better theoretical model of the physical system being studied.

We have been applying this approach to understanding the small- scale waves in the equatorial electrojet. Chapter 2 developed the theoretical model of the Farley-Buneman instability. Chapter 3 converted this model into a viable numerical method. Chapter 4 discussed the results of the simulations in terms of the observational data base and also conducted a number of theoretical experiments by turning off various components of the simulations.

This chapter applies what we have learned from the simulation to develop a more accurate physical model. If we believe the simulation results presented in the previous chapter, this model must be nonlinear. We have made substantial progress

primarily on one front: explaining the secondary waves caused by nonlinear electron drifts. The first section of this chapter develops this theory. In the second section, we speculate on other nonlinear analyses which may lead to an even more complete understanding of the Farley-Buneman instability.

5.1 Analysis of Secondary Waves

Motion pictures of the simulated two-stream waves show the development of secondary waves traveling in a direction roughly perpendicular to the direction in which the primary wave travels. Section 4.1.1 explains how to find and play these movies. This section also describes these images in detail. These motion pictures give the impression that the predominant nonlinear behavior of both the hybrid and fluid simulations results from these secondary waves. Removing the nonlinear electron drift term from the potential equation eliminates these secondary waves and increases the saturated density perturbation by an order of magnitude. Therefore, this term causes the secondary waves and plays a crucial role in saturation.

As a simulation begins, the primary waves grow, traveling principally horizontally. As the density perturbation level approaches its saturation level, secondary waves develop, traveling upward on the crests and downward in the valleys of each primary waves. These secondary waves result from the perturbed primary wave electric field, $\delta\mathbf{E}$, interacting with the Earth's magnetic field $\mathbf{B}_0$ to cause a $\delta\mathbf{E}\times\mathbf{B}_0$ electron drift. Quasineutrality causes this phenomenon to show up as the nonlinear term $\kappa\nabla\phi\times\nabla n\cdot\hat{z}$ in the electric potential equation, (2.15).

Our analysis of the secondary waves may be summarized as follows. First, we describe the horizontally traveling and growing primary wave as a simple plane wave with the relationships between density, potential and velocity described by linear theory. We represent these waves with sines and cosines and call them the first order waves. Second, we postulate the existence of secondary waves and assume we can describe them as simple traveling waves represented by complex exponential notation. The secondary waves respond to the nonlinear $\nabla\phi\times\nabla n\cdot\hat{z}$ term that does not affect the primary waves.

This analysis shows that waves localized in the vicinity of the extrema of the primary wave grow when a well-defined threshold is exceeded, while waves at the null points of the primary wave are highly damped for all appropriate parameter

regimes. The analysis assumes the existence of a small parameter, ϵ, which is used to order the equations. The large multiplier found in front of the nonlinear $\nabla\phi\times\nabla n\cdot\hat{z}$ term, κ, is assumed to be of order ϵ^{-1} in this ordering scheme.

This analysis suffers from a number of weaknesses. First, because the secondary waves travel upward along a moving wave front, they travel diagonally. Second, in the simulations, these secondary waves appear as patches implying that the spectra are far more complex than assumed by this essentially linear analysis. Third, a strange scaling is required to incorporate the nonlinear term. This will become apparent as we proceed with the analysis. Lastly, this type of perturbation analysis cannot explicitly explain the saturated state.

In spite of these weaknesses, this analysis helps explain the simulations and, hopefully, physical two-stream waves. It gives a threshold for the onset of secondary wave growth. It explains the resulting $\pm k_x$ asymmetry observed in the saturated state. It gives a mechanism by which the primary wave can turn from the direction of maximum linear growth. This in turn affects the growth of the primary wave and, most likely, plays a role in saturation.

This analysis begins by keeping only one nonlinear term from the fluid description of the Farley-Buneman instability as represented by equations (2.13) through (2.15). Reducing these equations to two dimensions leaves,

$$\frac{\partial\chi}{\partial t} = -\frac{e}{m_i}\phi - \frac{T_i}{m_i}s - \nu_i\chi \tag{5.1}$$

$$\frac{\partial s}{\partial t} = -\nabla^2\chi \tag{5.2}$$

$$\begin{aligned}\nabla^2\phi &= -\kappa\nabla\phi\times\nabla s\cdot\hat{z} + \mathbf{E}_0\cdot\nabla s + \kappa\mathbf{E}_0\times\nabla s\cdot\hat{z} \\ &\quad + \frac{\gamma T_e}{e}\nabla^2 s + (1+\kappa^2)\frac{m_e\nu_e}{e}\nabla^2\chi\end{aligned} \tag{5.3}$$

$$s \equiv \ln(n/n_0) \tag{5.4}$$

The only nonlinear term in these equations is $\kappa\nabla\phi\times\nabla s\cdot\hat{z}$ in equation (5.3). The simulations show the importance of this term. When we eliminate it, the density perturbations increase by almost an order of magnitude

We assume a steady-state primary wave, labeled with the subscript "1", and a secondary wave, labeled with subscript "2". We make the following assumptions about the forms of s, χ and ϕ:

$$s = s_1\sin(\mathbf{K}_1)e^{\alpha t} + s_2\exp[i(\mathbf{k}_2\cdot\mathbf{x} - \omega_2 t)] \tag{5.5}$$

$$\chi = [\chi_{1c}\cos(\mathbf{K}_1) + \chi_{1s}\sin(\mathbf{K}_1)]e^{\alpha t} + \chi_2 \exp[i(\mathrm{k}_2\cdot\mathbf{x} - \omega_2 t)] \quad (5.6)$$

$$\phi = [\phi_{1c}\cos(\mathbf{K}_1) + \phi_{1s}\sin(\mathbf{K}_1)]e^{\alpha t} + \phi_2 \exp[i(\mathrm{k}_2\cdot\mathbf{x} - \omega_2 t)] \quad (5.7)$$

$$\mathbf{K}_1 \equiv \mathrm{k}_1\cdot\mathbf{x} - \omega_1 t \quad (5.8)$$

where α is the growth rate of the first-order terms and ω_1 is real.

Substituting these expressions into the linear equations, (5.1) - (5.3) and ignoring the second-order terms gives us six equations defining relationships between s_1, χ_{1c}, χ_{1s}, ϕ_{1c}, ϕ_{1s}, ω and α. Combining these six equations defines ω and α in a dispersion relationship identical to the linear dispersion equation, (2.23).

We find a second-order dispersion relation by substituting the assumed form of s, χ and ϕ represented by (5.5) through (5.7) into equations (5.1) - (5.3). Also, assuming that the first-order terms remain unaffected by the second-order terms, we can cancel them, leaving only second-order terms. These include the crucial nonlinear term of equation (5.1). At first glance, it appears that the $\kappa\nabla\phi\times\nabla s\cdot\hat{z}$ is third-order, but we claim that the large κ constant effectively raises the order. Given these assumptions, we approximate the nonlinear term as

$$\kappa\nabla\phi\times\nabla n\cdot\hat{z} \approx i\kappa\exp[i(\mathrm{k}_2\cdot\mathbf{x} - \omega_2 t)]\mathrm{k}_1\times\mathrm{k}_2\cdot\hat{z} \times[(-\phi_{1c}\sin(\mathbf{K}_1) + \phi_{1s}\cos(\mathbf{K}_1))s_2 - s_1\cos(\mathbf{K}_1)\phi_2] \ . \quad (5.9)$$

At this point, we can approach the problem from a number of directions. We could generate a second-order dispersion relation, treating $\sin(\mathbf{K}_1)$ and $\cos(\mathbf{K}_1)$ as parameters. However, a more intuitive approach is to look at the extrema and nulls of the waves separately as is done in the following subsections.

5.1.1 Secondary Waves at the Extrema

Let us closely examine the crests and valleys of the primary waves. At these points, s_1 is either a maximum or minimum value, so $\sin(\mathbf{K}_1) = \pm 1$ and $\cos(\mathbf{K}_1) = 0$. Also, assuming that the primary wave travels in the $\hat{x}$ direction makes $\mathrm{k}_1 = k_1\hat{x}$. This reduces the nonlinear term to

$$\kappa\nabla\phi\times\nabla n\cdot\hat{z} \approx -i\kappa\exp[i(\mathrm{k}_2\cdot\mathbf{x} - \omega_2 t)](k_1\hat{x}\times\mathrm{k}_2\cdot\hat{z})\phi_{1c}s_2 \quad (5.10)$$

We can now write a set of linearized second-order relationship between s_2, χ_2 and ϕ_2 using this nonlinear term and the remaining components of equations (5.1)

through (5.3) as

$$\begin{aligned} -k_2^2\phi_2 &= -i\kappa(k_1\hat{x}\times\mathbf{k}_2\cdot\hat{z})\phi_{1c}s_2 + i\mathbf{E}_0\cdot\mathbf{k}_2 s_2 + i\kappa\mathbf{E}_0\times\mathbf{k}_2\cdot\hat{z}s_2 \\ &\quad -\frac{\gamma T_e}{e}k_2^2 s_2 - (1+\kappa^2)\frac{m_e\nu_e}{e}k_2^2\chi_2 \qquad (5.11) \\ i\omega_2\chi_2 &= \frac{e}{m_i}\phi_2 + \frac{T_i}{m_i}s_2 + \nu_i\chi_2 \qquad (5.12) \\ -i\omega_2 s_2 &= k_2^2\chi_2 \qquad (5.13) \end{aligned}$$

These can be combined to generate a single dispersion equation:

$$\begin{aligned} \omega_2^2 + i\nu_i\omega_2 - k_2^2C_s^2 &+ i\frac{e}{m_i}(-\kappa(k_1\hat{x}\times\mathbf{k}_2\cdot\hat{z})\phi_{1c} + \kappa\mathbf{E}_0\times\mathbf{k}_2\cdot\hat{z} + \mathbf{E}_0\cdot\mathbf{k}_2) \\ &+ (1+\kappa^2)\nu_e\frac{m_e}{m_i}i\omega_2 = 0 \qquad (5.14) \end{aligned}$$

This equation is almost identical to the linear equations of the original Farley-Buneman instability linear dispersion relation, (2.32), which in two dimensions is

$$\omega^2 + i\omega\nu_i - k^2C_s^2 + iE_{0y}\frac{e}{m_i}(k_y - \kappa k_x) + \frac{m_e}{m_i}\nu_e(1+\kappa^2)i\omega = 0 \qquad (5.15)$$

In the second-order equation, the driver for the instability is, primarily, the perturbed field from the principal wave. This appears as the nonlinear term, $\kappa(k_1\hat{x}\times\mathbf{k}_2\cdot\hat{z})\phi_{1c}$. In the first-order equation, the zero-order electric field drives the instability through the $E_{0y}(e/m_i)(i\kappa k_x)$ term. If we ignore, for the moment, the terms with E_0 in the second-order dispersion relation, we obtain the same form as in the original two dimensional dispersion relation except the effective drift speed is now determined by the primary wave amplitude,

$$V_d \Rightarrow V_s \equiv \frac{ik_1\phi_{1c}}{B_0} = \frac{E_1}{B_0}\,. \qquad (5.16)$$

Hence, when $V_s > C_s(1+\Psi_0)$, the secondary waves begin to grow.

It appears that we went through a fairly elaborate calculation to obtain a relatively obvious result. The perturbed field of the primary two-stream waves drives, upon reaching a sufficient magnitude, more two-stream waves, propagating perpendicularly to the original waves. However, this analysis ignores a few crucial points.

First, the secondary waves can never have the simple spectrum implied by this analysis. These waves travel both in the electron drift direction with the

primary wave, and upward on the crests and downward in the valleys of the primary waves. Since these horizontal and the vertical motions have similar velocities, the secondary waves actually travel diagonally. This makes their spectra in k-space more complex.

Second, the motion pictures show that the secondary waves are limited to the extrema of the primary waves and, therefore, have a limited spatial range. The analysis of the next subsection shows that when we move off the extrema of the primary waves these waves are highly damped. This explains the limited extent of the secondary waves and further complicates their spectra. Our analysis of secondary waves implicitly assumes that the conditions holding at the extrema of the primary wave extend far enough to either side of the actual local maxima that we can model the secondary waves as a plane wave structure. The assumption that the finite width of the secondary waves does not fundamentally alter their plane wave behavior is called a local approximation.

Third, the second-order dispersion relation has an additional term, $\kappa \mathbf{E}_0 \times \mathbf{k}_2 \cdot \hat{z}$, which also drives these secondary waves. Even though $\mathbf{E}_0$ and $\mathbf{k}_2$ are mostly parallel, the inevitable component of $\mathbf{k}_2$ perpendicular to $\mathbf{E}_0$ causes this term to affect the secondary waves.

Third, in the original two-stream equations, the $\mathbf{E}_0 \cdot \mathrm{k}$ term leads to the ik_y term in the first-order dispersion relationship. Because $\mathbf{E}_0$ and the dominant k are perpendicular and, because there is no κ multiplying this term, this term was largely inconsequential in the first-order dispersion relationship except when evaluating the behavior of the highly damped, vertical waves. In the secondary wave dispersion relation, $\mathbf{E}_0$ now points in the direction of the instability so that, even though the large κ factor is absent, this term may be relevant.

A close evaluation of these three points helps to explain the behavior of the waves seen in the simulation. When the primary waves reach a sufficient magnitude, the second-order dispersion relation predicts growing secondary waves. We can evaluate the saturated state of each simulation to determine if this threshold is achieved. For example in the low driving force simulation, we see horizontal primary wave electric fields of $30mV/m$ (see section 4.6). To see if secondary waves grow in such a system, we do not solve equation (5.14). Instead, we derive a second-order linear *kinetic* dispersion relationship by combining the kinetic methods of section 2.3 and the approach used to generate a second-order dispersion

relation. Since the secondary waves travel at roughly 45° degrees, the resulting relationship can be solved numerically, using techniques identical to those used to solve the two dimensional linear kinetic dispersion relation of section 2.3.1.

Additionally, those terms in the second-order dispersion relationship containing $\mathbf{E}_0$ may act to generate an asymmetry in the secondary waves. The $\mathbf{k}_2 \cdot \mathbf{E}_0$ and $\kappa \mathbf{k}_2 \times \mathbf{E}_0 \cdot \hat{z}$ terms affect negative k_{2y} differently from positive k_{2y} when added to the principal driver of the secondary waves, the horizontal electric field of the primary waves. These, in turn, may act on the primary waves to cause the apparent $\pm k_x$ asymmetry discussed in section 4.4.1.

These secondary waves also affect the primary waves in other ways. They remove energy from the primary wave. We have yet to explore this effect.

The perturbed fields of the secondary waves affect the D.C. field driving the primary wave, variously reducing and enhancing it in different locations. In places where the primary, zero-order, field is enhanced, we expect the primary wave to travel faster while in places where the primary field is reduced, we expect the primary wave to slow. This should lead to the bending of the primary wave. The secondary wave also bends the wave by effectively turning the perturbed electric field. When the electric field of the secondary wave is added to the horizontal electric field of the primary wave, the net field points diagonally. Figure 4.38 shows that, in the high resolution simulation, the perturbed vertical electric field attains magnitudes almost as large as the perturbed horizontal electric fields though only in regions having the largest density perturbations. We also see from this figure that the vertical electric field is asymmetric having larger negative values than positive ones. An asymmetry in the secondary waves should affect the bending of the primary wave causing preferential bending in one direction over another. These effects can add up to cause wave saturation, though this type of perturbation analysis does not directly address this question.

5.1.2 Secondary Waves on the Edges

A parametric analysis of secondary waves traveling vertically on the edges (at the null points) of the primary waves reveals that such waves are strongly damped. The calculation of a dispersion relationship for this case diverges from the previous case in the approximation applied to the nonlinear term in equation (5.9). Now, we choose values for s_1 that apply at the points where the density equals the

unperturbed density so, $\sin(\mathbf{K}_1) = 0$ and $\cos(\mathbf{K}_1) = 1$. This makes the nonlinear term

$$\kappa\nabla\phi\times\nabla n\cdot\hat{z} \approx i\kappa\exp[i(\mathrm{k}_2\cdot\mathbf{x}-\omega_2 t)](k_1\hat{x}\times\mathrm{k}_2\cdot\hat{z})(\phi_{1s}s_2 - s_1\phi_2) \quad (5.17)$$

This term, combined with the linearized versions of equations (5.1) through (5.3), generates a linear dispersion relationship analogous to the linear dispersion relationship (5.14). These dispersion relations differ in two important ways. First, the ϕ_{1c} term represents the component of the electric field coincident with the peak density while the ϕ_{1s} term represents the far smaller out of phase component of the electric field. This means the ϕ_{1s} term in (5.17) is small. Second, the s_1 term in (5.17) appears in the resulting dispersion relation for the complex frequency, ω_2, as a large negative imaginary term. This means that these secondary waves are heavily damped. This agrees with the simulation results which show that secondary traveling waves do not appear on the edges of the primary waves. However, a rippling effect appears on the wave edges for which we cannot account.

5.2 Nonlinear Speculation

Obtaining useful information from a truly nonlinear theory is extraordinarily difficult. In the last decades, substantial advances have been made in understanding simple, low dimensional, nonlinear systems [*Lorenz*, 1963; *Feigenbaum*, 1978; *Strogatz*, 1994]. However, higher dimensional nonlinear systems remain difficult to quantify. Numerous techniques and approaches to these problems exist but, unlike for linear systems, no relatively simple path has been mapped out which guides us through the study of nonlinear systems. This section proposes a number of approaches toward better understanding the dynamics seen in the simulations and in observations of the Farley-Buneman instability. While we have considered these techniques, we have not yet pursued them with any diligence.

The first nonlinear evaluation of our simulation should determine the correlation dimension of the system. This gives one a sense of the complexity of a dynamical system's behavior. If the correlation dimension is reasonably low, then we can consider studying the Farley-Buneman system with a small number of coupled modes. If the correlation dimension is high, then we can perhaps apply some of

the techniques used to study turbulence, or, perhaps, just argue that simple models do not apply.

To determine the correlation dimension of a system such as ours, we apply a method proposed by *Grassberger and Procaccia* [1983]. We begin with a vector of values, $\{\mathbf{x}_i, i = 1, ..., m\}$, generated by solving a time dependent set of equations where we have stored the output from each of the m timesteps in $\mathbf{x}_i$. Generally, $\mathbf{x}_i$ is a D dimensional vector with one dimension for each equation in the set. For the case of our 64 by 64 simulation with the variables, n, ϕ, J_{ix} and J_{iy}, we have a $64 \times 64 \times 4 = 16384$ dimensional system. Define a D dimensional sphere of radius ϵ and determine the number of points from the vector, $\mathbf{x}_i$, falls within this sphere. To make this work well, one needs a large number of points ($m \gg 1$). Expand the sphere to $\epsilon + \delta$ and determine the number of points within this larger sphere, $N(\epsilon + \delta)$. Determine how N scales with δ. Typically, the scaling goes as

$$N(\epsilon) \propto \epsilon^d \tag{5.18}$$

where d is called the pointwise dimension. When this analysis is repeated for many values of ϵ, the dimension d should remain constant as long as ϵ is within appropriate bounds. If ϵ is too small then no points or a statistically insignificant number of points lie within ϵ. If ϵ is too large, then the sphere may encompass most of the points, and x_i and N will not change with increasing ϵ. The average of the pointwise dimensions for those values of ϵ within the appropriate bounds is called the correlation dimension.

When this analysis is performed on the three coupled Lorentz equations, the correlation dimension turns out to be $d = 2.08$. When one plots the three variables of the Lorenz equation as they evolve in time, one sees that the trajectories mostly lie in a pair of planes. That is why the correlation dimension lies just above two. In spite of this low dimensionality, the Lorentz equations display chaotic behavior (i.e., non-periodic oscillations), behavior which a two dimensional system cannot have. In contrast, a good random number generator should always give one the number of dimensions matching the vector dimension D. An easily understood explanation of this and other low-dimensional nonlinear dynamical systems is given by *Strogatz* [1994].

Chapter 6

Discussions and Conclusions

Wisdom too often never comes, and so one ought not to reject it merely because it comes late.

– Supreme Court Justice Felix Frankfurter

The simulations presented here aim to provide us with a better understanding of the two-stream waves observed in the equatorial electrojet. By translating a physical model of these waves into a computer simulation and then using the results from these simulations to build a more detailed physical model, we have added to our understanding of the Farley-Buneman instability.

In the introductory chapter, we discussed the physical situation in which these waves arise, the measurements made of these waves, and the theories which explained them. The chapter on fluid and linear theory developed the detailed theoretical models of the Farley-Buneman instability used for our calculations. The following chapter applied numerical methods to convert this model into a viable simulation. The chapter after that discussed the results of the simulations in terms of the observational data base and also described a number of numerical experiments conducted by eliminating various components of the simulations. The next chapter developed a more detailed explanation of some of the important nonlinear features observed in the simulations and made a number of suggestions for additional nonlinear analyses.

This concluding chapter begins by summarizing the most important results from these simulations and theories. These results are then compared to the observational data, to existing theories and to other simulations. The last topic

addressed is our plan for continuing the theoretical and numerical investigations into the behavior of the Farley-Buneman instability in the equatorial electrojet.

6.1 Summary of Results

The first substantial achievement of this project, making all the subsequent research possible, was the development of a numerical technique which correctly solved the difficult potential equation, (2.12). This equation was solved by formulating the finite difference equations to eliminate an inherent singularity and by using a variant on the biconjugate gradient method, called the quasi-minimal residual method, to solve the resulting linear equations. Previous attempts at solving the same set of equations either failed to solve this system correctly [*Newman and Ott*, 1981] or had avoided this problem by using a particle model for electron behavior, a technique so computationally expensive that reaching saturation proved impossible [*Janhunen*, 1994*b*].

The results of our simulations agree well with the existing observations of type I waves. The dominant waves are quite monochromatic, a result agreeing with the rocket observation of *Pfaff et al.* [1987*a*]. During periods of two-stream activity, radars detect waves with Doppler shifts which exceed their spectral width. They also measure vertically oriented density irregularities with no net Doppler shift and a spectral width of approximately twice the acoustic velocity. Our simulation sees both these features. In addition, the simulation can simultaneously evaluate the relative magnitudes of waves having different elevation angles and wavelengths, a feature comparable to observations made by multiple radars looking at the same point simultaneously. This experiment has yet to be done.

The phase velocities obtained from our numerical solutions are below the velocities predicted by linear theory. This corroborates the observations of *Cohen and Bowles* [1967], *Hanuise and Crochet* [1981] and many others. We also see some evidence that the phase velocities remain constant within an extended range of elevation angles as first observed by *Cohen and Bowles* [1967].

The dominant nonlinear effect observed in the simulation arises from the nonlinear electron, $\delta\mathbf{E}\times B_0$, drift. This effect derives from the $\kappa\hat{\mathbf{z}}\cdot\nabla\phi\times\nabla n$ term of the potential equation, (2.12). Inclusion of this term in the simulations has a profound effect on the numerical solutions, an effect far greater than that caused by dra-

matically varying the electron temperature or even the electron drift speed. The nonlinear $\delta\mathbf{E}\times\mathbf{B}$ drift causes small scale waves to travel upward along the crests and downward in the valleys of the principal, mostly horizontal, waves. Nonlinear analysis of section 5.1 uses a perturbation theory to show that these secondary waves propagate and grow. It is likely that this effect accounts for the vertically propagating component of type I echoes *Kudeki et al.* [1987]. Coupling between these secondary waves and the principal waves plays a role in turning the primary waves off the drift direction and probably also plays a crucial role in saturating and reducing the effective phase velocity of the primary waves.

Our simulations indicate that ion kinetic effects are not essential for saturation of the instability. They also indicate that electron temperature effects have a minor impact on the final saturated state. However, because this code does not model electron kinetic effects, we cannot say for sure what role such kinetic effects might have. Many of the observed features of the two-stream instability are seen in these simulations without including the electron kinetic effects.

All simulations of the Farley-Buneman instability perpendicular to the magnetic field show that the dominant waves do not travel strictly parallel to the electron drift direction as predicted by linear theory, but develop a vertical component instead. This was first observed in the simulations of *Newman and Ott* [1981]. *Janhunen* [1994*b*] pointed out that the primary wave always satisfies $\mathbf{E}_0\cdot\mathrm{k}_{wave} < 0$. He also pointed out a number of possible effects of this wave turning, though he provided no explanation of its causes. Our simulations also display wave turning and show that the angle between the drift direction and the dominant wave of the system becomes larger as the driving electric field, $|E_{0y}|$, becomes larger. The nonlinear theory presented in Chapter 5 shows that electron $\delta\mathbf{E}\times\mathbf{B}$ drift causes secondary waves with asymmetric vertical propagation. These waves can interact with the primary waves causing them to turn away from a directly horizontal direction, with a preference for upward propagation over downward.

Electron temperature effects have a minor impact on the final saturated state. Dropping the electron temperature from $208°K$ to $0°K$ increases the root-mean-squared saturation level, $\langle|\delta n/n_0|^2\rangle 0$, from 5% to 6% while doubling the temperature to $408°K$ reduces the saturation level to 3%. This contrasts with the effect of linearizing the electron dynamics which raises the saturation level from 5% to 35%. Doubling the electron drift velocity from $800m/s$ to $1600m/s$ raises the level

from 5% to 17%.

Ion kinetic effects play an important role in damping waves with short wavelengths (high |k|). The fact that both the hybrid simulation and the two-fluid code reach similar saturation levels implies that nonlinear ion kinetic effects, which are not modeled in the two-fluid code, do not play a crucial role in saturating the waves. Nonlinear electron $\mathbf{E} \times \mathbf{B}$ drifting appears to be the dominant mechanism which saturates the waves. Section 4.2 demonstrates this by showing the dramatic effect on the saturation level caused by eliminating the nonlinear term from the equations responsible for this electron drift.

6.2 Comparison with Published Observations

In the introduction, we discussed the results of a large number of observational studies of the E region ionosphere. In this section, we will explicitly compare our results with these observations. While many of the points have been previously discussed along with the specific result or observation, we summarize these discussions here.

6.2.1 Comparison with Radar Measurements

Type I radar echoes from the E-region show pronounced Doppler shifts with spectral widths considerably smaller than the Doppler shift. The simulations produce similar spectra making it easy to believe that the same physical basis underlies both the observations and the simulation spectra.

During periods of intense type I activity in the upper electrojet, radars detect vertical echoes with a large spectral width and little Doppler shift. The simulation spectra also show vertical irregularities with no net Doppler shift and similar spectral widths. As discussed in sections 4.1.1 and 5, secondary waves probably cause these vertical irregularities.

The Doppler shift of type I radar echoes indicates that two-stream waves travel with a phase velocity close to the acoustic speed. These observations have the problem that they cannot determine the difference between the electron drift speed and the plasma acoustic speed, leaving the expected phase velocity quite ambiguous [*Pfaff et al.*, 1987*b*]. Nonetheless, the simulation shows waves traveling at velocities between that predicted by linear theory and the acoustic velocity.

Radars also observe two-stream waves traveling at the acoustic velocity independent of the elevation angle at which they are measured [*Cohen and Bowles*, 1967; *Farley*, 1985]. This does not agree with the predictions of linear theory. Some controversy exists about this conclusion since the radars which make these measurements may inadvertently average together both secondary and primary two-stream waves. Our simulations show some evidence that the phase velocities remain constant within an extended range of elevation angles.

Radars have observed an up-down asymmetry in two-stream waves. For instance, when a radar points westward with a 45° degree elevation angle, the power of the reflected waves is less than when the radar points eastward with the same elevation angle [*Dougherty and Farley*, 1967].It was proposed that this resulted from geographic features of the area surrounding the radar, but this difference has now been observed at enough locations to make this explanation unlikely. Both our simulations and those of *Janhunen* [1994*b*] show this same effect. It appears to result from the preponderance of upward-traveling secondary waves arising from the $\pm k_x$ asymmetry discussed in section 4.4.1 and explained in section 5.1.

6.2.2 Comparison with the CONDOR Rocket Campaign

The CONDOR combined rocket and radar campaign provided some of the most recent and detailed observations of two-stream waves of the upper electrojet [*Kudeki et al.*, 1987]. *Pfaff et al.* [1987*a*] lists seven points or observations which we paraphrase as follows:

1. At 108km altitude, coherent, narrow banded oscillations of δE and $\delta n/n$ indicate the presence of two-stream waves.

2. These waves traveled principally horizontally and have an altitude range of 2km.

3. The measured wavelength having the most power matches the wavelength which linear theory shows to have the highest growth rate.

4. The phase velocity of these waves "may have been comparable to the linear theory result."

5. "Because the primary two-stream waves propagate horizontally along the direction of the current, VHF backscatter radars looking vertically or obliquely

would not be expected to observe these waves, as predicted by *Lee et al.* [1974]. The constant phase velocities observed by the radars are not in contradiction with the horizontal phase velocities observed *in-situ*, since the oblique and vertical two-stream waves observed by backscatter radars at the equator could originate primarily in the interaction region of large-scale waves and not from the laminar topside layer."

6. The phase velocity of the horizontally propagating waves does not decrease at the highest elevations where the current density drops. "This remains puzzling ..."

7. "The strong component of vertical waves associated with the "pure" two-stream region observed both by the Jicamarca radar and the *in-situ* electric field instruments may possibly be explained by coupling to the gradient drift instability as explained in *Pfaff et al.* [1987*b*]..."

Our results corroborate the first three points with the qualification that the principal modes travel almost, but not quite, horizontally. For those simulations with realistic equatorial electrojet parameters (i.e., when the electron drift velocity is less than double the acoustic speed), we observe that the dominant mode matches the wavelength with the highest growth rate predicted by linear kinetic theory. Our simulations, as well as those of *Newman and Ott* [1981] and *Janhunen* [1994*b*], show that the dominant waves do not travel exactly parallel to the electron drift direction. However, for the simulations with modest electron drift velocities, this angle is small enough that the *in-situ* instruments may not be capable of measuring them.

Point 6 seems to contradict point 4 and may corroborate our simulation results which indicate that two-stream waves travel at a velocity between the acoustic velocity and the velocity predicted by linear theory. Further, *Crochet et al.* [1979], *Hanuise and Crochet* [1981] and *Prakash et al.* [1974] all claim to have seen two-stream waves traveling at velocities close to the acoustic speed, instead of the velocity predicted by linear theory.

Points 5 and 7 call for a gradient drift mechanism to generate vertical waves. However, our simulations show that these can result from nonlinear $\delta\mathbf{E}\times\mathbf{B}_0$ electron drifts produced by the primary two-stream waves as discussed in section 5.1. Our simulations indicate that it is unnecessary to invoke coupling to longer scale waves

to generate vertical density irregularities with characteristics matching those of numerous radar observations.

Farley [1985] states that "It is especially difficult to understand how the vertically propagating waves can be excited by horizontal electron flow. Some sort of nonlinear mode coupling is obviously required." His speculation about mode coupling was correct. However, we argue that the mechanism for these waves is the relatively simple nonlinear $\mathbf{E}\times\mathbf{B}$ electron two-stream wave described in section 5.1.

6.3 Comparisons with Published Theories

The testing ground for any theory of type I irregularities must be its ability to explain the features of the saturated state. The longstanding problem with both the linear and nonlinear theories of the two-stream instability has been their failure to account for the observed spectral Doppler shifts, which are always close to the ion acoustic speed, regardless of the strength of the driving electron drift. A promising attempt at solving this puzzle was put forward by *Sudan* [1983]. Briefly, random perturbations to the electron orbits caused by large amplitude, short wavelength waves are shown to act much as electron-neutral collisions do. This 'anomalous' or 'wave-induced' diffusion can cause nonlinear modification of the phase velocity of the waves and eventually bring about their saturation. Our simulations do not directly test this theory because a fluid model of electron behavior cannot have its orbits perturbed in the manner described in that paper. However, without allowing this type of anomalous diffusion, we still see a reduced phase velocity and a number of other features of the experimental data. This indicates that the *Sudan* [1983] mechanism may not be crucial, but it cannot be ruled out until we perform a comparable simulation allowing electron orbit perturbations.

6.4 Comparisons with Other Simulations

Earlier simulations of the Farley-Buneman instability were quite limited, but their results generally agree with our own. *Newman and Ott* [1981] presented simulations containing a serious numerical flaw, as described in section 3.1.2, and were hampered by limited computational resources. The *Machida and Goertz* [1988]

simulations successfully compared various electron heating mechanisms proposed to explain observations in the polar two-stream instability. They modeled wave behavior in the plane parallel to the Earth's magnetic field, neglecting the nonlinear effects which play a crucial role in our simulations. The simulations most comparable to our own, by *Janhunen* [1994*b*], focused on the polar electrojet and modeled wave behavior in the plane perpendicular to the magnetic field. These simulations had noise problems typical of pure particle-in-cell codes as described in section 3.2.1. Nonetheless Janhunen observed that the primary waves traveled at an angle with respect to the electron drift direction, such that $\mathbf{k} \cdot \mathbf{E}_0 < 0$. Our simulations also show this behavior as described in section 4.4. Janhunen also observed waves traveling at the velocity predicted by linear theory but this may result from his simulation's failure to reach a saturated state. Our results show waves traveling with phase velocities between the linear velocity and the acoustic speed.

6.5 Future work

Our simulations of the Farley-Buneman instability have not exhausted the possible benefits of such studies. Much remains to be done and we plan to extend this work in the future. Our existing simulation results can be evaluated further. Section 5.2 describes a number of possible approaches to further understanding the nonlinear behavior of these simulations.

A code that generates less particle noise than our current hybrid code could simulate the Farley-Buneman instability of the equatorial electrojet more accurately, allowing smaller values of the electron drift velocity. To achieve this, we could attempt to further reduce the particle noise of the hybrid code. Since adding significantly to the number of particles is likely to remain beyond our computational capacity for the next few years, we could apply a technique called orbit averaging or time-averaging. This method determines the ion densities and currents based on an average of the ion positions and velocities from the previous few timesteps and leads to smoother density profiles. Alternately, we could improve the two-fluid code to better match the hybrid code's short wavelength behavior. This should enable us to model waves driven by a slightly unstable electrojet. The results from these quieter simulations may enable us to determine what other nonlinear

processes affect the saturated state. Those results might also suggest a theoretical analysis of other nonlinear mechanisms.

Two-stream waves also occur in the polar electrojet. In these regions, the driving electrical field is much larger and electron heating has been observed [*Schlegel and St.-Maurice*, 1981; *Nielsen and Schlegel*, 1985]. In order to perform simulations of those waves, electron temperature effects need to be represented with a more sophisticated model than the adiabatic or isothermal models we currently use. We envision three possible approaches to this.

First, a heat equation can be used to determine electron thermal energy. The theories of *Schunk* [1977] or *Dimant and Sudan* [1995] can be used to generate moments of the Boltzmann equation. This is probably the most computationally efficient method. Second, electron dynamics can be simulated using a PIC method. To avoid electron cyclotron behavior, a guiding center algorithm could be used. To eliminate numerical noise problems, described in section 3.2.1, one can use the information gathered from these particles to determine only the electron pressure effects. Our current code assumes a simple isothermal or adiabatic approximation for electron temperature effects and then applies this information as a scalar pressure term in the potential equation. This technique would be computationally less expensive than a pure PIC code, because it requires far fewer electrons. Also, by using the potential equation, (2.12), this method would maintain quasineutrality and inertialess electrons. Third, an implicit numerical method can be used to model electron behavior. Any one of these techniques should enable an evaluation of the causes and effects of electron heating on two-stream waves in both the polar and equatorial electrojet.

Another open question concerns the importance of the previously neglected third dimension. While the effect of coupling energy into and out of the direction parallel to the magnetic field is expected to be small, we feel it is necessary to test this hypothesis thoroughly. A three dimensional two-fluid code with wavelength-dependent damping could evaluate this question.

Yet another possibility involves extending this work to the study of coupling between the gradient drift instability and the two-stream instability. The observational data shows that this coupling occurs and is probably important to the large scale structuring of the bottom side equatorial *E*-region. This is a difficult problem because waves ranging from kilometric scales down to meter scales need to

be analyzed together. While the detailed approach to this problem is still unclear, the assorted tools we have developed should be useful in analyzing this problem further.

6.6 Concluding remarks

The objective of this work was to answer a number of longstanding questions about the behavior in the Farley-Buneman instability. Within the limits of our simulation techniques, we have made some appreciable progress toward that end. Our hybrid code displays much of the behavior of the observed Farley-Buneman instability. The detailed information obtained from these simulations has enabled us to put forward a simple theory explaining a number of important aspects of two-stream waves. We believe that both the simulations themselves and this theory contribute to the understanding of plasma physics in the equatorial ionosphere.

Appendix A

Applying Finite Difference Equations: A Minimal Explanation

Solving partial differential equations (PDE) numerically is a topic which fills many books. The following paragraphs present enough information so that a reader with a limited numerical background can understand the material of Chapter 3. We will not touch on many of the limitations and failings of these methods, nor will we discuss any of the alternate methods which exist. For this, I recommend reading *Press et al.* [1988], chapter 19, or, for even more detail, *Ames* [1977].

We wish to convert a PDE which we cannot solve analytically into a set of linear algebraic equations which we can solve. As a simple example, we will convert the one dimensional Poisson's equation, $\partial_x^2 \phi = \rho$. First, we discretize space into a finite set of points, called nodes. The simplest and most commonly used method of dividing physical space into a set of nodes uses evenly spaced nodes. Let's say we want to place a hundred nodes between 0 an 1 meters. Then the first node should be placed at 0.0m, the second at .01m, the third at .02m and so forth. For some problems a non-uniformly spaced set of nodes works better. The only requirements on a one dimensional mesh is that no two nodes can be located at the same point. The simplest 2 dimensional mesh simply draws a uniform grid over the 2D physical space and designates a node at the intersection points of that grid. More complicated grids are possible though, for the simulations of this thesis, a

uniform grid is adequate.

After the nodes are defined, we seek an approximate solution of Poisson's equation for ϕ_i at each of these nodes. Taking two successive finite difference approximations for ϕ gives us a linear algebraic expression for the potential at each node,

$$\frac{\partial^2\phi}{\partial x^2} \approx \frac{\partial}{\partial x}\left(\frac{\phi_{i+1/2} - \phi_{i-1/2}}{\Delta x}\right) \approx \frac{\phi_{i-1} - 2\phi_i + \phi_{i+1}}{\Delta x^2} = \rho_i \ . \tag{A.1}$$

This approximation is considered "consistent" with Poisson's equation because, in the limit as $\Delta x \to 0$, this expression becomes Poisson's equation. At each node, we have an equation for ϕ_i which is a function of the potential at the neighboring nodes, ϕ_{i+1} and ϕ_{i-1}, and ρ_i which we know. A complete set of these equations for all the nodes plus two boundary condition equations form a linear matrix problem which can be written in the matrix notation, $\mathsf{Ax} = \mathsf{b}$. The solution of this matrix problem gives us a good approximate solution of Poisson's equation only when one uses enough nodes to allow a gradual change in the solution from node to node.

This method applies for numerous types of boundary conditions. If we fix the potential at the two endpoints, we then have Dirichlet boundary conditions. If we assume the system wraps around, so that the potential on one side of the system equals the potential on the other side, then we have periodic boundary conditions. A simple four-node Poisson's equation system, where the potential at the boundaries (ie. outside the four nodes) is zero, generates a matrix equation of the following form:

$$\begin{bmatrix} -2/\Delta x & 1/\Delta x & 0 & 0 \\ 1/\Delta x & -2/\Delta x & 1/\Delta x & 0 \\ 0 & 1/\Delta x & -2/\Delta x & 1/\Delta x \\ 0 & 0 & 1/\Delta x & -2/\Delta x \end{bmatrix} \begin{bmatrix} \phi_{1,1} \\ \phi_{1,2} \\ \phi_{1,3} \\ \phi_{1,4} \end{bmatrix} = \begin{bmatrix} d_{1,1} \\ d_{1,2} \\ d_{1,3} \\ d_{1,4} \end{bmatrix} \ . \tag{A.2}$$

This matrix is sparse, tridiagonal and symmetric and is easily solved using any one of a numerous set of algorithms (see Chapter. 2, *Press et al.* [1988]).

These ideas are easily extended to two or three dimensions. Instead of just discretizing the space along a line into a set of points, one must discretizing the entire area or volume. In two dimensions, one usually places a simple, rectangular mesh over the area of interest and labels each node with two indices. For Poisson's equation, one does the same substitution of finite difference equations for differential equations as we done for one dimension. This results in an equation for the

potential at each node which is a function of the potentials above, below, to the left and to the right. Typically, one writes

$$\frac{\phi_{i+1,j} - 2\phi_{i,j} + \phi_{i-1,j}}{\Delta x^2} + \frac{\phi_{i,j+1} - 2\phi_{i,j} + \phi_{i,j-1}}{\Delta y^2} = \rho_{i,j} \tag{A.3}$$

A complete set of these equations generates a pentadiagonal matrix which can be solved with either the same techniques applied to the one dimensional system or a more sophisticated method, optimized for pentadiagonal matrices.

This brief tutorial should enable the reader to understand the arguments made in chapter 3. However, we have just scratched the tip of a large iceberg which hides much below the surface. We have not discussed any of numerous potential hazards and pitfalls which are endemic in numerical methods. Again, we recommend reading the references cited at the beginning of this appendix.

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Glossary

Terminology

E-**region** Electrically active region of the ionosphere, typically at altitudes from 90 to 130 km.

HF High frequency radio waves: 3MHz-30MHz

in-situ Measurment of phenomena made in its natural setting, such as by a current detector on a rocket, as opposed to a remote measurement such as those made by radar.

FFT Fast Fourier Transform

PDE Partial differential equation

Periodic Boundary Conditions The left side of the simulation connects to the right side and, in 2 dimensions, the top also connects to the bottom.

PIC "Particle-in-cell" A method for determining densities and velocities in a simulation given the positions and velocities of simulated particles (see *Birdsall and Langdon* [1985]).

SOR Successive-over-relaxation technique for solving PDEs (see Chapter 19, *Press et al.* [1988])

Type I echoes Radar reflections from Farley-Buneman (two-stream) waves.

Type II echoes Radar reflections from gradient drift waves.

VHF Very high frequency radio waves: 30MHz-300MHz

Variables

$\mathbf{B_0}$ The Earth's static magnetic field of .25 Gauss. Points in the $\hat{z}$ direction.

C_s The acoustic velocity, $\sqrt{k_b(T_i + \gamma T_e)/m_i}$

E_{0y} or $\mathbf{E_0}$ Mean DC vertical $\mathbf{E}$ field in the electrojet; typically 10mV/m to 20mV/m

k_b Boltzmann's constant, $1.38 \times 10^{-23} J/^\circ K$

k Wavenumber vector

m_i Effective ion mass (mostly O_2^+ and NO^+ in the *E*-region), about $4.6 \times 10^{-26} g$

m_e electron mass; $9.11 \times 10^{-31} kg$

n Perturbed ion or electron density. These are equal to about one part in a million for the waves described in this thesis.

n_0 Mean ion and electron density; roughly $10^{11}m^{-3}$

n_n Neutral density; roughly $6 \times 10^{18}m^{-3}$

P_i, P_e Ion or electron pressure tensors

s The log of the density perturbation, $\ln(n)$

T_e, T_i, T_n Initial ion, electron and neutral temperatures

$\mathbf{V}_d$ Electron drift velocity of $\mathbf{E}_0 \times \mathbf{B}_0/B_0^2$

$\mathbf{V}_{id}$ The ion streaming velocity of $q\mathbf{E}_0/(m_i\nu_i)$, also called the ion Pederson drift speed

ϕ Perturbed electric potential

ν_e e^--neutral collision frequency; roughly $4.0 \times 10^4 s^{-1}$

ν_i Ion-neutral collision frequency; roughly $2.8 \times 10^3 s^{-1}$

κ Ω_e/ν_e; typically around 100 in the E-region

ω Frequency in radians/second

Ω_e The electron cyclotron frequency in the earth's magnetic field (eB_0/m_e)

χ Log of the normalized density, $ln(n/n_0)$

Ψ_0 $\nu_e\nu_i/(\Omega_e\Omega_i)$